



Specification

C-10 and C-12 Truck Engines

S/N: 8YF1-UP



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Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

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i02896831

Accelerator Pedal Position Sensor

SMCS - 1913

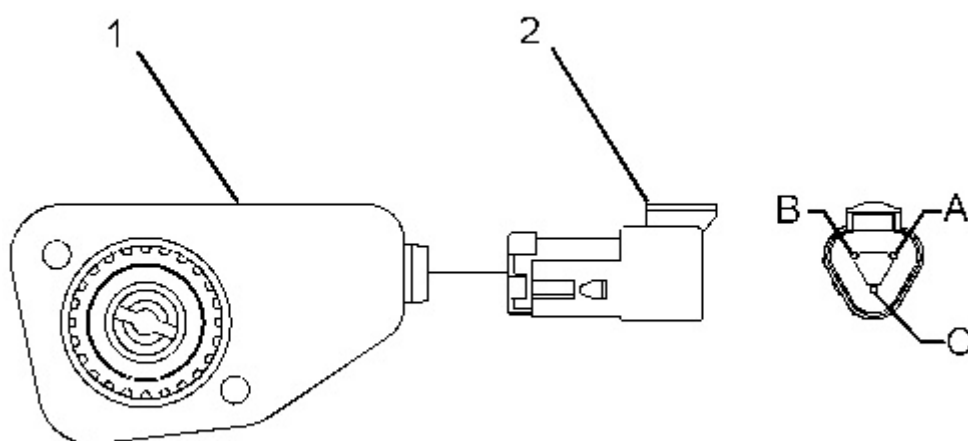


Illustration 1

g01442314

(1) Sensor assembly

(2) The plug has socket A, socket B, and socket C.

CATERPILLAR®

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i02887858

Accessory Drive (Air Compressor)

SMCS - 1207

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

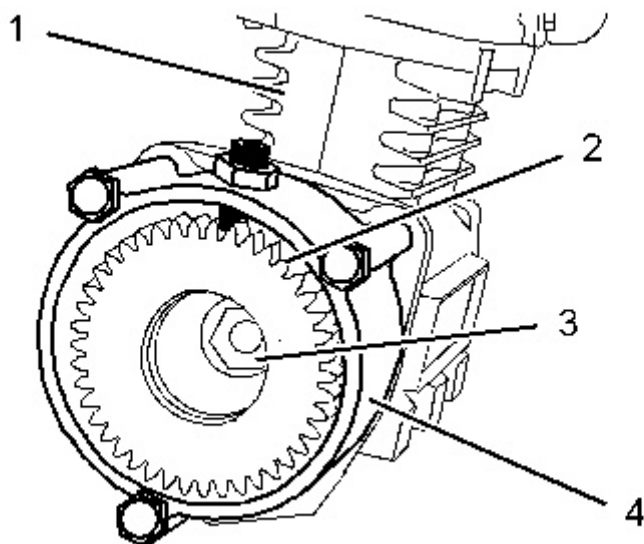


Illustration 1

g01437396

(1) Air compressor

(2) Accessory drive gear (Air compressor)

(4) Housing

Note: Caterpillar may use air compressors that have a tapered shaft or a straight shaft. The torque for the nut on the air compressor crankshaft is different for each type of crankshaft. See the following specifications for the proper torque values.

Note: Identify the manufacturer of the air compressor for your particular application.

(3) Tighten the air compressor shaft nut to the following torque.

Bendix Air Compressors

Tapered shaft

Torque the nut for the air compressor crankshaft (1) to the following torque when the air compressor crankshaft is tapered. $160 \pm 15 \text{ N}\cdot\text{m}$ ($118 \pm 11 \text{ lb ft}$) Tap the gear axially with a soft faced hammer in order to seat the gear onto the tapered crankshaft. Torque the nut for the air compressor crankshaft again to $160 \pm 15 \text{ N}\cdot\text{m}$ ($118 \pm 11 \text{ lb ft}$).

Straight shaft

Torque the nut for the air compressor crankshaft (1) to the following torque when the air compressor crankshaft is straight. $460 \pm 60 \text{ N}\cdot\text{m}$ ($340 \pm 44 \text{ lb ft}$)

Wabco 250 Air Compressors

Torque the nut for the air compressor crankshaft (1) to the following torque. $150 \pm 5 \text{ N}\cdot\text{m}$ ($111 \pm 4 \text{ lb ft}$)

Place an index mark on the nut and turn an additional 60 ± 5 degrees.



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i02896835

Air Compressor

SMCS - 1803

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

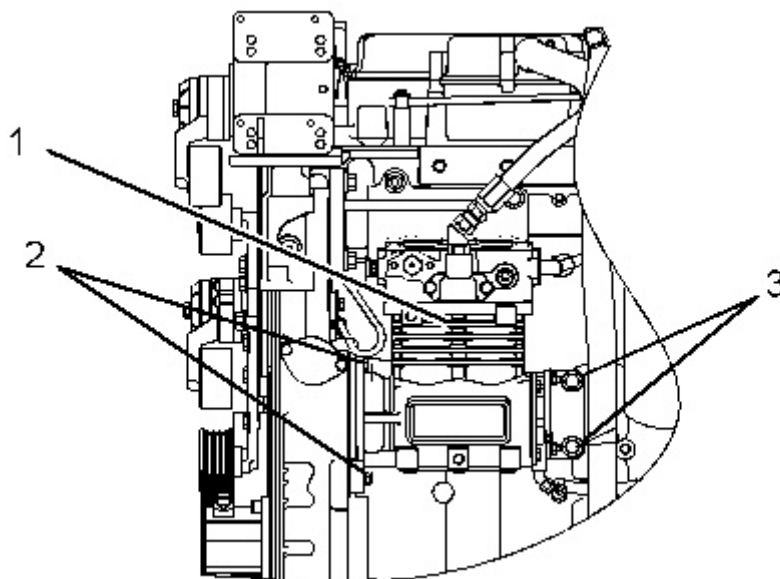


Illustration 1

g01442319

(1) Air compressor

(2) Bolts

Apply **9S-3263** Thread Lock Compound to the threads of the bolts. Tighten the bolts to the following torque. ... $100 \pm 20 \text{ N}\cdot\text{m}$ ($75 \pm 15 \text{ lb ft}$)

(3) Bolts

Tighten the bolts to the following torque. ... $55 \pm 10 \text{ N}\cdot\text{m}$ ($41 \pm 7 \text{ lb ft}$)



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i02888351

Alternator and Regulator

SMCS - 1405; 1410

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

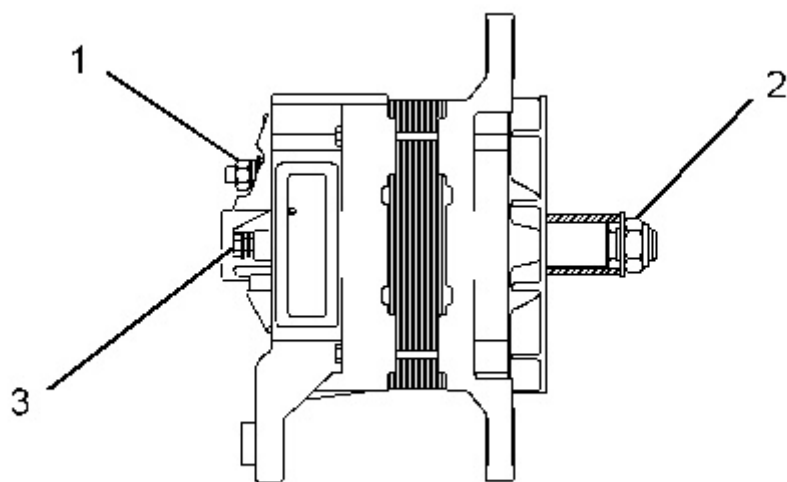


Illustration 1

g01437930

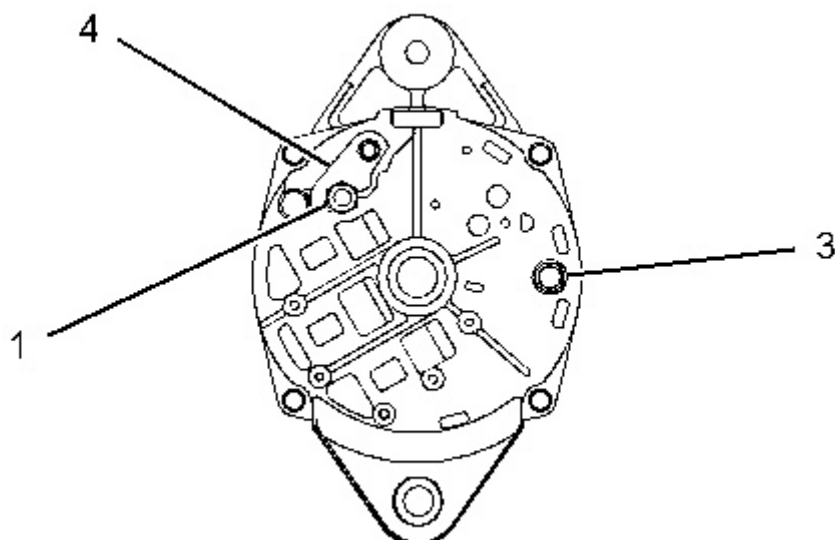


Illustration 2

g01437935

Note: Load the battery with a carbon pile **4C-4911** Battery Load Tester in order to get the maximum alternator output.

Polarity ... Negative Ground

Rotation ... Clockwise

Minimum full load current at 5000 rpm ... 107 Amp

Minimum full load current at 1500 rpm ... 45 Amp

Output voltage ... 14.0 ± 0.5 V

(1) Positive terminal torque ... 11.3 ± 2.3 N·m (100 ± 20 lb in)

(2) Shaft nut torque ... 102 ± 7 N·m (75 ± 5 lb ft)

(3) Negative terminal torque ... 6.2 ± 0.6 N·m (55 ± 5 lb in)

(4) Regulator

Voltage setting ... No Adjustment

Permissible voltage range ... 13.5 to 14.5 V



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i02841226

Atmospheric Pressure Sensor

SMCS - 1923

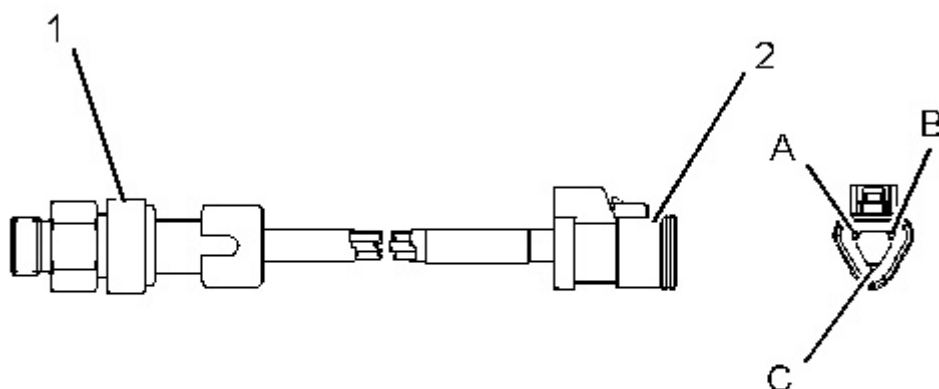


Illustration 1

g01417163

(1) Sensor assembly

Tighten the sensor assembly to the following torque. ... 10 ± 2 N·m (90 ± 18 lb in)

(2) The plug has socket A, socket B, and socket C.



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i02888251

Auxiliary Drive Pulley

SMCS - 1205

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

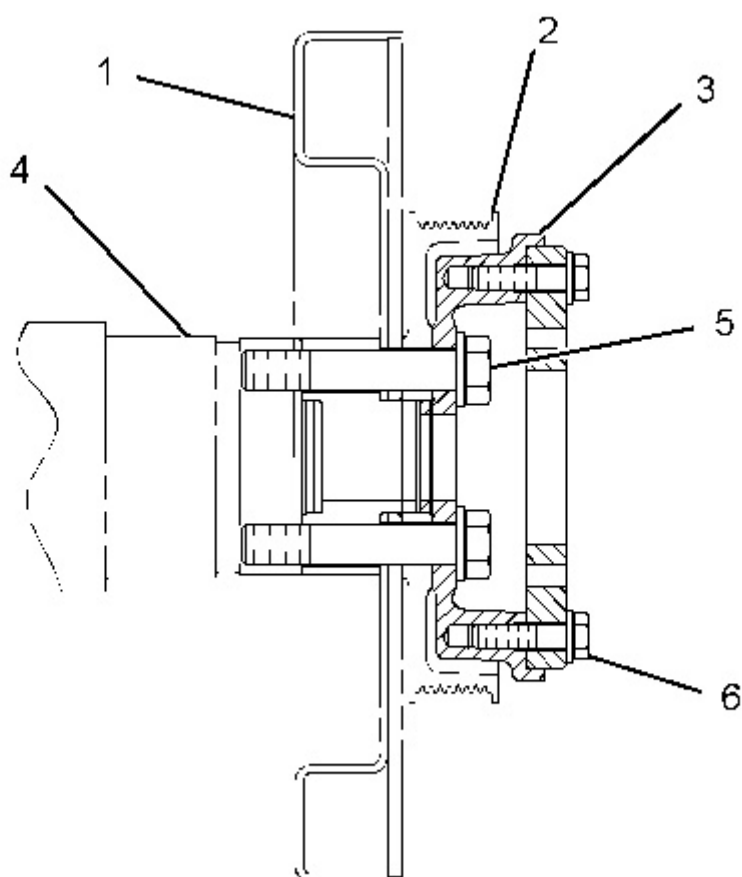


Illustration 1

g01437892

(1) Damper assembly

(2) Pulley

(3) Adapter

(4) Crankshaft

(5) Tighten the bolts to the following torque. ... $300 \pm 40 \text{ N}\cdot\text{m}$ ($220 \pm 30 \text{ lb ft}$)(6) Tighten the bolts to the following torque. ... $70 \pm 15 \text{ N}\cdot\text{m}$ ($50 \pm 11 \text{ lb ft}$)



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i05084397

Belt Tension Chart

SMCS - 1357

Table 1

Specification for the 7N-9693 Tension Chart			
Item	Qty	Part	Specification Description
-	-	-	The new belt tension chart has been standardized. The belt tension chart eliminates confusion about the correct belt tension to use. Studies were completed with different belt suppliers and the information that was gathered from the studies was developed into a new belt tension for each belt size. The chart does not apply to belts that use a spring loaded tensioner. For more accurate results, a Clavis frequency gauge should be used in order to measure the natural frequency of the belt. By the use of a formula, the natural frequency is converted into the belt tension force. If the Clavis frequency gauge cannot be used, then use the appropriate Kent-Moore belt tension gauge to measure the belt tension force. When matched sets of belts are used, check the tension of all the belts in order to verify that the lowest belt is in the acceptable tolerance range. Variations in tension between the belts can vary by as much as 65 pounds. In order to determine the tension of a "banded" belt, refer to the chart above. Multiply the tension value of the belt by the number of belts that are banded together. Two 1/2 inch belts would require a used belt tension equal to twice the value of the used belt tension for one 1/2 inch belt (2 X 80 = 160 lb).

Table 2

Belt Tension Chart				
SAE or RMA Belt Size	Width of Belt (Top Reference)	Belt Tension "Initial" ⁽¹⁾	Belt Tension Used ⁽²⁾	Belt Tension "Used" ^{(3) (4)}

IN	MM	IN	MM	LB	N	LB	N	LB	N
3/8 (0.380)	10A	0.42	10.72	145±10	645±44	115±10	510±44	70±10	311±44
7/16 (0.440)	11A	0.46	11.68	155±10	689±44	125±10	560±10	75±10	333±44
1/2 (0.500)	13A	0.55	13.89	165±10	734±44	130±10	580±10	80±10	356±44
5V/5VX	5V	0.63	15.88	180±10	801±44	145±10	645±10	85±10	378±44
11/16 (0.600)	15A	0.69	17.48	180±10	801±44	145±10	645±10	85±10	378±44
3/4 (0.660)	17A	0.75	19.05	205±10	912±44	165±10	730±44	95±10	422±44
7/8 (0.790)	20A	0.88	22.23	205±10	912±44	165±10	730±44	95±10	422±44
15/16	N/A	0.98	23.83	205±10	912±44	165±10	730±44	95±10	422±44
4K	4PK	0.54	13.72	145±10	645±44	115±10	510±44	60±10	267±44
6K	6PK	0.82	20.94	215±10	956±44	170±10	755±10	90±10	400±44
8K	8PK	1.1	27.82	285±10	1179±44	230±10	1025±10	115±10	511±44
10K	10PK	1.38	35.05	355±10	1579±44	285±10	1270±10	145±10	645±44
12K	12PK	1.66	42.16	425±10	1891±44	340±10	1510±10	175±10	778±44
14K	14PK	1.94	49.28	495±10	2202±44	395±10	1755±10	205±10	912±44
15K	15PK	2.08	52.83	530±10	2358±44	425±10	1890±44	215±10	956±44

⁽¹⁾ Belt Tension "Initial " is for a new belt.

⁽²⁾ Belt tension with less than 10 hours of operation

⁽³⁾ Belt tension "Used" is for a belt that has operated for more than 30 minutes of operation at the rated speed..

⁽⁴⁾ If a belt falls below the "Used" belt tension, the belt should be tightened again to 80% of the "Initial" belt tension.



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i02888195

Belt Tensioner

SMCS - 1358

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

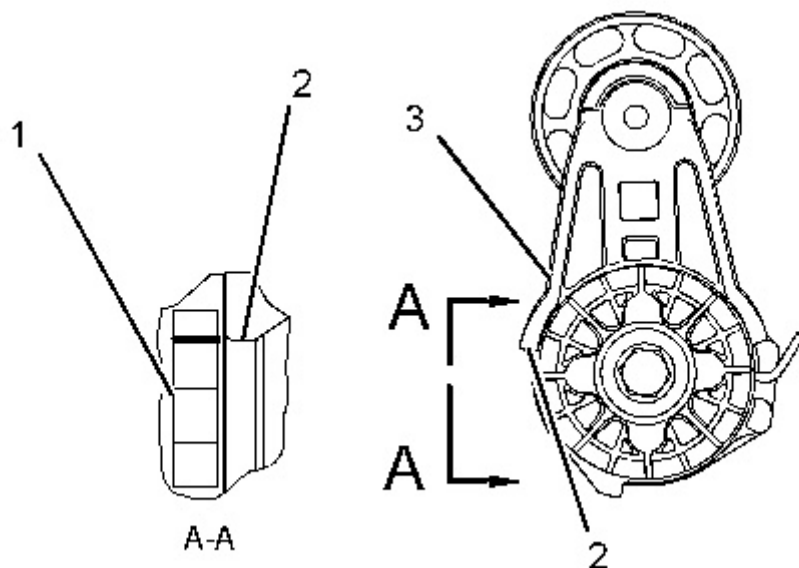


Illustration 1

g01437844

Use the following procedure in order to adjust the belt tensioner:

1. Remove the belt so that there is no tension on the belt tensioner.

2. Align the thick black line on **134-1018** Film (1) with shoulder (2) of the tensioner's arm (3) .
 3. Install the belt. Shoulder (2) should be in the green zone of **134-1018** Film (1) .
-



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Belt Tensioner

SMCS - 1358

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

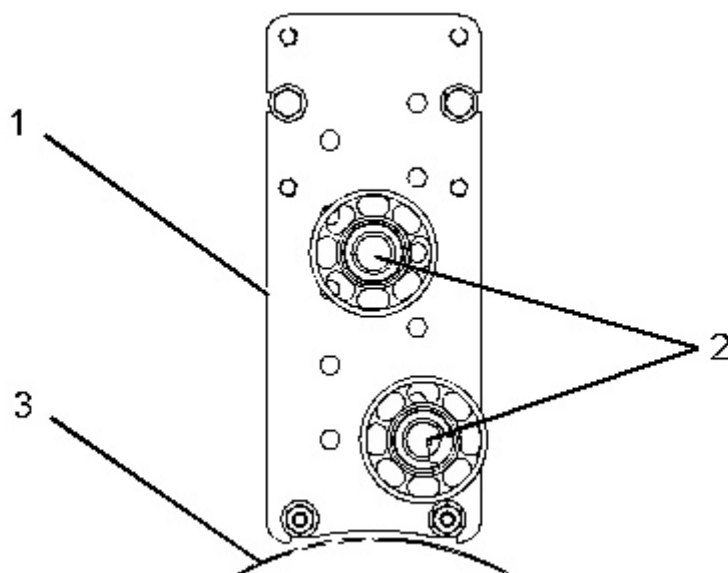


Illustration 1

g01437845

(3) Damper

(1) Plate

Note: Install plate (1) with "SIDE A" toward the front of the engine.

(2) Tighten the bolts to the following torque. ... $85 \pm 5 \text{ N}\cdot\text{m}$ ($65 \pm 4 \text{ lb ft}$)



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i02888210

Belt Tensioner

SMCS - 1358

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

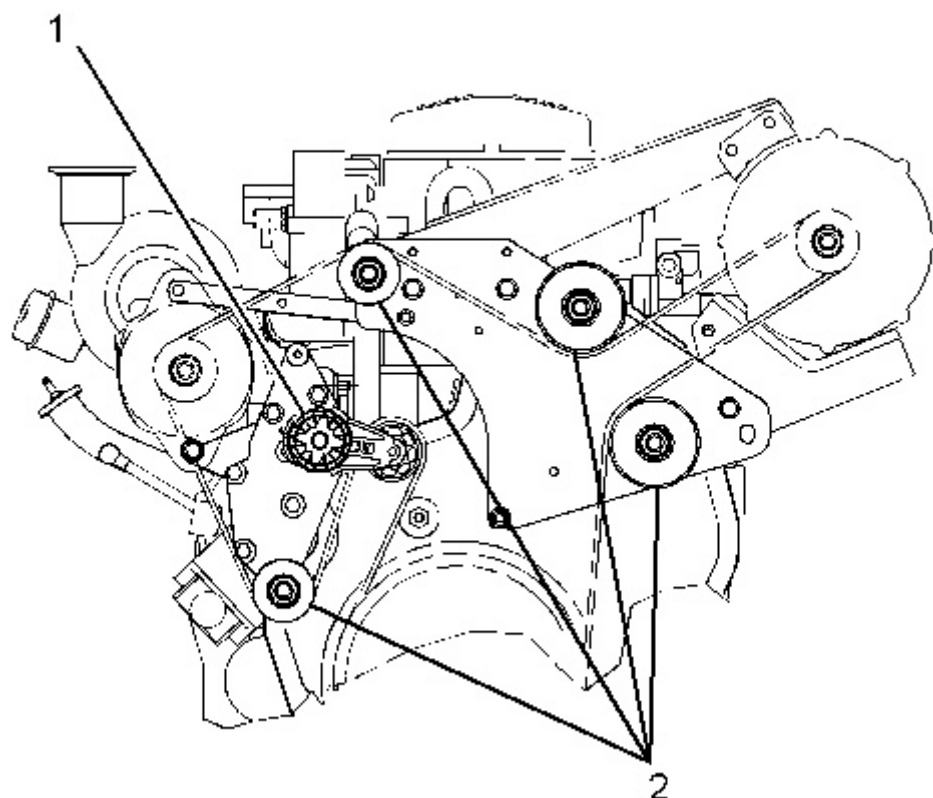


Illustration 1

g01437862

Front view of engine

(1) Belt tensioner

Tighten the bolt that fastens the belt tensioner to the bracket to the following torque. ... $50 \pm 5 \text{ N}\cdot\text{m}$ ($37 \pm 4 \text{ lb ft}$)

(2) Idler pulley assembly

Tighten the idler pulley assembly to the following torque. ... $50 \pm 5 \text{ N}\cdot\text{m}$ ($37 \pm 4 \text{ lb ft}$)



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i02888194

Belt Tensioner

SMCS - 1358

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

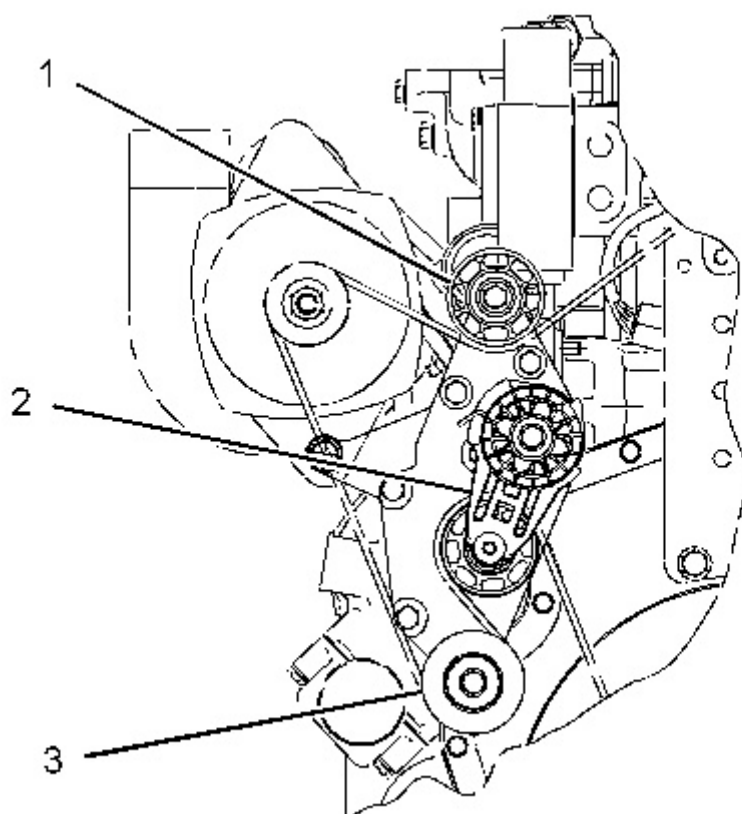


Illustration 1

g01437842

(2) Belt tensioner

(1) Idler pulley assembly

Tighten the idler pulley assembly to the following torque. ... $50 \pm 5 \text{ N}\cdot\text{m}$ ($37 \pm 4 \text{ lb ft}$)

(3) Idler pulley assembly

Tighten the idler pulley assembly to the following torque. ... $50 \pm 5 \text{ N}\cdot\text{m}$ ($37 \pm 4 \text{ lb ft}$)



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i02888240

Belt Tightener

SMCS - 1358

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

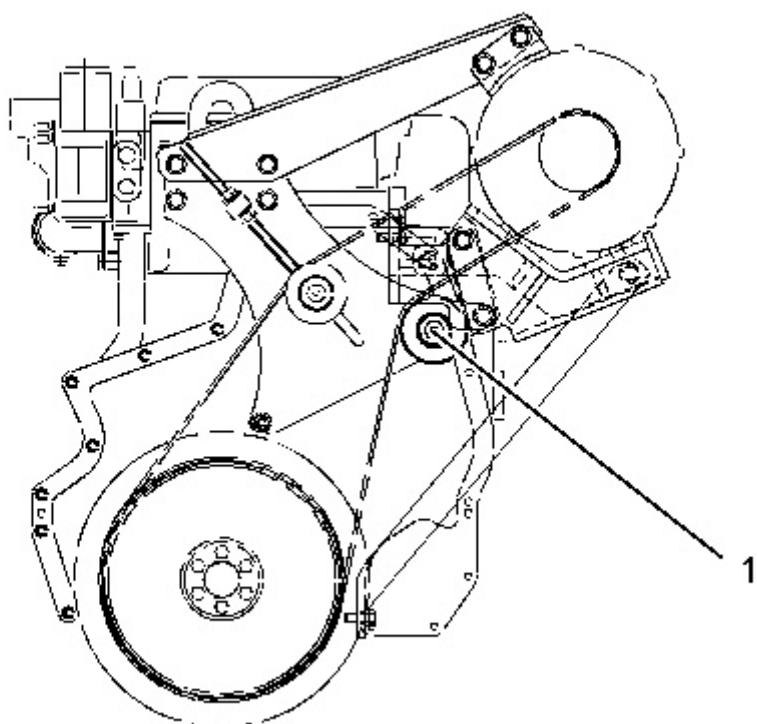


Illustration 1

g01437872

(1) Tighten the bolt to the following torque. ... $85 \pm 5 \text{ N}\cdot\text{m}$ ($65 \pm 4 \text{ lb ft}$)



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i02888504

Boost Pressure Sensor

SMCS - 1917

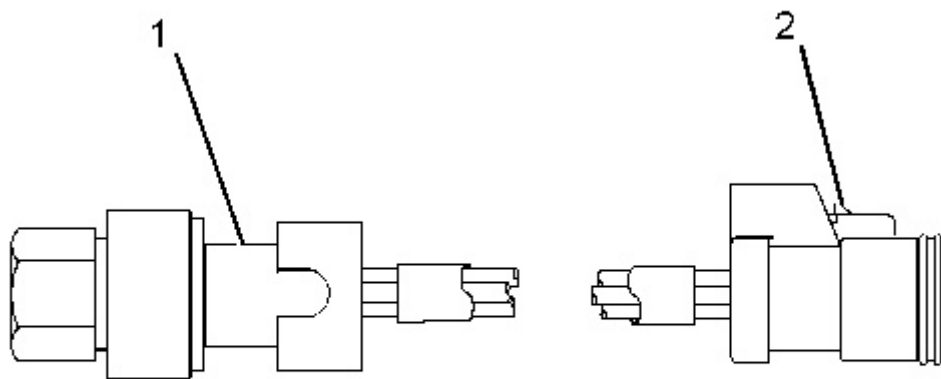


Illustration 1

g01438045

(2) Plug

(1) Sensor assembly

Tighten the sensor assembly to the following torque. ... $10 \pm 2 \text{ N}\cdot\text{m}$ ($88 \pm 18 \text{ lb in}$)



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i02886415

Camshaft

SMCS - 1210

S/N - 2KS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

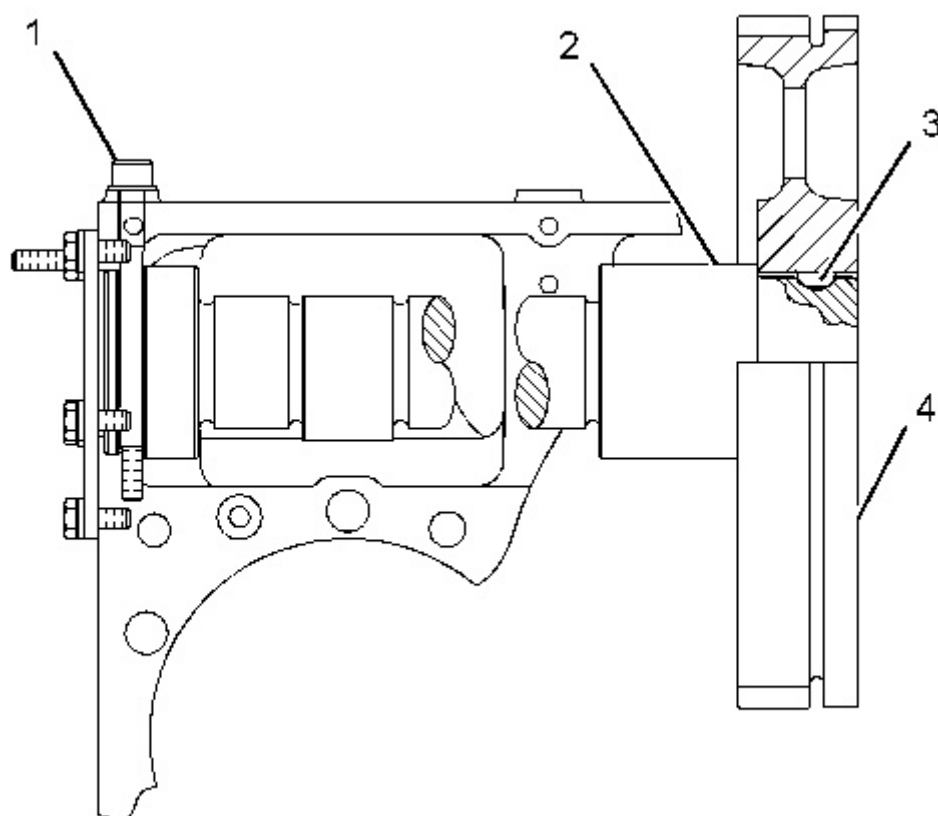


Illustration 1

g01436825

(3) Key

(4) Gear

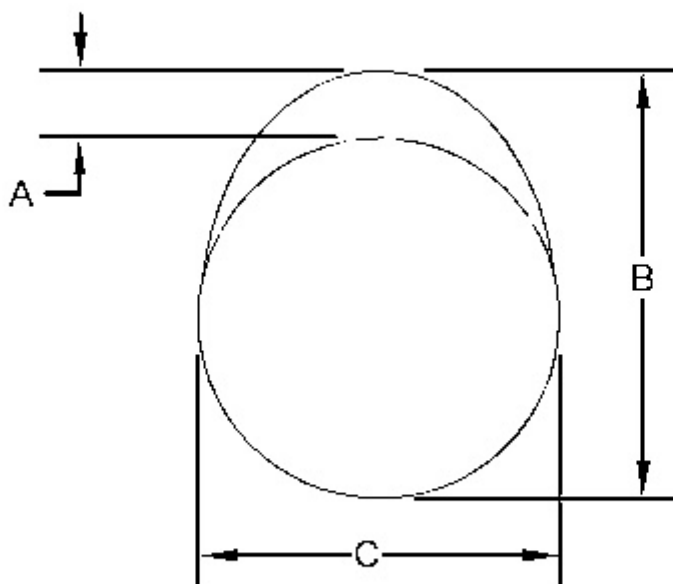
(1) Torque for thrust pin ... $28 \pm 7 \text{ N}\cdot\text{m}$ ($20 \pm 5 \text{ lb ft}$)(2) Diameter of new camshaft journals ... $74.850 \pm 0.025 \text{ mm}$ ($2.9468 \pm .0010 \text{ inch}$)

Illustration 2

g01436826

(A) Camshaft lobe lift

(B) Camshaft lobe height

(C) Base circle

(A) Camshaft lobe lift

Specified camshaft lobe lift (A)

Exhaust lobe ... 9.550 mm (.3760 inch)Inlet lobe ... 11.371 mm (.4477 inch)Injector lobe ... 11.275 mm (.4439 inch)

Specified base circle (C)

Exhaust lobe ... 55.0 ± 0.4 mm ($2.17 \pm .02$ inch)

Inlet lobe ... 51.0 ± 0.4 mm ($2.01 \pm .02$ inch)

Injector lobe ... $51.0 \pm .4$ mm ($2.01 \pm .02$ inch)

To find the lobe lift, use the procedure that follows:

1. Measure the camshaft lobe height (B) .
2. Measure the base circle (C) .
3. Subtract the base circle that is found in Step 2 from the lobe height that is found in Step 1.
The difference is the actual camshaft lobe lift (A) .

Maximum permissible difference between actual lobe lift and the specified lobe lift of a new camshaft ... ± 0.13 mm (.005 inch)



Specifications

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i05398656

Compression Brake

SMCS - 1119; 1129

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

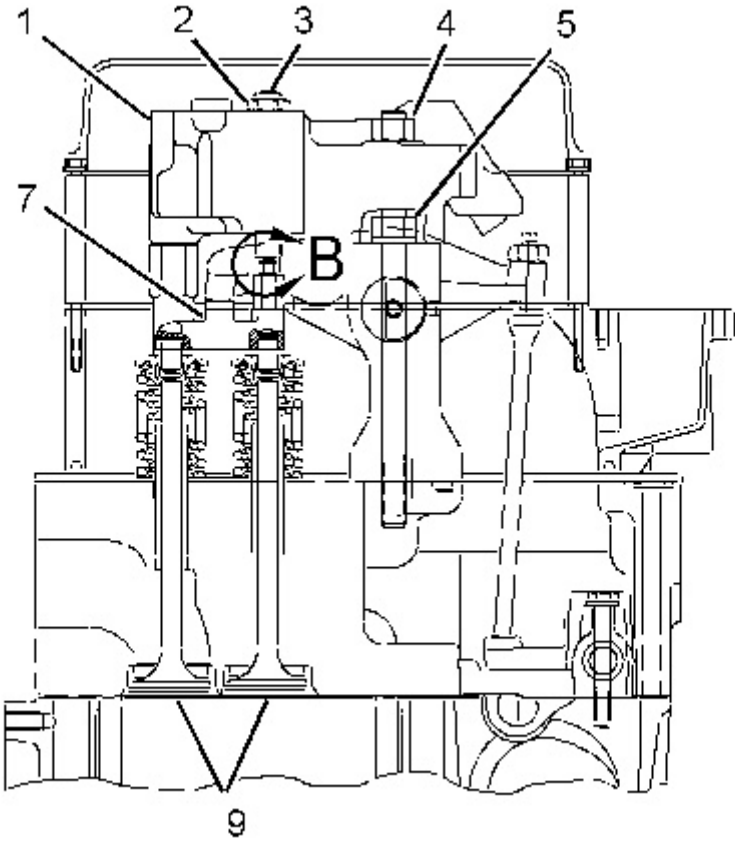


Illustration 1

g01439044

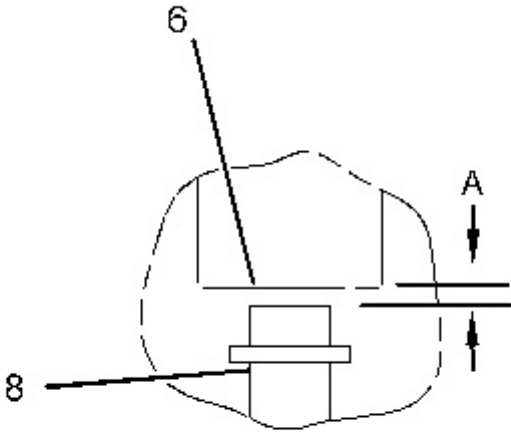


Illustration 2

g01439048

Detail B

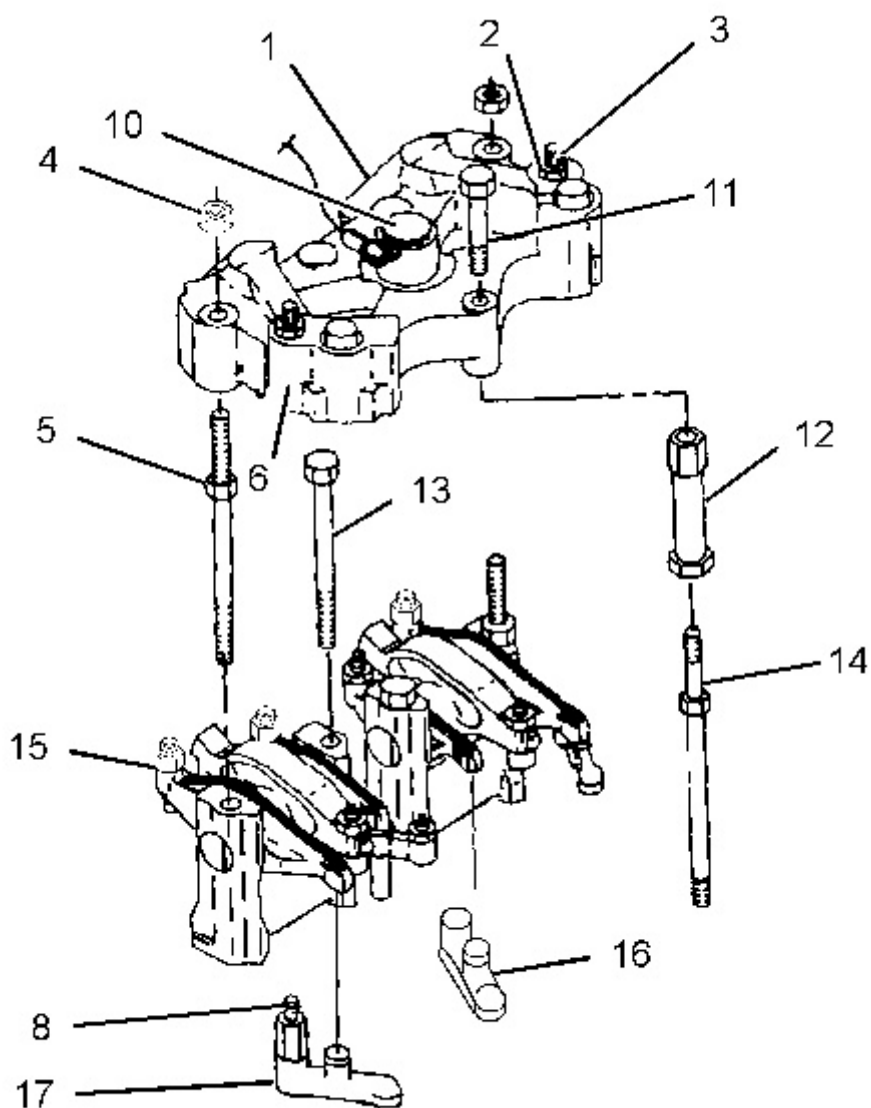


Illustration 3

g01439066

- (1) Compression brake
- (3) Slave piston adjustment screw
- (6) Slave piston
- (7) Exhaust valve bridge
- (9) Exhaust valves
- (16) Inlet valve bridge
- (17) Exhaust valve bridge

(2) Locknut

Tighten the locknut to the following torque. ... $35 \pm 8 \text{ N}\cdot\text{m}$ ($26 \pm 6 \text{ lb ft}$)

(4) Hold-down nut

Tighten the hold-down nuts to the following torque. ... $80 \pm 15 \text{ N}\cdot\text{m}$ ($60 \pm 11 \text{ lb ft}$)

(5) Stud

Tighten the studs to the following torque. ... $100 \pm 20 \text{ N}\cdot\text{m}$ ($75 \pm 15 \text{ lb ft}$)

Note: Prior to installation of the compression brake, install stud (5) and tighten stud (5) to the recommended torque.

(A) Slave piston lash setting

Note: The valve lash must be adjusted before adjusting the slave piston lash.

Table 1

Slave Piston Lash Setting Jake Brake Compression Brakes	
Serial Number Prefix	Slave Piston Lash Setting
3CS1-UP 2PN1-UP 8YS1-UP	$0.64 \pm 0.08 \text{ mm}$ ($0.025 \pm 0.003 \text{ inch}$)
CPD1-UP 9SM1-UP 2KS1-UP SRK1-UP 9NS1-UP 8YF1-UP	$1.02 \pm 0.08 \text{ mm}$ ($0.040 \pm 0.003 \text{ inch}$)
1YN1-UP	$1.14 \pm 0.08 \text{ mm}$ ($0.045 \pm 0.003 \text{ inch}$)
MBJ1-UP MBL1-UP	$1.40 \pm 0.08 \text{ mm}$ ($0.055 \pm 0.003 \text{ inch}$)

Adjust the slave piston lash while the engine is stopped and the exhaust valves are closed.

Refer to the Testing and Adjusting, "Compression Brake" for the adjustment procedure.

(8) Screw assembly (if equipped)

Apply **4C-9507** Retaining Compound to the threads of the screw assembly.

Tighten the locknut to the following torque. ... $25 \pm 7 \text{ N}\cdot\text{m}$ ($18 \pm 5 \text{ lb ft}$)

(10) Solenoid valve

There are two different designs of solenoid valves. The difference can be determined by the number of points on the head of the solenoid valve. Use the correct torque value for your solenoid valve.

Tighten the solenoid valve with a twelve point head to the following torque. ... $12.5 \text{ N}\cdot\text{m}$ (110 lb in)

Tighten the solenoid valve with a six point head to the following torque. ... $20 \text{ N}\cdot\text{m}$ (15 lb ft)

(11) Bolt

Tighten the bolt to the following torque. ... $55 \pm 10 \text{ N}\cdot\text{m}$ ($41 \pm 7 \text{ lb ft}$)

(12) Spacer

Tighten the spacer to the following torque. ... $100 \pm 20 \text{ N}\cdot\text{m}$ ($75 \pm 15 \text{ lb ft}$)

(13) Bolt

Tighten the bolt to the following torque. ... $100 \pm 20 \text{ N}\cdot\text{m}$ ($75 \pm 15 \text{ lb ft}$)

(14) Head bolt stud

Refer to the Specifications, "Cylinder Head" topic for the tightening procedure.

(15) Adjustment screw locknut for the valve lash clearance adjustment

Refer to the Specifications, "Valve Mechanism" for the adjustment procedure.



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i01003510

Connecting Rod Bearing Journal

- SMCS - 1202; 1219; 1225
- S/N - 2KS1-UP
- S/N - 3CS1-UP
- S/N - 8YF1-UP

Table 1

Connecting Rod Bearing Journal	
Original size journal	82.000 ± 0.020 mm (3.2283 ± .0008 inch)
Undersize journal 0.508 mm (.0200 inch)	81.492 ± 0.020 mm (3.2083 ± .0008 inch)
Undersize journal 0.762 mm (.0300 inch)	81.238 ± 0.020 mm (3.1983 ± .0008 inch)

The clearance between a new bearing and a new journal is the following value. ... 0.046 to 0.116 mm (.0018 to .0046 inch)



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i02845742

Connecting Rod

SMCS - 1218

S/N - 2KS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

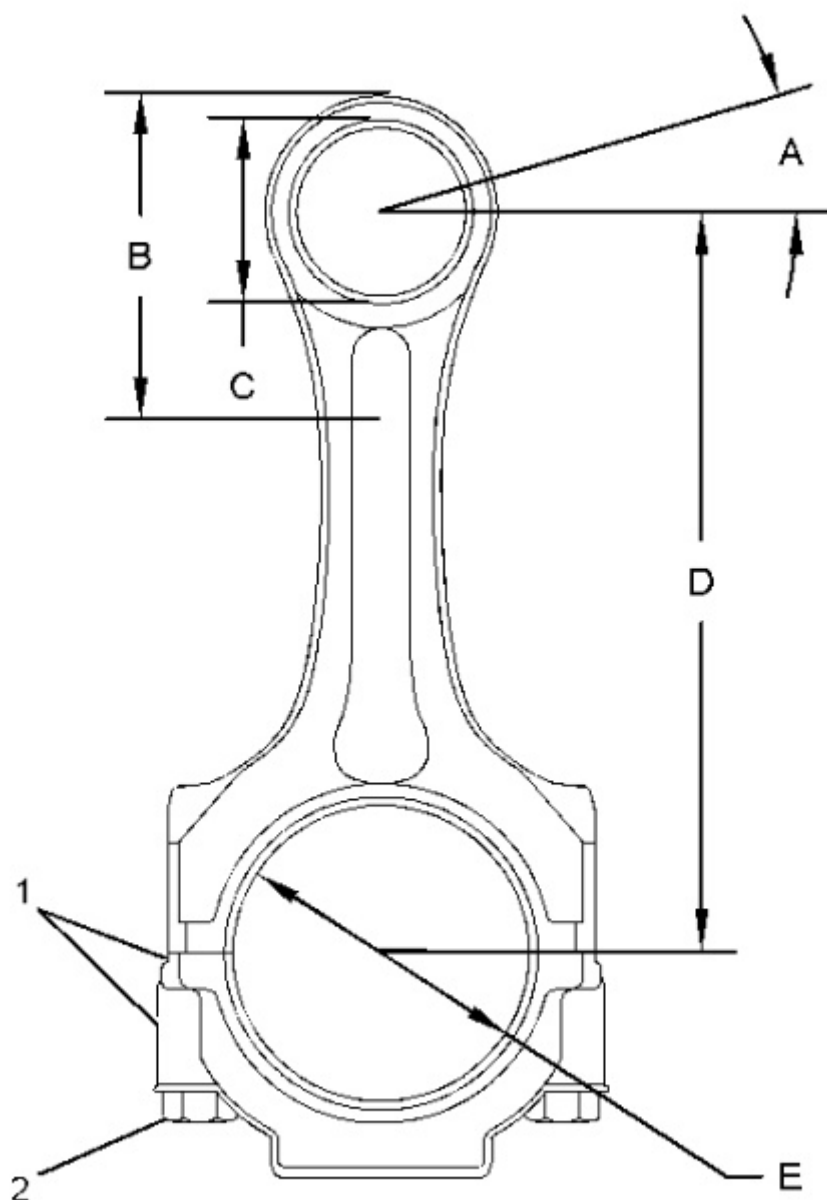


Illustration 1

g01418662

(A) The bearing joint for the piston pin bearing must be positioned to the following angle above the centerline on either side. ... 12.5 ± 5.0 degrees

Note: The connecting rod must be heated for the installation of the piston pin bearing. Do not use a torch.

(B) The connecting rod may be heated from 175 to 260°C (347 to 500°F) for the installation of the piston pin bearing. Maximum distance for heating the connecting rod ... 85.0 mm (3.35 inch)

Note: Thoroughly lubricate the piston pin with clean engine oil prior to assembly of the piston and connecting rod.

(C) Bore in the piston pin bearing after installation of the bearing ... 53.205 ± 0.008 mm (2.0947 ± 0.0003 inch)

Bore in the connecting rod for the piston pin bearing ... 57.810 ± 0.013 mm (2.2760 ± 0.0005 inch)

Diameter of the piston pin ... 53.170 ± 0.005 mm (2.0933 ± 0.0002 inch)

(D) Distance between the center of the bearings ... 242.5 mm (9.55 inch)

(E) Bore in the connecting rod for the bearing for the crankshaft connecting rod journal ... 93.800 ± 0.013 mm (3.6929 ± 0.0005 inch)

(1) Etch the cylinder number on the connecting rod and the cap in this location. Mark the connecting rod and the cap with a number 1 through 6. Mark the numbers on the same side of the connecting rod as the bearing retainer notch.

Note: Install the connecting rod in the engine with the part number to the rear of the engine.

Use the following procedure to tighten the connecting rod bolts (2):

1. Lubricate the threads of the bolts and the seating faces of the caps with **4C-5592** Anti-Seize Compound .
 2. Torque the bolts to 130 ± 7 N·m (95 ± 5 lb ft).
 3. Place an index mark on each connecting rod cap bolt.
 4. Rotate each bolt for an additional 60 ± 5 degrees ($1/6$ of a turn).
-



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i02886810

Coolant Conditioner Base

SMCS - 1352

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

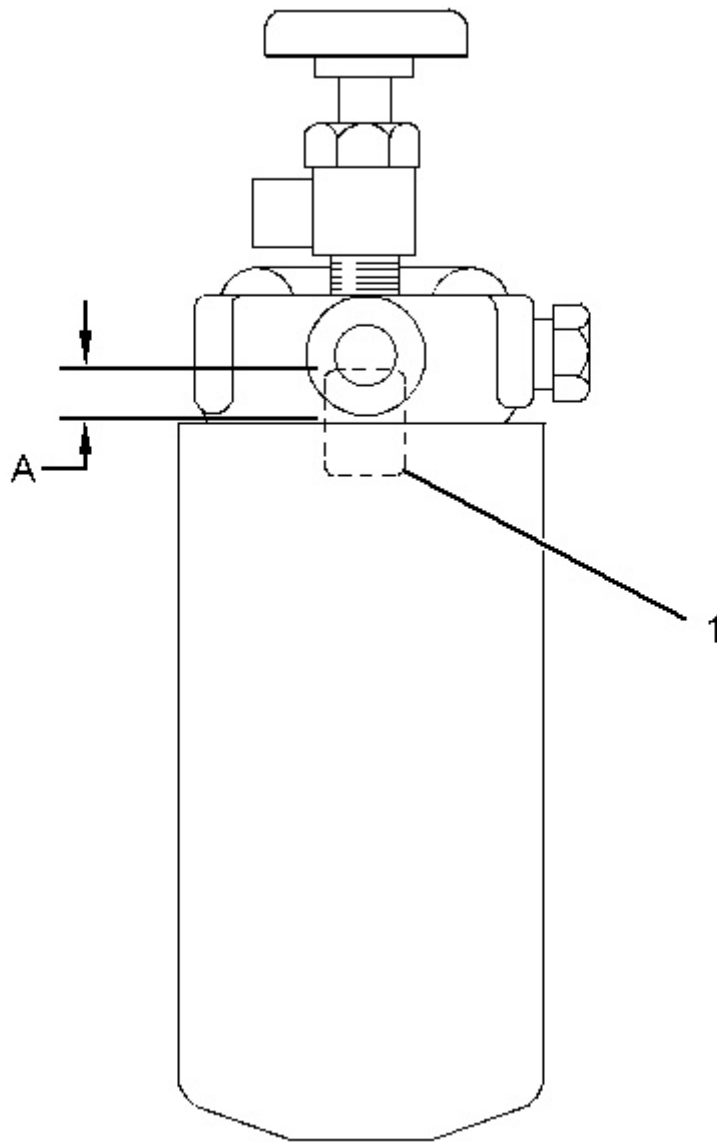


Illustration 1

g01437078

(A) Apply **9S-3263** Thread Lock Compound on the tapered end of the stud to the following distance. ... 7.6 mm (.30 inch)

(1) Tighten the stud to the following torque. ... 60 ± 7 N·m (44 ± 5 lb ft)



Specifications
C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i04448859

Coolant Temperature Sensor

SMCS - 1906

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

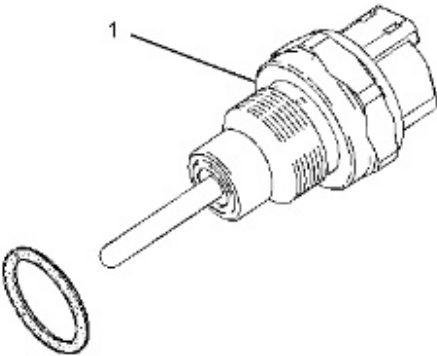


Illustration 1

g02574738

Table 1

Specification for 130-9811 Temperature Sensor Gp , 274-7396 Temperature Sensor Gp , 197-8392 Temperature Sensor Gp, and 256-6453 Temperature Sensor			
Item	Qty	Part	Specification Description
1	1	-	Torque to 20 ± 3 N·m (175 ± 25 lb in).
			Output type is passive.



Specifications
C-10 and C-12 Truck Engines
Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i02650568

Crankshaft Seals

SMCS - 1160; 1161

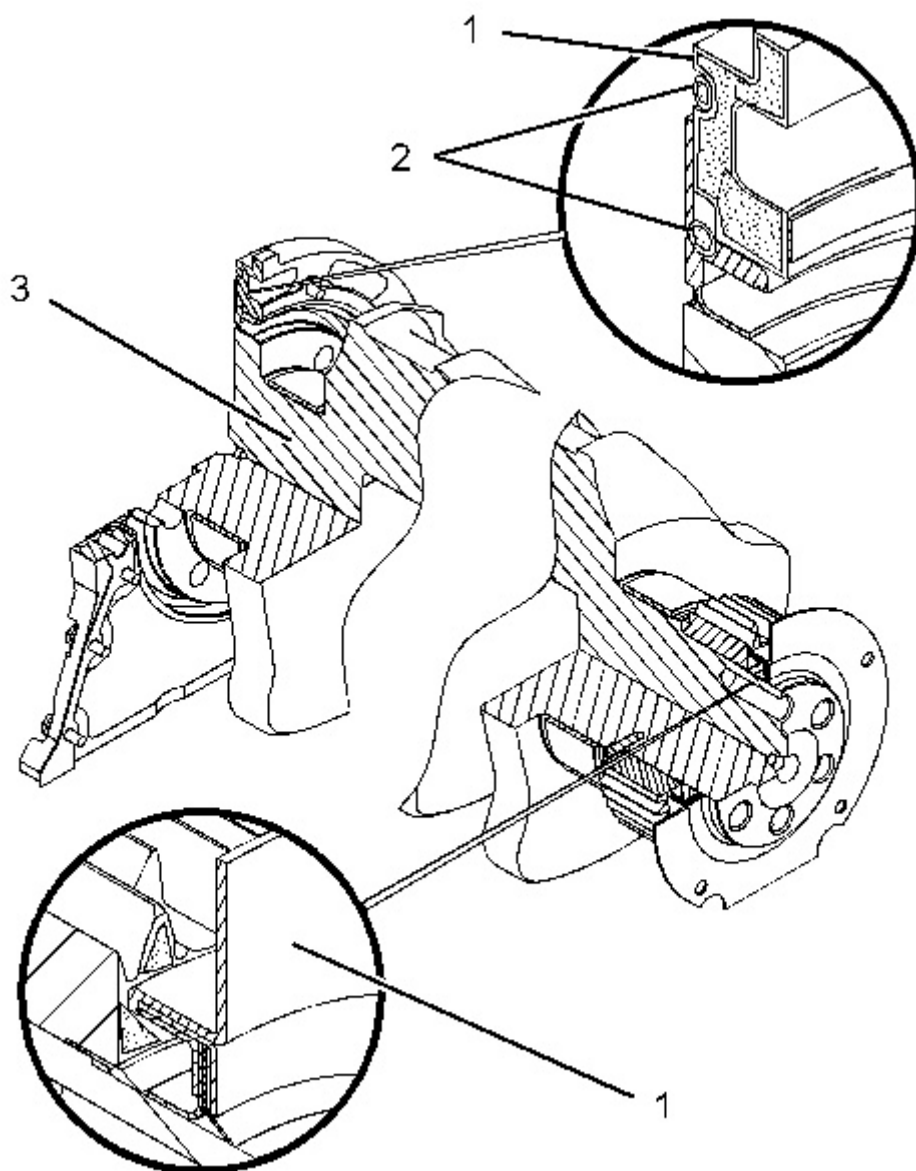


Illustration 1

g01332511

(1) Crankshaft seal group

(3) Crankshaft

Note: Install the front crankshaft seals and the rear crankshaft seals dry.

(2) O-ring seal ... Lightly lubricate the seal with the lubricant that is being sealed.

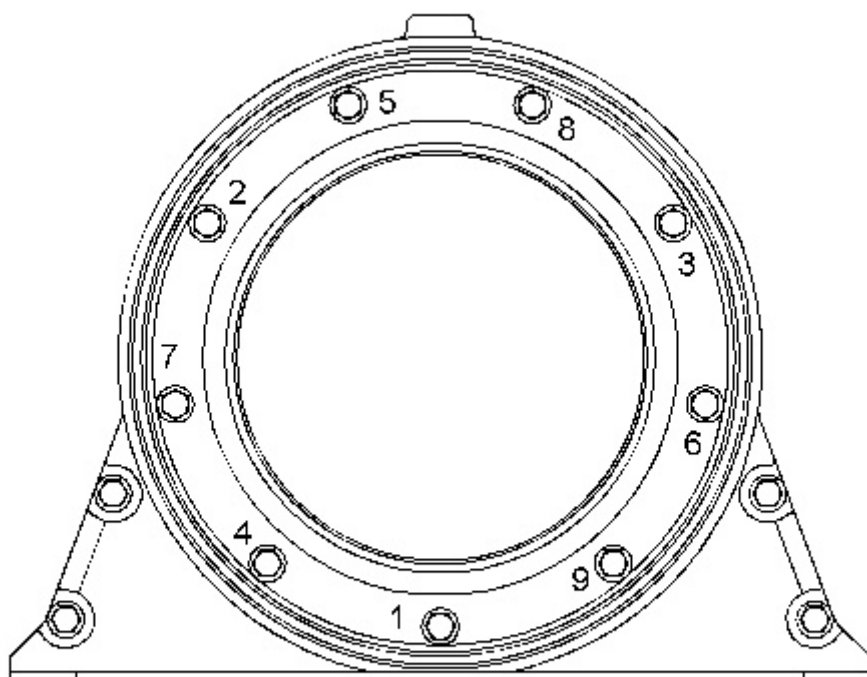


Illustration 2

g01331026

Note: Use the alternating sequence that is shown when tightening the front and rear seal bolts.



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i02886857

Crankshaft

SMCS - 1202

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

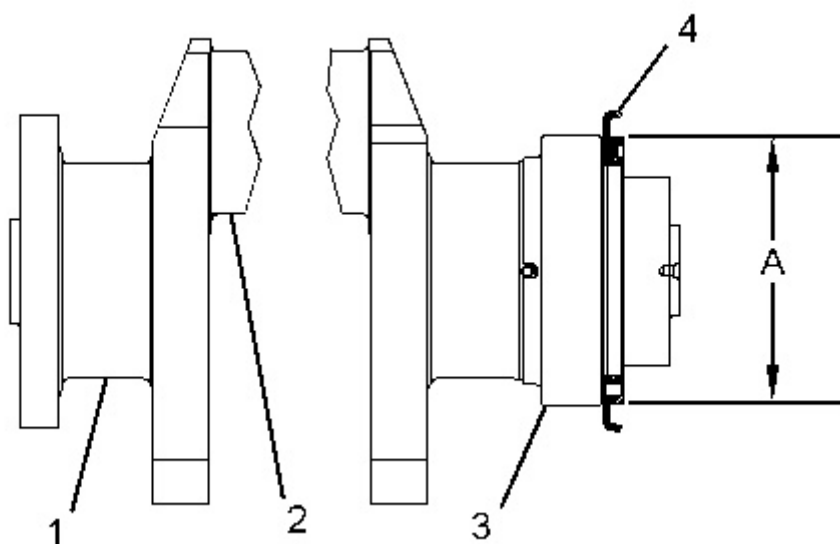


Illustration 1

g01437084

(1) Main bearing journals

(2) Connecting rod bearing journals

(4) Plate assembly (sensor)

Note: The thrust plate is used on the center main bearing only.

(3) Gear

In order to install the gear, heat the gear to the following temperature. ... 210 °C (410 °F)

Note: Heat the gear for no more than one hour.

(A) Average diameter of gear after assembly ... 136.20 ± 0.11 mm ($5.362 \pm .004$ inch)

Maximum diameter of gear after assembly ... 136.36 mm (5.368 inch)

Note: After the crankshaft is assembled into the block, the crankshaft end play should be a maximum of 0.50 mm (.020 inch) and a minimum of 0.10 mm (.004 inch).



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

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i02971360

Cylinder Block

SMCS - 1201

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

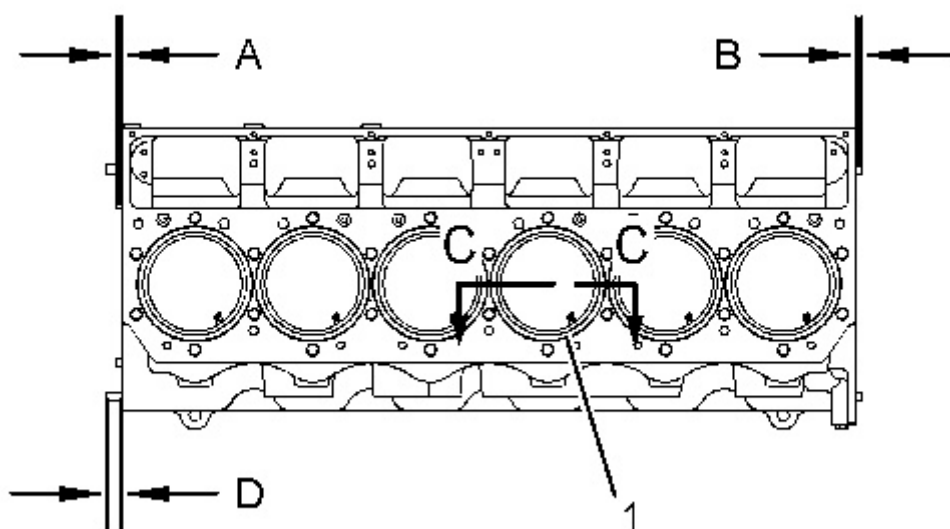


Illustration 1

g01413469

(1) Bore for the cylinder liner

(A) The dowels extend past the end of the cylinder block by the following distance. ... 10.0 ± 0.5 mm (0.39 ± 0.02 inch)

(B) The dowels extend past the end of the cylinder block by the following distance. ... 10.0 ± 0.5 mm (0.39 ± 0.02 inch)

(D) The dowels extend past the end of the cylinder block by the following distance. ... 28.0 ± 0.5 mm (1.10 ± 0.02 inch)

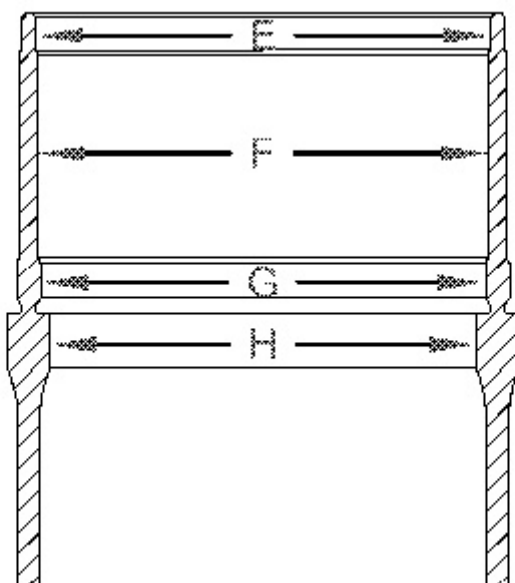


Illustration 2

g01413742

Section C-C

Bore sizes in the cylinder block prior to installation of the cylinder liner:

Dimension (E) ... 151.50 ± 0.03 mm (5.965 ± 0.001 inch)

Dimension (F) ... 149.80 ± 0.30 mm (5.898 ± 0.012 inch)

Dimension (G) ... 148.00 ± 0.03 mm (5.827 ± 0.001 inch)

Dimension (H) ... 141.90 ± 0.15 mm (5.587 ± 0.006 inch)

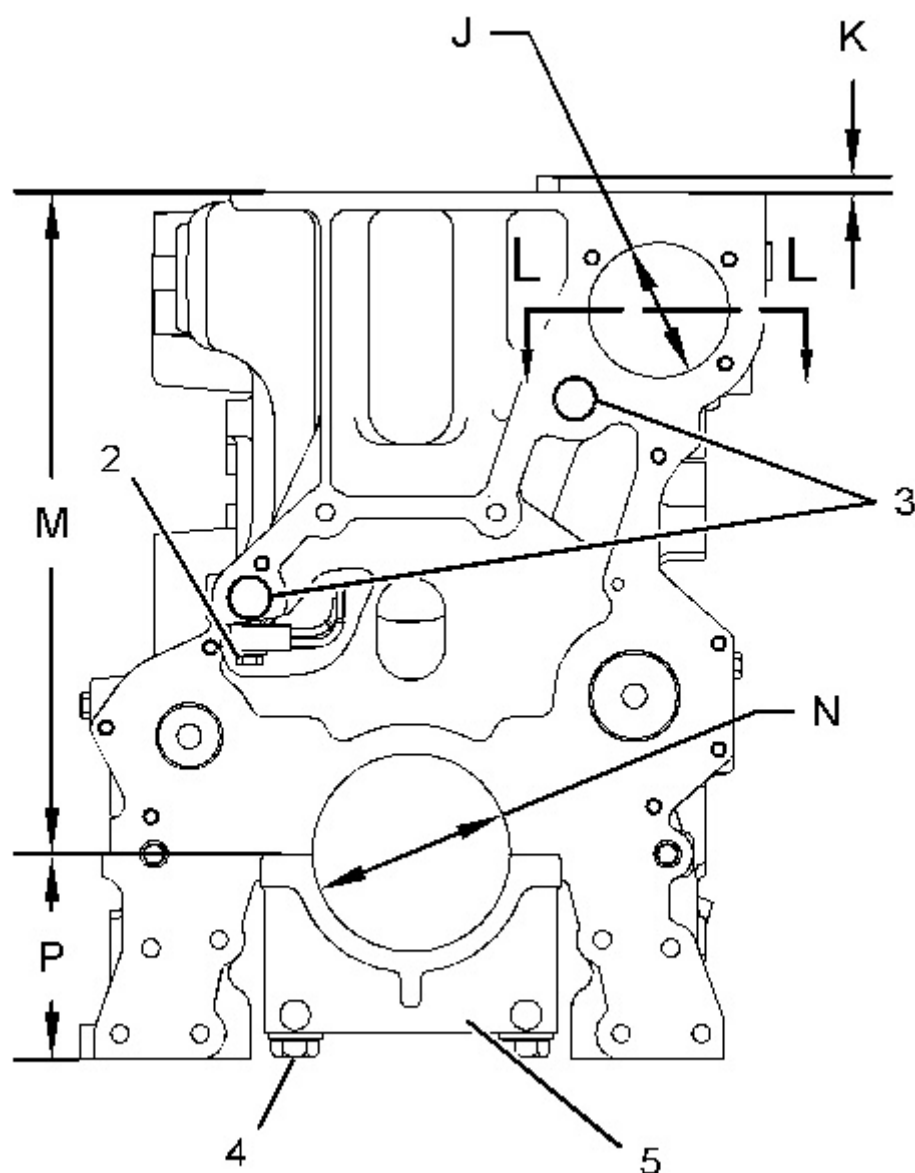


Illustration 3

g01413748

Front face of the cylinder block

The flatness of the top contact surface of the cylinder block must be within 0.05 mm (0.002 inch) for any 150 mm (5.9 inch) section of the surface.

(2) Torque for the bolt that holds the piston cooling jet ... $30 \pm 7 \text{ N}\cdot\text{m}$ ($22 \pm 5 \text{ lb ft}$)

(3) Apply **6V-6640** Sealant to the outside diameter of the cup plug prior to installation into the cylinder block. Install the cup plug to the following depth. ... $1.25 \pm 0.25 \text{ mm}$ ($0.049 \pm 0.010 \text{ inch}$)

Use the following procedure in order to install the main bearing cap bolts (4) :

1. The main bearing caps are numbered 1 through 7. Ensure that the main bearing caps are in the correct location. The part number on the main bearing cap must face to the right side and toward the front of the block. Orient the main bearing cap correctly in the cylinder block. If

caps are oriented correctly, the tab slots that are in the block and in the main bearing caps will be adjacent.

Note: Use SAE 30 oil or molybdenum grease to lubricate the threads and the washer face of the main bearing cap bolts.

2. Torque the main bearing cap bolts to 95 N·m (70 lb ft).
3. Put an alignment mark on each cap and bolt.
4. Rotate the bolts for an additional turn in the clockwise direction to 90 degrees.

(5) Main bearing cap

Width of the main bearing cap ... 175.000 ± 0.020 mm (6.8897 ± 0.0008 inch)

Width of the cylinder block for the main bearing cap ... 175.000 ± 0.018 mm (6.8897 ± 0.0007 inch)

(J) Diameter of the camshaft bores ... 81.000 ± 0.015 mm (3.1890 ± 0.0006 inch)

(K) The dowels extend past the top surface of the cylinder block by the following distance. ... 10.0 ± 0.5 mm (0.39 ± 0.02 inch)

(M) Distance from the centerline of the crankshaft bore to the top surface of the cylinder block ... 387.00 ± 0.15 mm (15.236 ± 0.006 inch)

(N) Bore in the cylinder block for the seven main bearings ... 116.000 ± 0.013 mm (4.5669 ± 0.0005 inch)

(P) Distance from the centerline of the crankshaft to the bottom surface of the cylinder block ... 120.0 mm (4.72 inch)

Camshaft Bearing Position

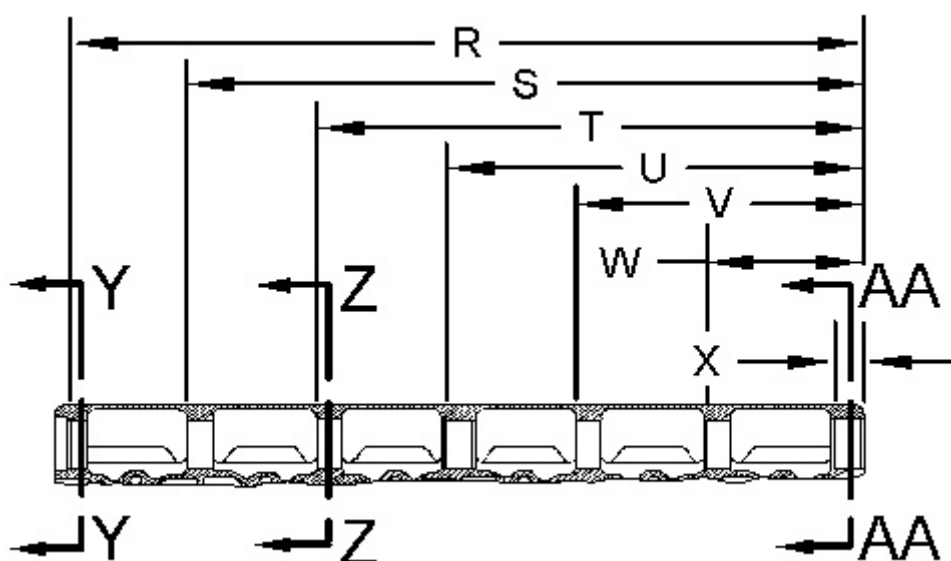


Illustration 4

g01413770

Section L-L

Install the sleeve bearings into the cylinder block at the location that is specified. The distance that is listed below is measured from the front face of the cylinder block.

Dimension (R) ... 954.1 mm (37.56 inch)

Dimension (S) ... 815.2 mm (32.09 inch)

Dimension (T) ... 659.2 mm (25.95 inch)

Dimension (U) ... 503.2 mm (19.81 inch)

Dimension (V) ... 347.2 mm (13.67 inch)

Dimension (W) ... 191.2 mm (7.53 inch)

Dimension (X) ... 35.2 mm (1.39 inch)

NOTICE

Camshaft bearings must be installed into their correct position. Failure to do so will result in engine damage.

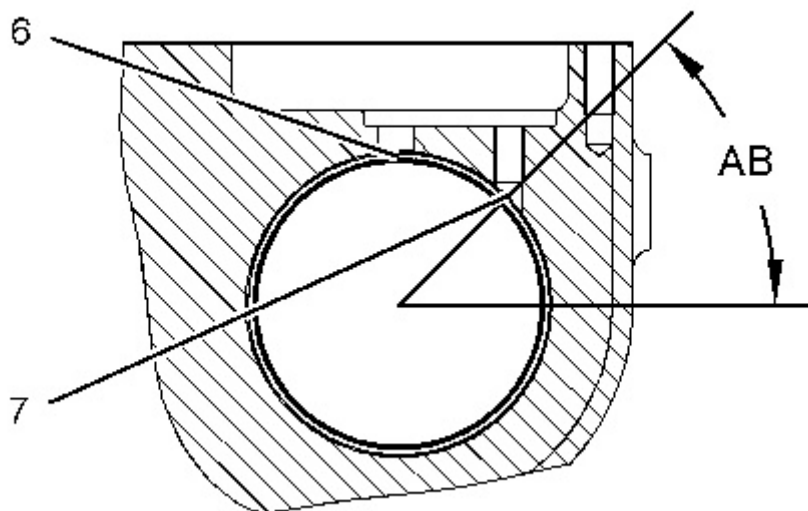


Illustration 5

g01413787

Section Y-Y

(6) Bearing joint

(AB) Orient the oil hole of the sleeve bearing (7) to the following angle. ... 45 degrees

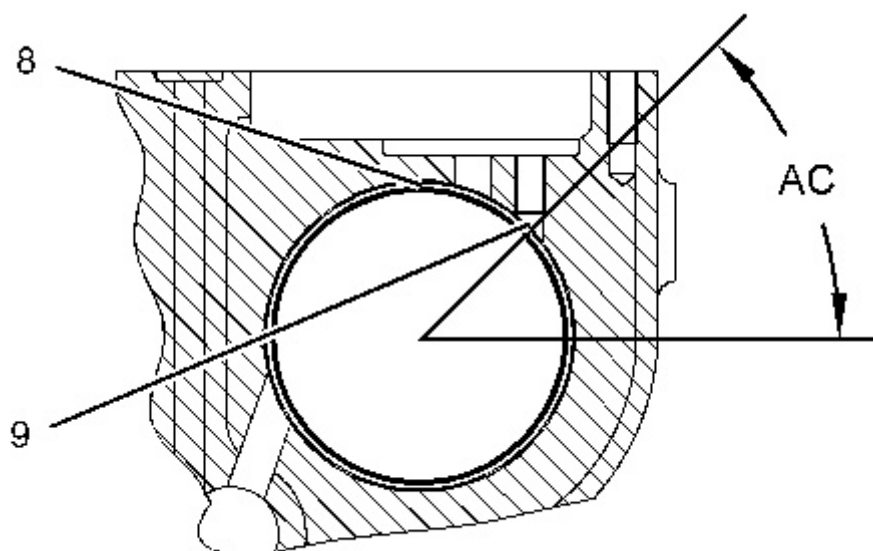


Illustration 6

g01413792

Section Z-Z

(8) Bearing joint

(AC) Orient the oil hole of the sleeve bearing (9) to the following angle. ... 45 degrees

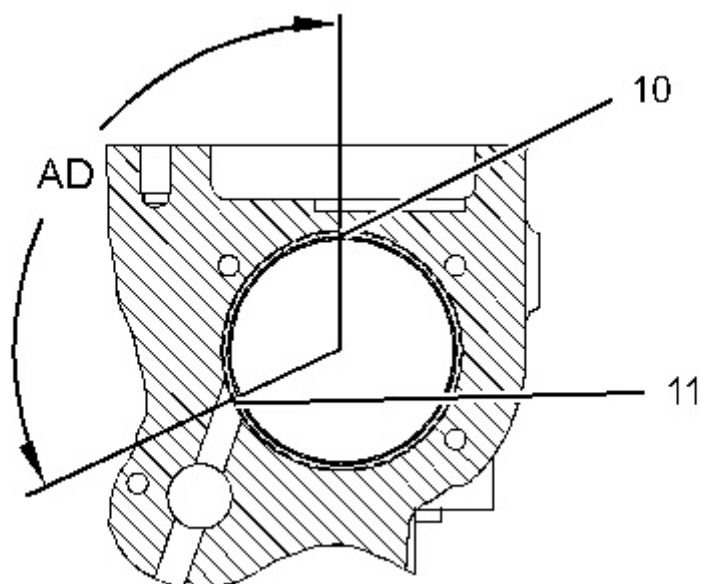


Illustration 7

g01413796

Section AA-AA

(10) Bearing joint

(AD) Orient the oil hole of the sleeve bearing (11) to the following angle. ... 115 degrees

Note: The width of the number 1 sleeve bearing to the number 6 sleeve bearing is 33.4 ± 0.5 mm (1.31 ± 0.02 inch). The width of the number 7 sleeve bearing is 28.0 ± 0.5 mm (1.10 ± 0.02 inch). The number 1 sleeve bearing has a 5.0 ± 0.5 mm (0.20 ± 0.02 inch) wide single groove on the inside diameter. The width of the single groove on the outside diameter of the number 2 sleeve bearing to the number 6 sleeve bearing is 8.0 ± 0.5 mm (0.32 ± 0.02 inch). The number 7 sleeve bearing has a 3.2 ± 0.5 mm (0.13 ± 0.02 inch) wide single groove on the outside diameter.



Specifications

C-10 and C-12 Truck Engines

Media Number -RENR2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i04146590

Cylinder Head Valves

SMCS - 1105

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

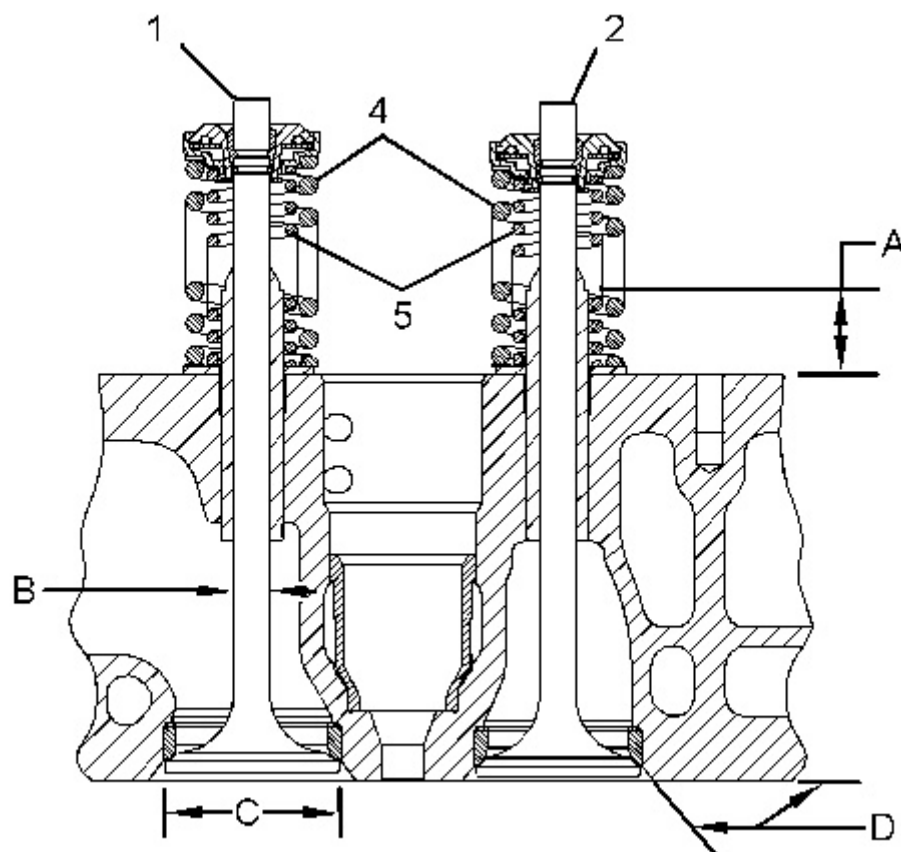


Illustration 1

g01433246

Note: Coat the inlet valve stems and the exhaust valve stems with **8T-2998** Lubricant prior to installation into the cylinder head.

(1) Inlet valve

(2) Exhaust valve

(A) Height to the step that is in the valve guide ... 22.00 ± 0.50 mm ($.866 \pm .020$ inch)

(3) **224-4441** Spring

Assembled Length ... 50.47 mm (1.987 inch)
 Load at assembled length ... 160 ± 13 N (36 ± 3 lb)
 Operating length (minimum) ... 34.92 mm (1.37 inch)
 Load at minimum operating length ... 385 ± 18 N (87 ± 4 lb)
 Free length after test ... 64.1 mm (2.52 inch)
 Outside diameter ... 23.24 mm (0.915 inch)

(4) **214-0586** Spring

Assembled Length ... 52.95 mm (2.085 inch)
 Load at assembled length ... 320 ± 31 N (72 ± 7 lb)
 Operating length (minimum) ... 37.40 mm (1.472 inch)
 Load at minimum operating length ... 860 ± 34 N (190 ± 7 lb)
 Free length after test ... 62.7 mm (2.47 inch)

Outside diameter ... 34.0 mm (1.33 inch)

(B) New valve stem diameter ... 9.441 ± 0.008 mm ($.3717 \pm .0003$ inch)

The valves can be reused if the minimum dimensions for the valve stems are met.

Inlet ... 9.309 mm (0.3665 inch)

Exhaust ... 9.309 mm (0.3665 inch)

Specifications for the valve guides

After installation into the cylinder head, the valve guide bore is the following value. ...

9.484 ± 0.026 mm ($.3734 \pm .0010$ inch)

The valve guide can be reused. The valve guide bore must not be larger than the following value. ... 9.538 mm (.3755 inch)

(C) Diameter of valve head

Inlet valve ... 45.00 ± 0.13 mm ($1.772 \pm .005$ inch)

Exhaust valve ... 42.00 ± 0.13 mm ($1.654 \pm .005$ inch)

(D) Angle of face of valve seat insert

Inlet valve ... $29.25^\circ \pm .25^\circ$

Exhaust valve ... $45.25^\circ \pm .25^\circ$

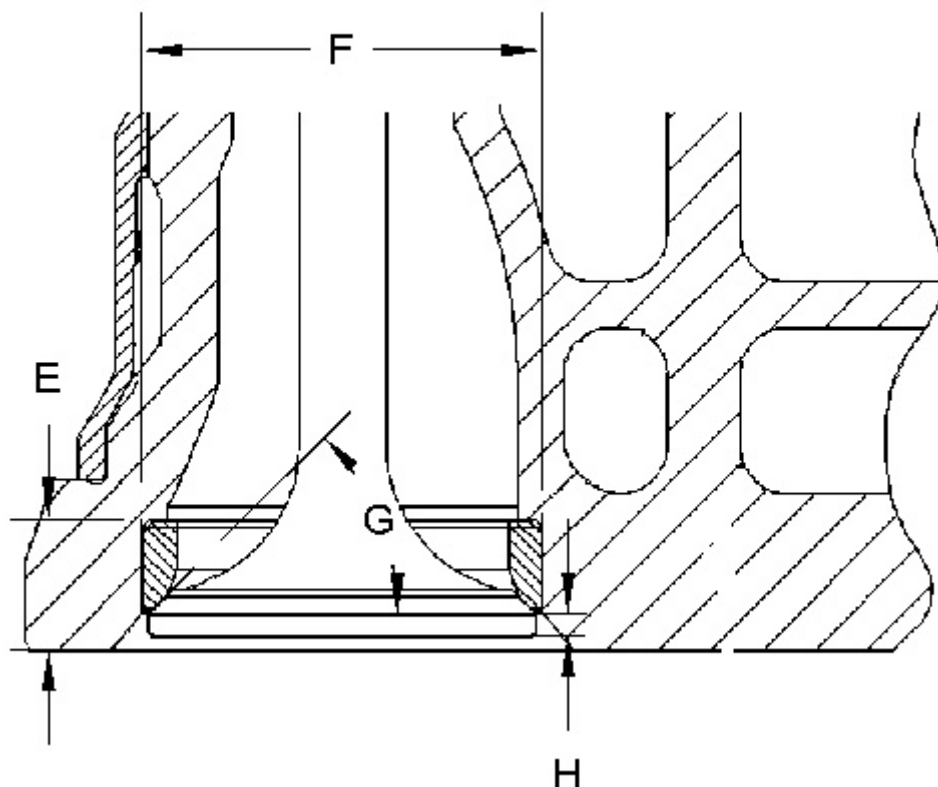


Illustration 2

g01416218

(E) Depth of the bore in the cylinder head for the valve seat insert

Inlet ... 15.00 ± 0.05 mm ($.591 \pm .004$ inch)Exhaust ... 14.10 ± 0.05 mm ($.555 \pm .004$ inch)

Note: Shrink the valve seat inserts by reducing the temperature. Shrinking the valve seat inserts allows placement into the counterbore.

(F) Valve seat insert

Diameter of valve seat insert (inlet valve) ... 46.025 ± 0.013 mm ($1.8120 \pm .0005$ inch)Bore in cylinder head for valve seat insert (inlet valve) ... 45.961 ± 0.013 mm ($1.8095 \pm .0005$ inch)Diameter of valve seat insert (exhaust valve) ... 43.390 ± 0.015 mm ($1.7083 \pm .0006$ inch)Bore in cylinder head for valve seat insert (exhaust valve) ... 43.320 ± 0.025 mm ($1.7055 \pm .0010$ inch)

(G) Angle of valve face

Angle of the face of the inlet valve ... $29 \frac{1}{4} \pm \frac{1}{4}$ degreesAngle of the face of the exhaust valve ... $44 \frac{1}{4} \pm \frac{1}{4}$ degrees

(H) Minimum thickness of the valve lip

Inlet valve ... 2.51 mm (.099 inch)

Exhaust valve ... 2.03 mm (.080 inch)

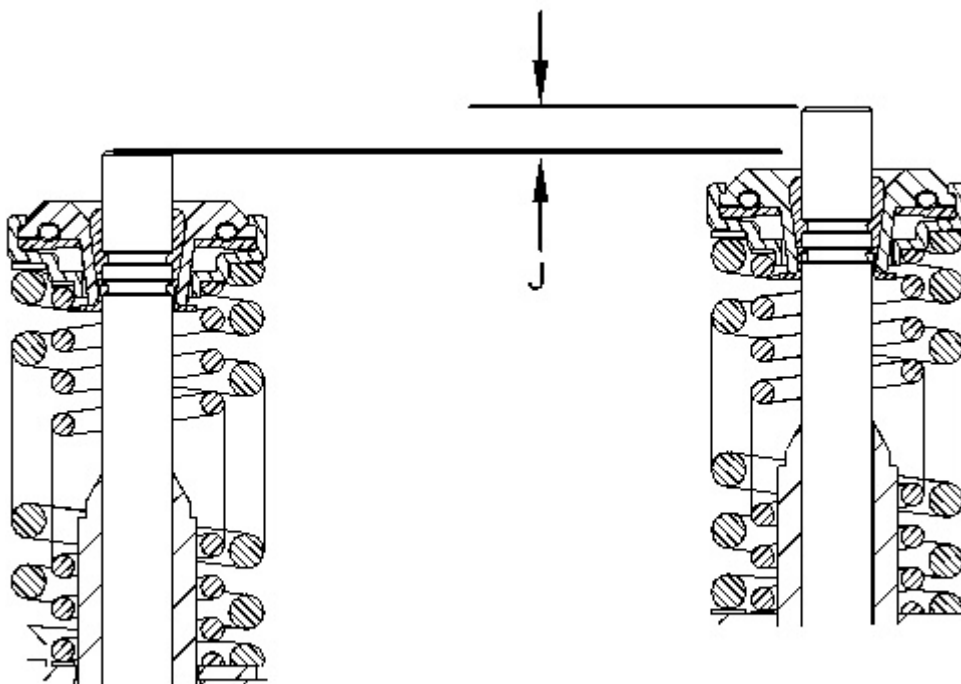


Illustration 3

g01433254

(J) Maximum variation of the valve stem length

Inlet valve to inlet valve ... 0.66 mm (.026 inch)

Exhaust valve to exhaust valve ... 0.66 mm (0.026 inch)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i03475342

Cylinder Head

SMCS - 1100

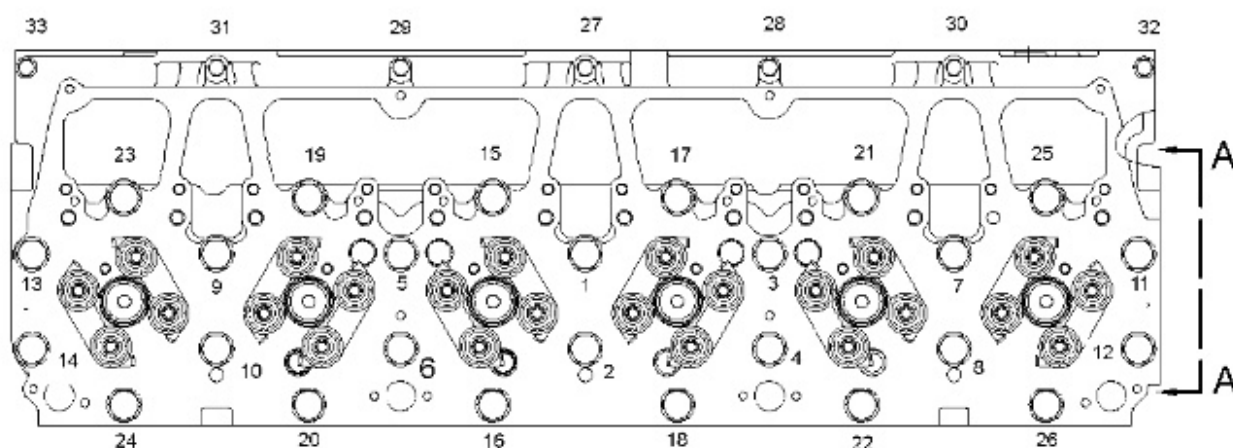


Illustration 1

g01436665

Note: When the bolts are removed from the cylinder head the bolts must be replaced with new cylinder head bolts.

Note: Lubricate the bolt threads, the underside of the bolt heads, and the washers with **6V-4876** Lubricant prior to assembly.

Use the following procedure in order to tighten the cylinder head bolts:

1. Tighten bolt (1) through bolt (26) in a numerical sequence.

Tighten the bolts to the following torque. ... $160 \pm 15 \text{ N}\cdot\text{m}$ ($120 \pm 11 \text{ lb ft}$)

2. Tighten bolt (1) through bolt (26) again in a numerical sequence.

Tighten the bolts again to the following torque. ... $160 \pm 15 \text{ N}\cdot\text{m}$ ($120 \pm 11 \text{ lb ft}$)

3. Place a mark on bolt (1) through bolt (26) . Rotate bolt (1) through bolt (26) in a numerical sequence.

Rotate the bolts in the clockwise direction. ... 90 degrees (1/4 turn)

4. **Loosen bolt (1) through bolt (26) until the washers are loose under the bolt heads.**

5. Tighten bolt (1) through bolt (26) in a numerical sequence.

Tighten the bolts to the following torque. ... $160 \pm 15 \text{ N}\cdot\text{m}$ ($120 \pm 11 \text{ lb ft}$)

6. Tighten bolt (1) through bolt (26) again in a numerical sequence.

Tighten the bolts again to the following torque. ... $160 \pm 15 \text{ N}\cdot\text{m}$ ($120 \pm 11 \text{ lb ft}$)

7. Place a mark on bolt (1) through bolt (26) . Rotate bolt (1) through bolt (26) in a numerical sequence.

Rotate the bolts in the clockwise direction. ... 90 degrees (1/4 turn)

8. Tighten bolt (27) through bolt (33) in a numerical sequence.

Tighten the bolts to the following torque. ... $28 \pm 7 \text{ N}\cdot\text{m}$ ($20 \pm 5 \text{ lb ft}$)

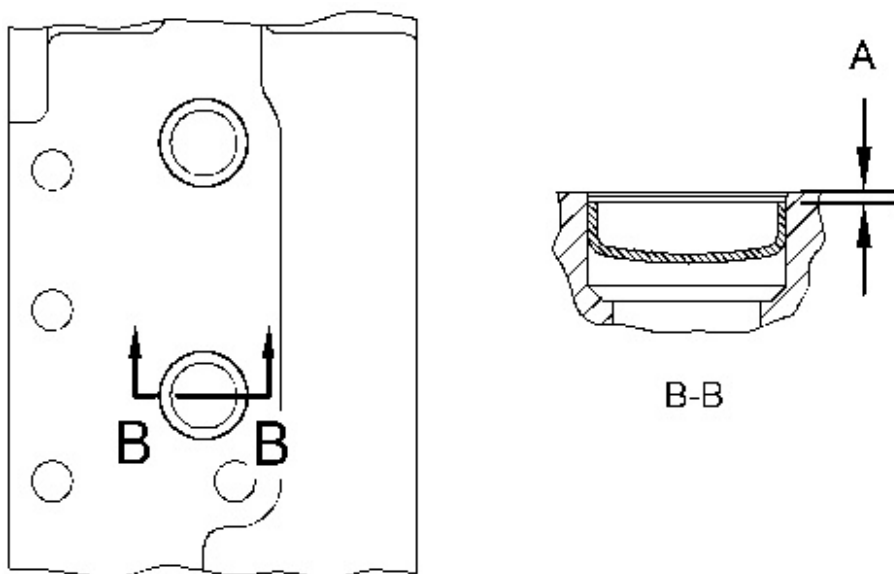


Illustration 2

g01436674

View A-A

(A) The cup plugs are measured from the head face to the top edge of the plug. Depth of installation ... $1.25 \pm 0.25 \text{ mm}$ ($0.049 \pm 0.010 \text{ inch}$)

Height of new cylinder head ... $105.00 \pm 0.15 \text{ mm}$ ($4.134 \pm 0.006 \text{ inch}$)

Minimum permissible thickness of cylinder head ... 104.35 mm (4.108 inch)

Note: The flatness of the cylinder head should be within 0.15 mm (0.006 inch). The flatness should also be a maximum of 0.05 mm (0.002 inch) for any 150.00 mm (5.906 inch) span.



Specifications
C-10 and C-12 Truck Engines
Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i02212928

Cylinder Liner Projection

SMCS - 1216

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

Table 1

Specifications	
Liner Projection	0.060 to 0.180 mm (0.0024 to 0.0071 inch)
Maximum Variation in Each Liner	0.050 mm (0.0020 inch)
Maximum Average Variation Between Adjacent Liners	0.080 mm (0.0031 inch)
Maximum Variation Between All Liners	0.100 mm (0.0040 inch)

For more information on measuring the cylinder liner projection, refer to the Testing and Adjusting manual.



Specifications C-10 and C-12 Truck Engines

Media Number -RENR2234-08

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i02586390

Cylinder Liner

SMCS - 1216

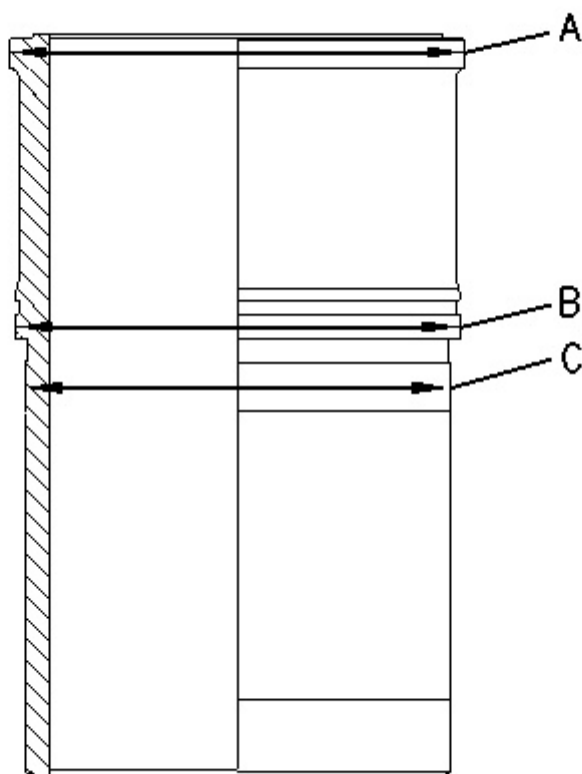


Illustration 1

g00517737

Outside diameters of the cylinder liner

Position (A) ... 151.42 ± 0.05 mm (5.9614 ± 0.002 inch)

Position (B) ... 147.90 ± 0.03 mm (5.823 ± 0.001 inch)

Position (C) ... 141.38 ± 0.08 mm (5.566 ± 0.003 inch)

Bore diameter for the cylinder liner ... 130.025 ± 0.025 mm (5.1191 ± 0.0010 inch)

Note: Apply **5P-3975** Rubber Lubricant to the cylinder liner seal prior to assembly.



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i02888421

Electric Starting Motor

SMCS - 1453

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

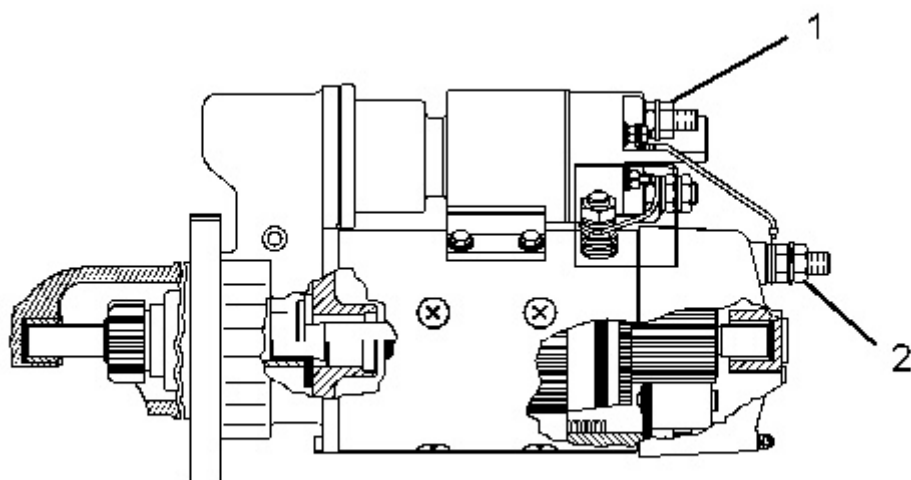


Illustration 1

g01437986

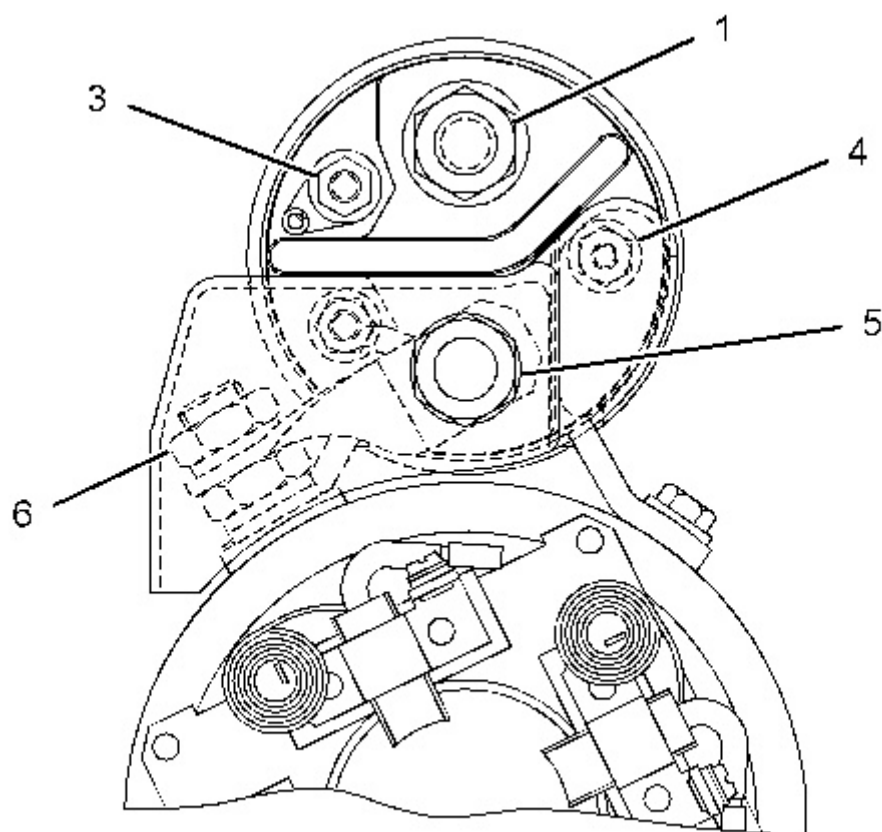


Illustration 2

g01437990

When the electric starting motor is viewed from the pinion end, the motor rotates in the following direction. ... Clockwise

No load conditions at 25 °C (77 °F)

Speed ... 5500 rpm minimum

Current draw ... 140 amp maximum

Voltage ... 24 V

(1) Battery terminal nut

A maximum of three cable or wire terminals should be between the nuts.

Torque ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(2) Ground terminal nut

Torque ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(3) Ground terminal nut

Torque ... 2.25 ± 0.25 N·m (20 ± 2 lb in)

(4) Switch terminal nut

The wire terminal that is on the switch terminal must be insulated with heat shrink tubing. Do not use molded terminals.

Torque ... $2.25 \pm 0.25 \text{ N}\cdot\text{m}$ ($20 \pm 2 \text{ lb in}$)

(5) Motor terminal nut

Torque ... $30.5 \pm 3.5 \text{ N}\cdot\text{m}$ ($22 \pm 3 \text{ lb ft}$)

(6) Nut

Torque ... $30.5 \pm 3.5 \text{ N}\cdot\text{m}$ ($22 \pm 3 \text{ lb ft}$)

Clearance between the pinion and the housing ... 8.3 to 9.9 mm (0.33 to 0.39 inch)

Solenoid

Current draw at 24 V and 25 °C (77 °F)

Pull-in windings ... 51 Amp

Hold-in windings ... 7 Amp maximum



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i02612946

Electric Starting Motor

SMCS - 1453

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

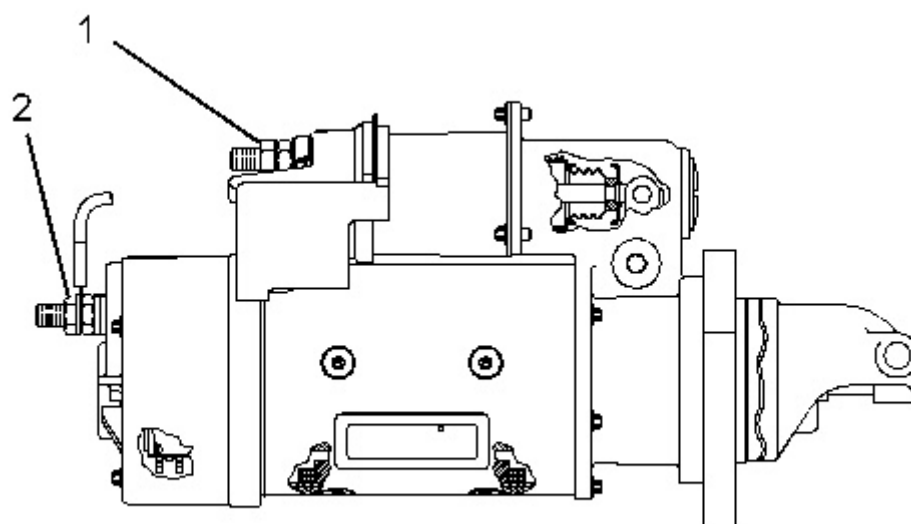


Illustration 1

g01122305

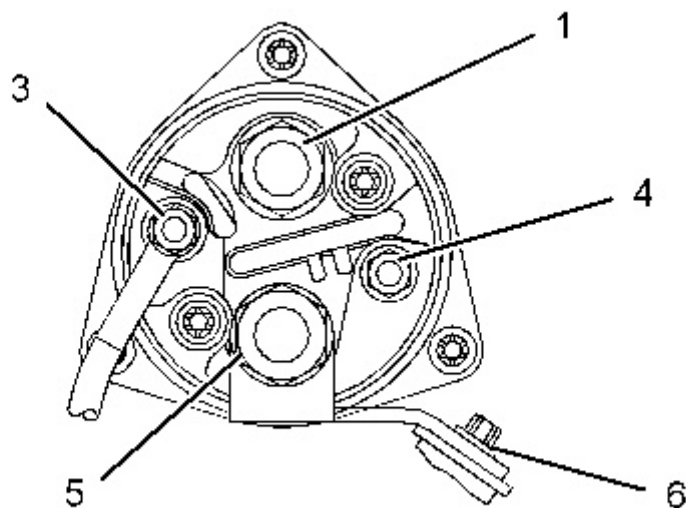


Illustration 2

g01122306

When the electric starting motor is viewed from the drive end, the motor rotates in the following direction. ... Clockwise

No load conditions at 25 °C (77 °F)

Speed ... 6900 ± 1800 rpm

Current draw ... 127.5 ± 27.5 amp

Voltage ... 11 V

(1) Battery terminal nut

A maximum of three cable or wire terminals should be between the nuts.

Torque ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(2) Ground terminal nut

Torque ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(3) Ground terminal nut

Torque ... 2.25 ± 0.25 N·m (20 ± 2 lb in)

(4) Switch terminal nut

Torque ... 2.25 ± 0.25 N·m (20 ± 2 lb in)

The wire terminal that is on the switch terminal must be insulated with heat shrink tubing. Do not use molded terminals.

(5) Motor terminal nut

Torque ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(6) Motor frame terminal nut

Torque ... $8 \pm 3 \text{ N}\cdot\text{m}$ ($70 \pm 27 \text{ lb in}$)

Clearance between the pinion and the housing ... 9.1 mm (0.36 inch)

Solenoid

Current draw at 10 V and 25 °C (77 °F)

Pull-in windings ... $92.5 \pm 7.5 \text{ Amp}$

Hold-in windings ... 23 Amp



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i02836290

Electronic Unit Injector Rocker Arm

SMCS - 1123

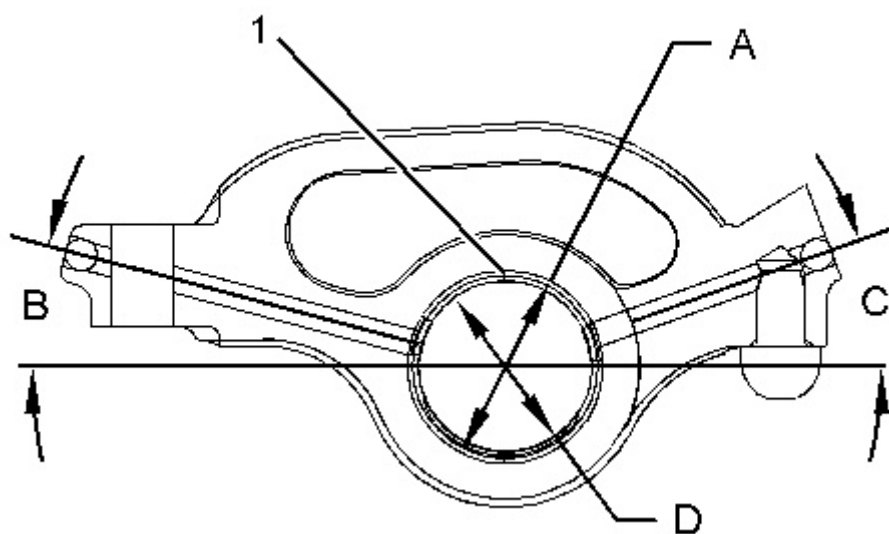


Illustration 1

g01414209

(1) Bearing joint

Note: The bearing must not extend beyond either face of the arm assembly.

(A) Bore in the arm assembly for the bearing ... 37.000 ± 0.020 mm (1.4567 ± 0.0008 inch)

Note: The oil hole openings that are in the bearing must be centered over the oil passages that are in the arm assembly. After assembly of bearing, minimum diameter of opening in the oil hole must be 2.3 mm (0.09 inch).

(B) Oil passage angle ... 15 degrees

(C) Oil passage angle ... 20 degrees

(D) Bore in bearing for shaft after installation ... 34.050 ± 0.015 mm (1.3405 ± 0.0006 inch)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

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i02837394

Electronic Unit Injector Wiring

SMCS - 1290

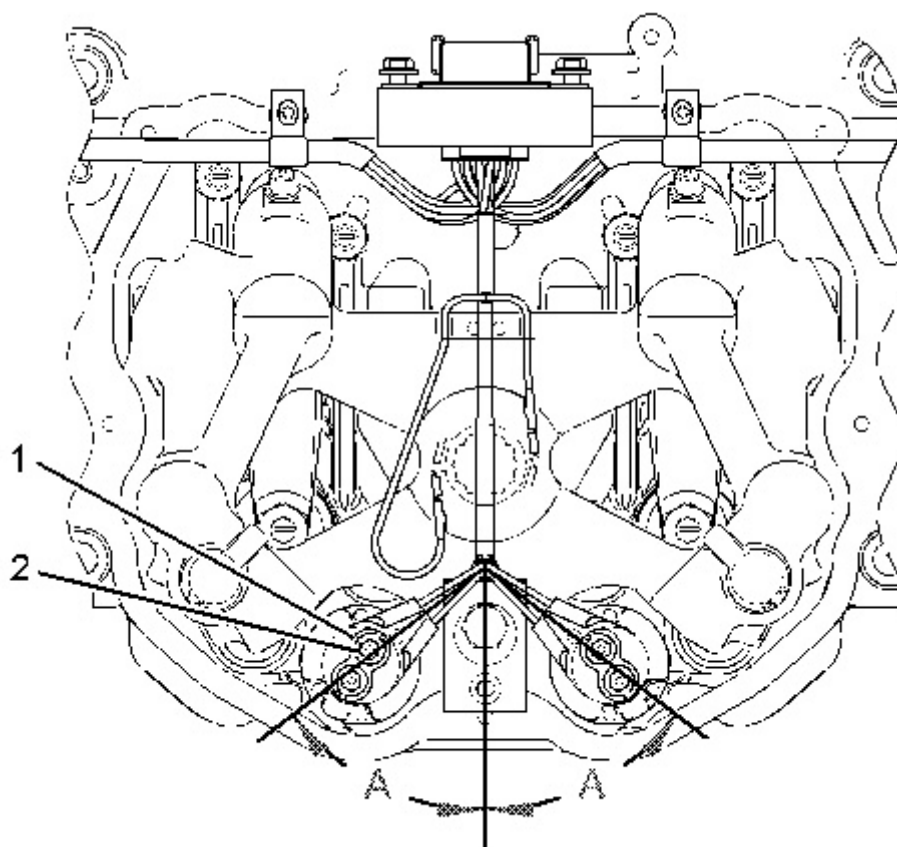


Illustration 1

g01414667

(1) Cap assembly

Note: All the tie straps should be removed prior to placing the valve cover base on the engine when the engine retarder is used.

Note: Orient the connectors properly so that wires are not contacting any other surfaces.

(A) Approximate angle for the connectors ... 52 degrees

(2) Torque for the retainer nuts ... $2.50 \pm 0.25 \text{ N}\cdot\text{m}$ ($22 \pm 2 \text{ lb in}$)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i02878472

Electronic Unit Injector

SMCS - 1290

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

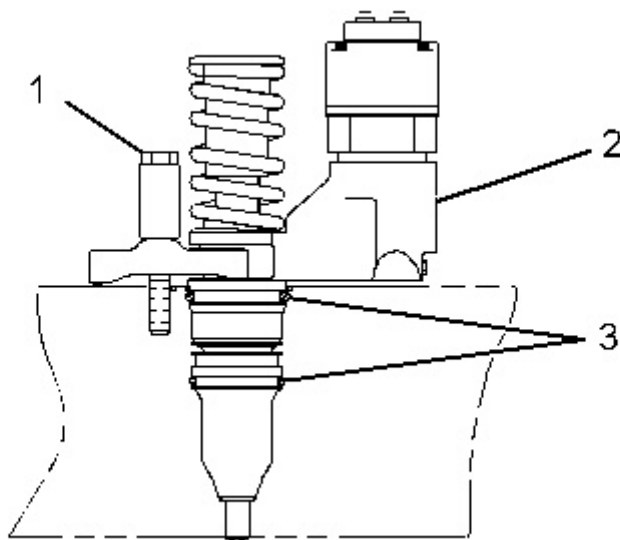


Illustration 1

g01432760

(2) Fuel injector group

Use the following procedure to tighten the bolt (1) :

1. Tighten the bolt to $30 \pm 7 \text{ N}\cdot\text{m}$ ($22 \pm 5 \text{ lb ft}$).
2. Loosen the bolt until the fuel injector clamp is loose.
3. Again tighten the bolt to $30 \pm 7 \text{ N}\cdot\text{m}$ ($22 \pm 5 \text{ lb ft}$).

(3) Lubricate the O-ring seals and the bore with a 50 percent mix of **8T-2998** Lubricant and clean engine oil.



Specifications

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i02600159

Engine Design

SMCS - 1201

S/N - 2KS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

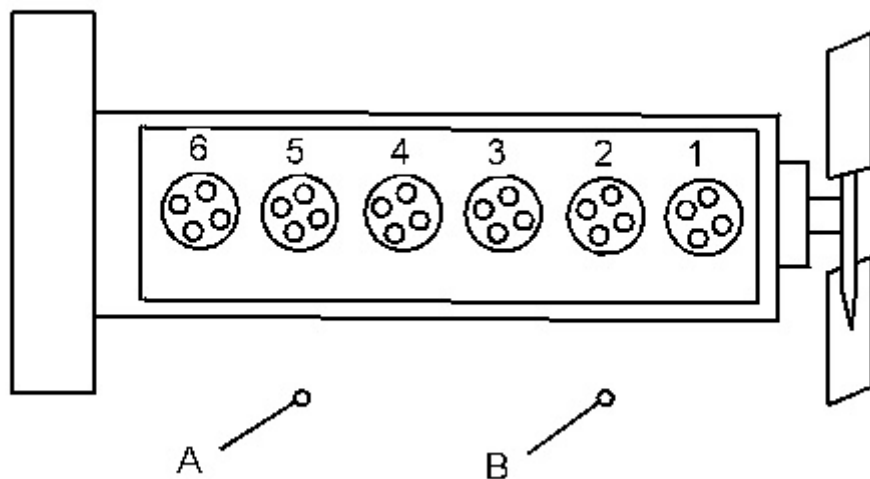


Illustration 1

g01301317

Cylinder And Valve Location

(A) Exhaust valves and (B) Inlet valves

Bore ... 130.0 mm (5.12 inch)

Stroke ... 150.0 mm (5.91 inch)

Displacement ... 12 L (732 cu in)

Cylinder arrangement ... In-line

Valves per cylinder ... 4

In order to check the engine valve lash setting, the engine must be cold and the engine must be stopped. Engine valve lash settings

Inlet ... 0.38 mm (.015 inch)

Exhaust ... 0.64 mm (.025 inch)

Engine compression brake ... 1.02 mm (.040 inch)

Type of combustion ... Direct Injection

Firing Order ... 1 - 5 - 3 - 6 - 2 - 4

The crankshaft rotation is viewed from the flywheel end of the engine. Crankshaft rotation ... Counterclockwise

Note: The front end of the engine is opposite of the flywheel end of the engine. The left side of the engine and the right side of the engine are viewed from the flywheel end of the engine. The No. 1 cylinder is the front cylinder.



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i02886517

Engine Oil Cooler

SMCS - 1378

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

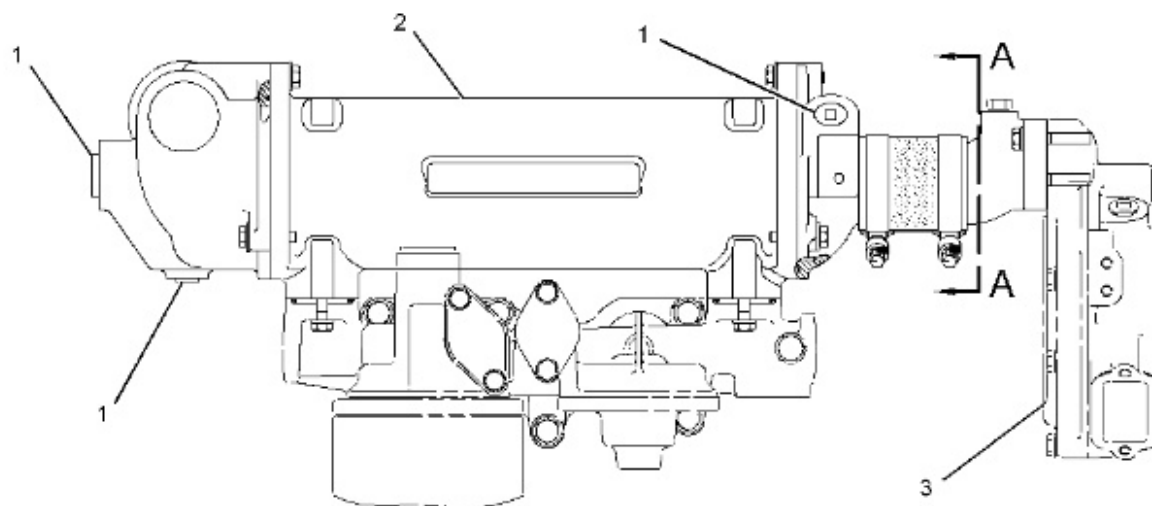


Illustration 1

g01436898

(2) Core assembly

(3) Water pump

(1) Apply **5P-3413** Pipe Sealant to the plug threads.

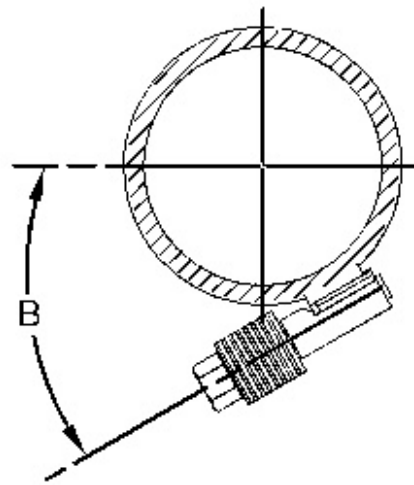


Illustration 2

g01436902

Section A-A

(B) Install the hose clamps at the following angle. ... 30 ± 5 degrees

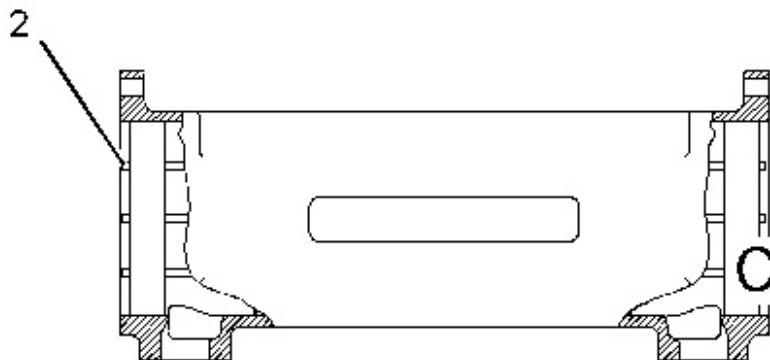


Illustration 3

g01436903

(2) Core assembly

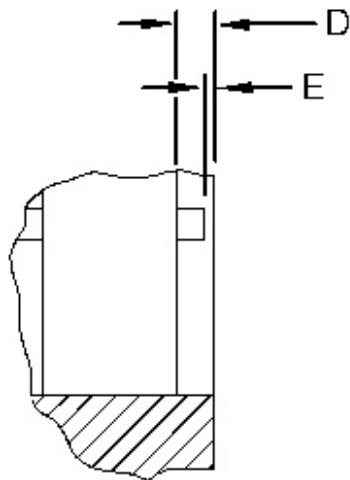


Illustration 4

g01436906

Detail C

The oil flow rate through the cooler assembly is 123 L/min (32 US gpm). The maximum water flow through the cooler assembly is 262 L/min (69 US gpm).

Lubricate the core with engine oil. Press the core into the housing to the following dimensions at both ends.

(D) The depth of the plate is the following value. ... 5.5 ± 0.5 mm (0.22 ± 0.02 inch)

(E) The depth of the tube is the following value. ... 1.5 ± 0.5 mm (0.06 ± 0.02 inch)



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i02886419

Engine Oil Filter Base

SMCS - 1306

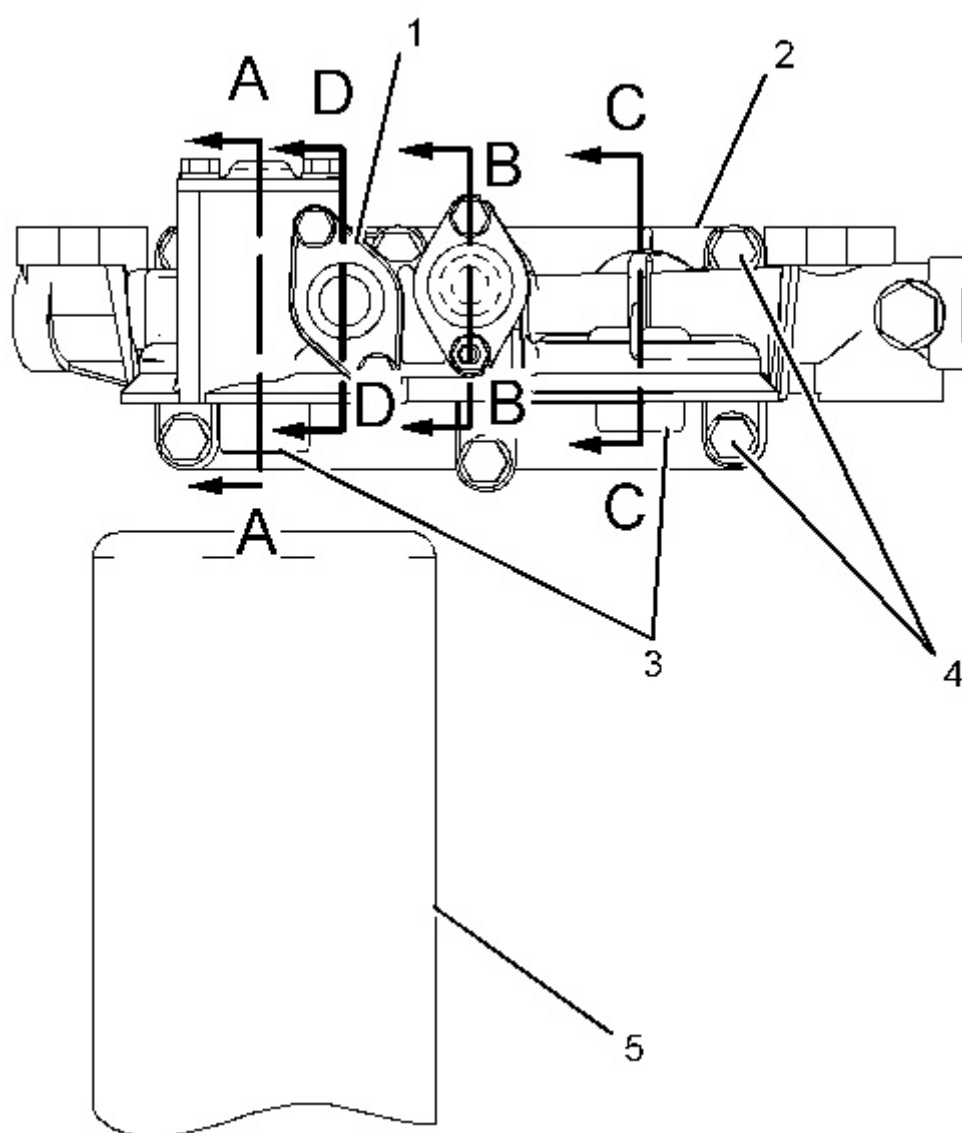


Illustration 1

g01436835

- (1) Cover
- (2) Oil filter base
- (5) Oil filter
- (6) O-ring seal
- (8) O-ring seal
- (9) Valve plunger
- (10) O-ring seal
- (12) Pump bypass valve
- (13) O-ring seal
- (14) O-ring seal
- (15) Spacer

- (17) O-ring seal
- (19) Valve plunger
- (20) O-ring seal

(3) Studs

Prior to assembly, apply **155-0695** Thread Lock Compound to the studs. The primary oil filter mounting stud is longer than the taperlock stud.

Tighten the studs into the oil filter base to the following torque. ... $68 \pm 7 \text{ N}\cdot\text{m}$ ($50 \pm 5 \text{ lb ft}$)

(4) Mounting bolts

Apply **9S-3263** Thread Lock Compound to each of the six mounting bolts.

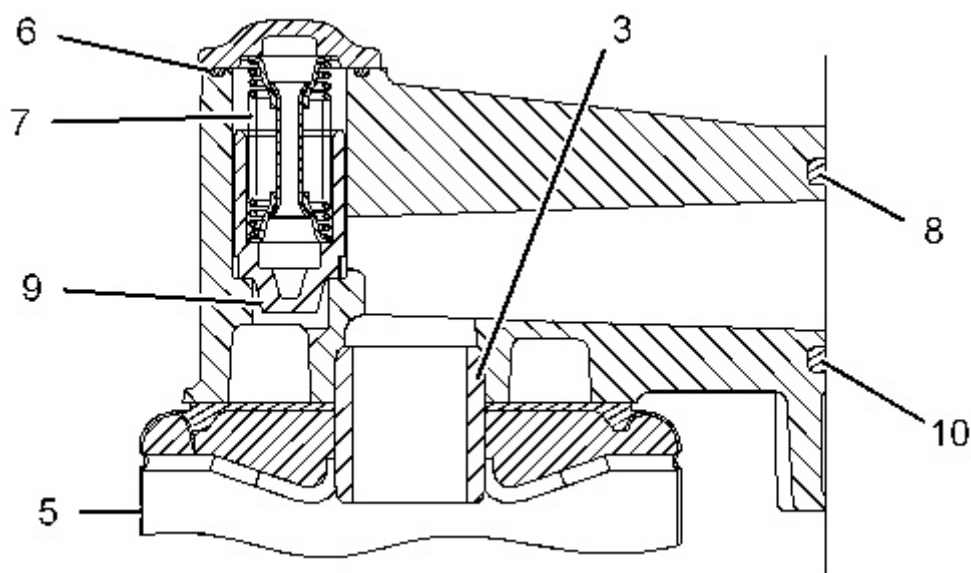


Illustration 2

g01116558

Section A-A

(7) 4N-8150 Spring

- Assembled length ... 55.25 mm (2.175 inch)
- Load at assembled length ... $75 \pm 6 \text{ N}$ ($17.0 \pm 1.3 \text{ lb}$)
- Free length after test ... 93.7 mm (3.69 inch)
- Outside diameter ... 20.6 mm (0.81 inch)

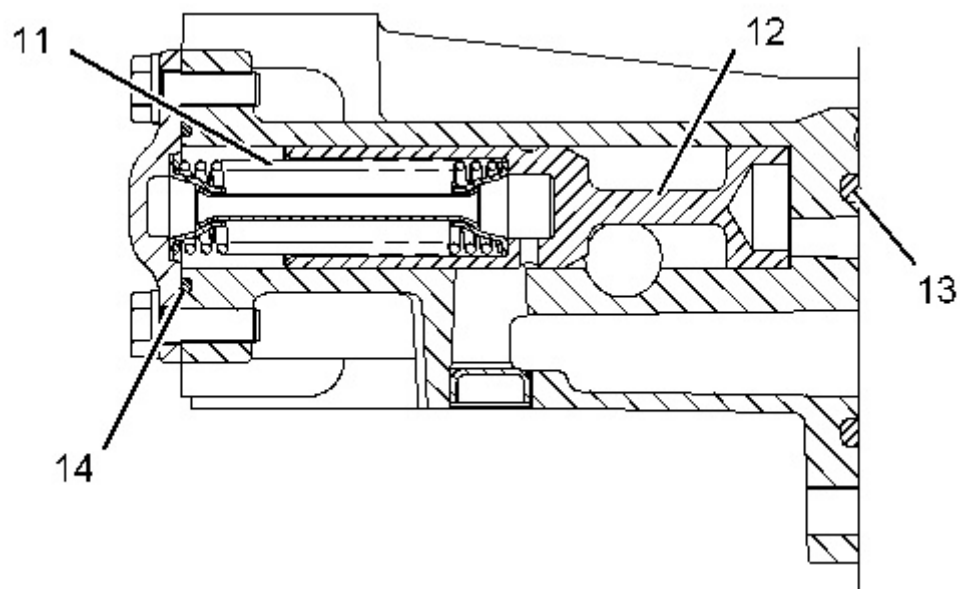


Illustration 3

g01116562

Section B-B

(11) 2N-6005 Spring

Assembled length ... 96.52 mm (3.800 inch)
Load at assembled length ... 92.52 N (21 lb)
Minimum operating length ... 45.72 mm (1.800 inch)
Load at minimum operating length ... 257.55 N (58 lb)
Free length after test ... 124.71 mm (4.910 inch)
Outside diameter ... 21.84 mm (0.860 inch)

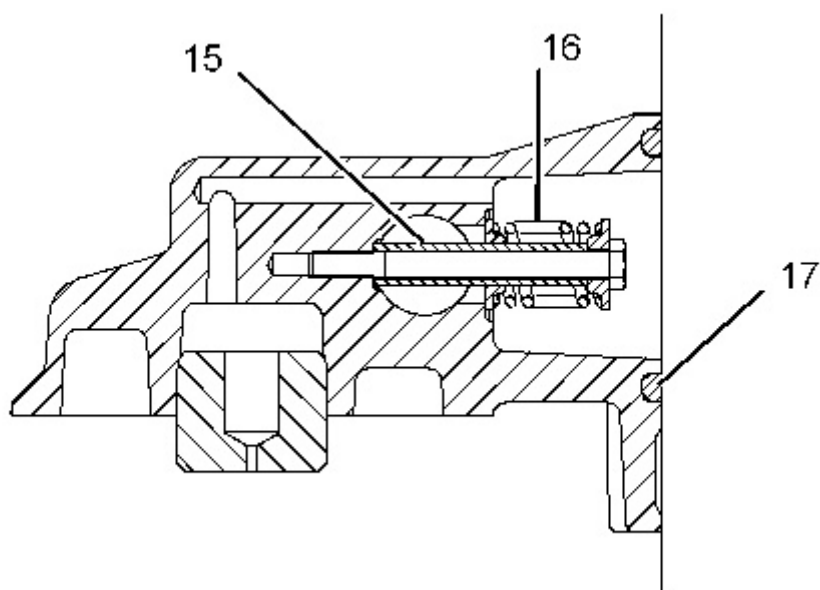


Illustration 4

g01116568

Section C-C

Relief pressure ... 696 kPa (101 psi)

(16) **1A-2170** Retainer Spring

Assembled length ... 21.4 mm (0.84 inch)

Load at assembled length ... 200 ± 15.5 N (45 ± 3.5 lb)

Free length after test ... 31.0 mm (1.22 inch)

Outside diameter ... 19.6 mm (0.77 inch)

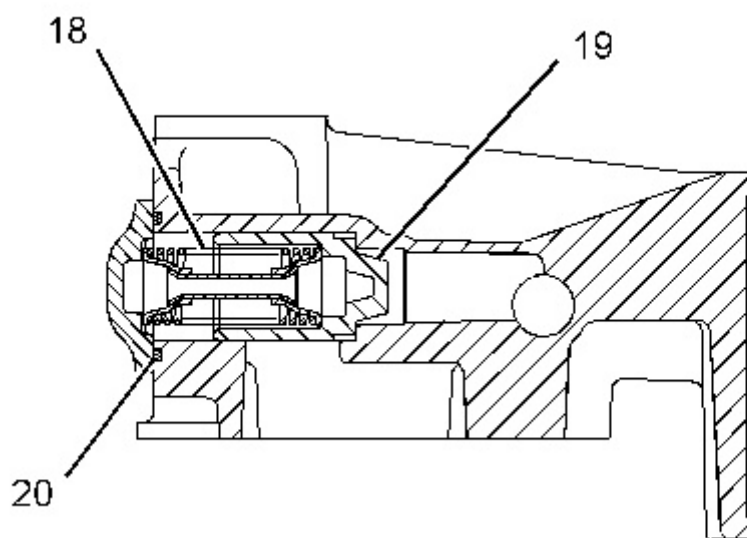


Illustration 5

g01116570

Section D-D

(18) **4N-8150** Spring

Assembled length ... 55.25 mm (2.175 inch)

Load at assembled length ... 75 ± 6 N (17.0 ± 1.3 lb)

Free length after test ... 93.7 mm (3.69 inch)

Outside diameter ... 20.6 mm (0.81 inch)



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i02886733

Engine Oil Pan

SMCS - 1302

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

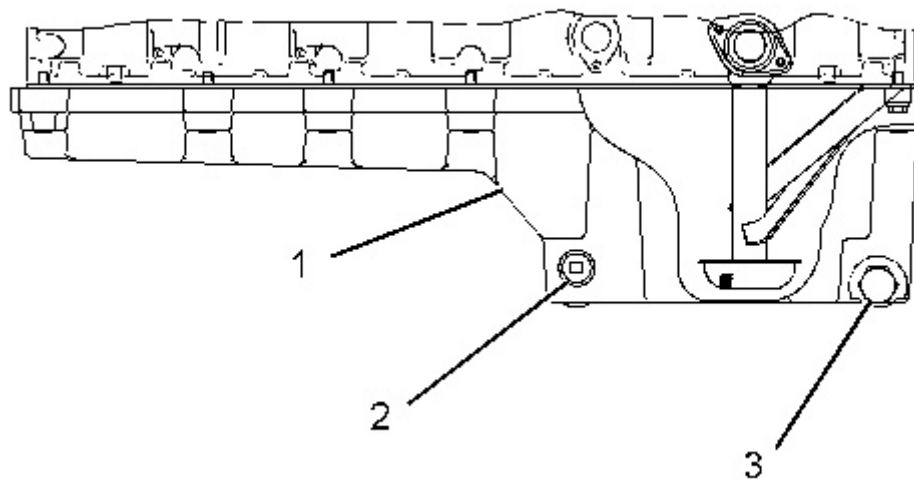


Illustration 1

g01437070

Shallow Front Sump

(1) Engine oil pan

- (2) Apply **9S-3263** Thread Lock Compound to the plug threads. Tighten the plugs to the following torque. ... $80 \pm 11 \text{ N}\cdot\text{m}$ ($60 \pm 8 \text{ lb ft}$)
- (3) Tighten the plugs to the following torque. ... $70 \pm 15 \text{ N}\cdot\text{m}$ ($50 \pm 11 \text{ lb ft}$)
-



Specifications

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i02838517

Engine Oil Pan

SMCS - 1302

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

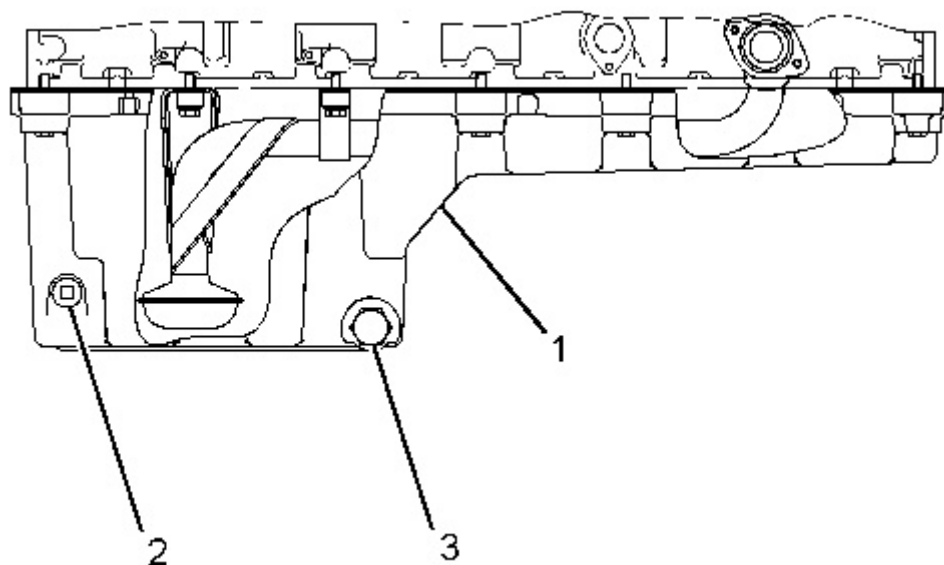


Illustration 1

g01415517

(1) Oil pan

Note: Apply **5P-3413** Pipe Sealant to the threads of the pipe plug.

(2) Torque for the pipe plug ... $80 \pm 11 \text{ N}\cdot\text{m}$ ($60 \pm 8 \text{ lb ft}$)

(3) Torque for the drain plug ... $70 \pm 15 \text{ N}\cdot\text{m}$ ($50 \pm 11 \text{ lb ft}$)



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i02895284

Engine Oil Pan

SMCS - 1302

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

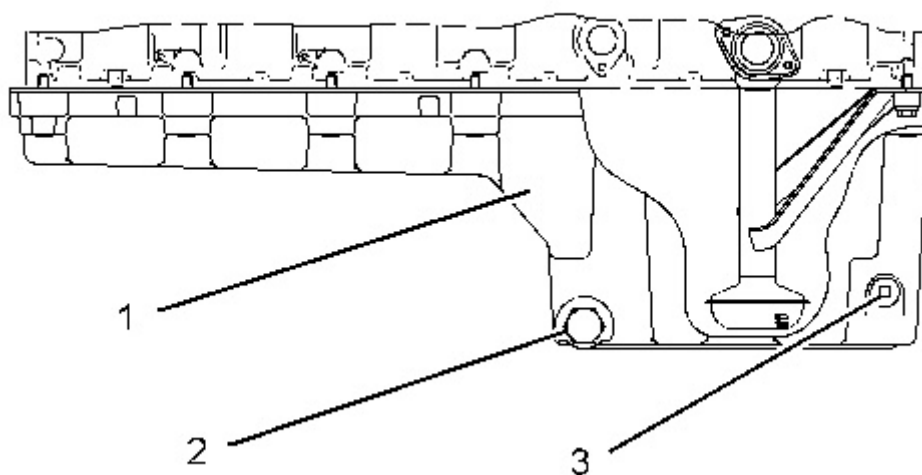


Illustration 1

g01441390

Deep Front Sump

(1) Engine oil pan

(2) Tighten the plugs to the following torque. ... $70 \pm 15 \text{ N}\cdot\text{m}$ ($50 \pm 11 \text{ lb ft}$)

(3) Apply **9S-3263** Thread Lock Compound to the plug threads. Tighten the plugs to the following torque. ... $80 \pm 11 \text{ N}\cdot\text{m}$ ($60 \pm 8 \text{ lb ft}$)



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i02886718

Engine Oil Pan

SMCS - 1302

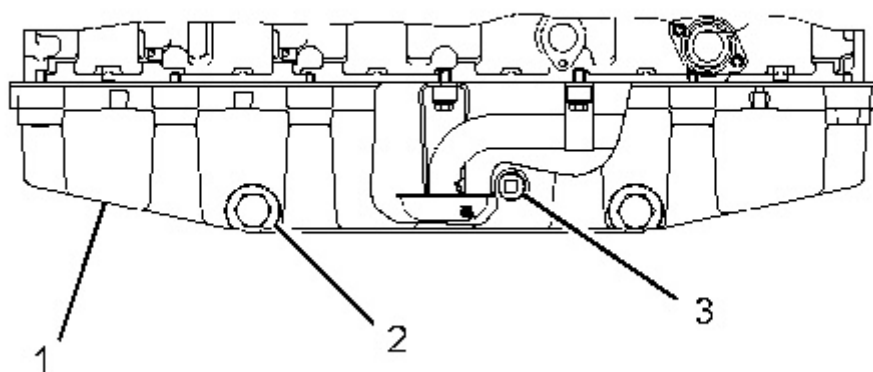


Illustration 1

g01437069

Center Sump

(1) Oil pan

(2) Tighten the plugs to the following torque. ... $70 \pm 15 \text{ N}\cdot\text{m}$ ($50 \pm 11 \text{ lb ft}$)

(3) Apply **9S-3263** Thread Lock Compound to plug threads. Tighten the plug to the following torque. ... $80 \pm 11 \text{ N}\cdot\text{m}$ ($60 \pm 8 \text{ lb ft}$)



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i04753349

Engine Oil Pressure Sensor

SMCS - 1924

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

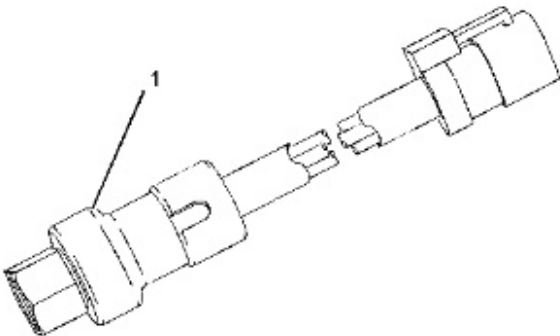


Illustration 1

g02554200

Table 1

Specification for the 194-6725 Pressure Sensor Gp , 161-9931 Pressure Sensor Gp , 163-8523 Pressure Sensor Gp , 299-3222 Pressure Sensor Gp , 161-9926 Pressure Sensor Gp , 239-2396 Pressure Sensor Gp , 239-3478 Pressure Sensor Gp , and 3E-8652 Pressure Sensor Gp			
Item	Qty	Part	Specification Description
1	1	-	Torque to 10 ± 2 N·m (90 ± 18 lb in).
			Operating voltage is 5.00 ± 0.25 VDC.



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i02886591

Engine Oil Pump

SMCS - 1304

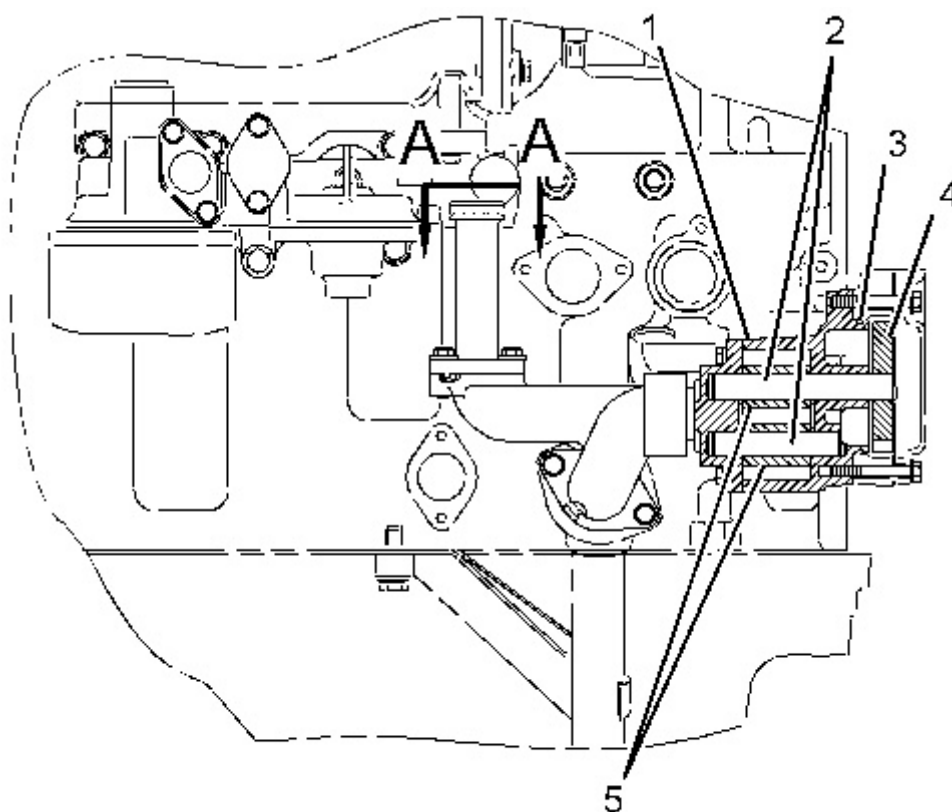


Illustration 1

g01436962

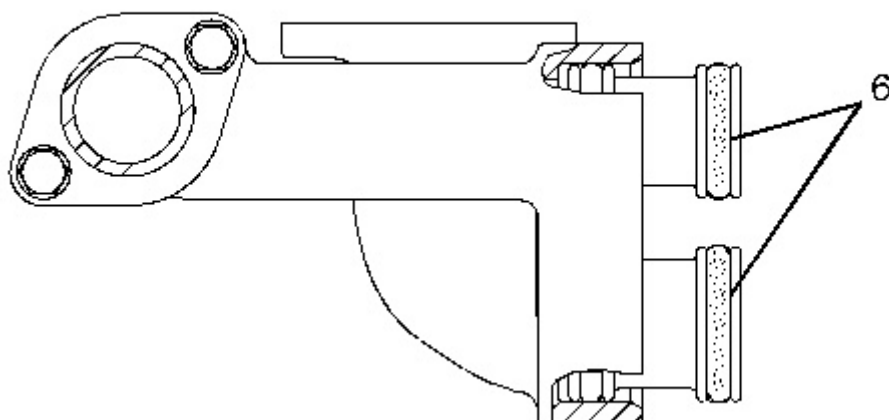


Illustration 2

g01436964

View A-A

NOTICE

Before operating the engine, the oil pump must be lubricated with clean engine oil. The oil pump must turn freely by hand. Damage to the drive gear and internal pump damage can occur if the oil pump is not lubricated with clean engine oil.

(1) Engine oil pump

Pump discharge pressure ... 359 kPa (52 psi)

(2) Shafts

Diameter of two shafts ... 18.000 ± 0.005 mm (0.7087 ± 0.0002 inch)

Bores in pump body for two shafts ... 18.050 ± 0.010 mm (0.7106 ± 0.0004 inch)

(3) Seal groove

Prior to installation, lubricate the bore in the engine front cover. This will allow the seal groove and the seal to slide.

(4) Drive gear

In order to install the drive gear, heat the drive gear to the following temperature. ... 316 °C (600 °F)

(5) Pump gears

In order to install the pump gears, heat the pump gears to the following temperature. ... 316 °C (600 °F)

Length of new gears ... 50.000 ± 0.025 mm (1.9685 ± 0.0010 inch)

Depth of bores in pump body for the gears ... 50.13 ± 0.02 mm (1.974 ± 0.001 inch)

(6) Seals

Prior to installation, lubricate the bore lightly with the lubricant that is being sealed.



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i02888147

Engine to Transmission Adapter

SMCS - 1001

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

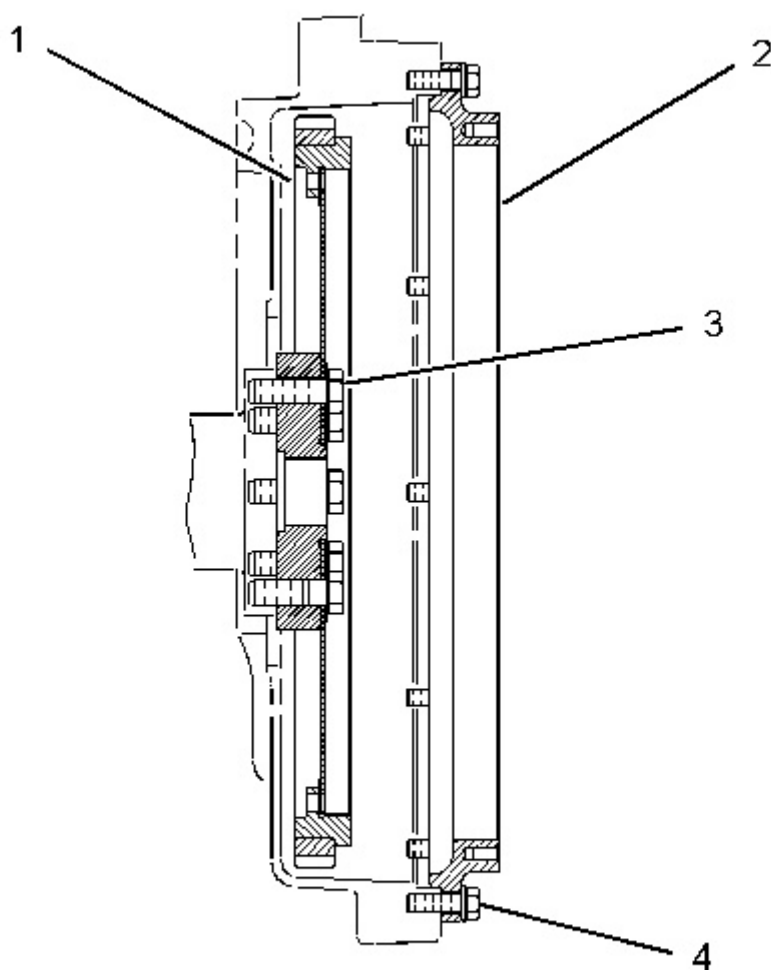


Illustration 1

g01437779

(1) Plate

(2) Adapter

(3) Apply **9S-3263** Thread Lock Compound to the bolts. Tighten the bolts to the following torque.
... $300 \pm 40 \text{ N}\cdot\text{m}$ ($220 \pm 30 \text{ lb ft}$)

(4) Tighten the bolts to the following torque. ... $70 \pm 15 \text{ N}\cdot\text{m}$ ($51 \pm 11 \text{ lb ft}$)



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i02854761

Exhaust Manifold

SMCS - 1059

S/N - 2KS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

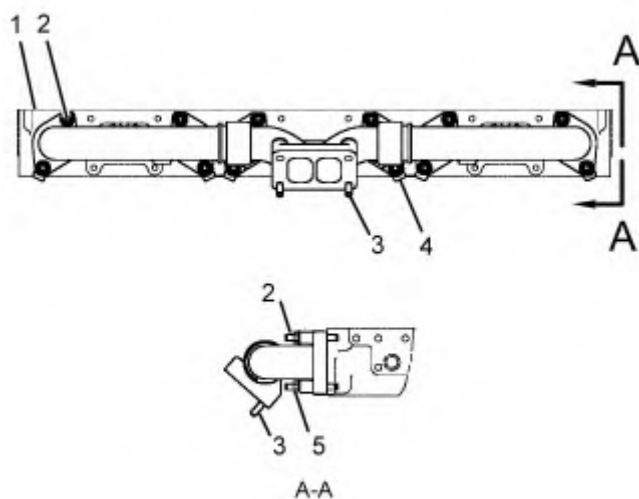


Illustration 1

g01421837

(1) Cylinder head

(2) Torque for 12 taperlock studs ... 35 ± 5 N·m (26 ± 4 lb ft)(3) Torque for two taperlock studs ... 35 ± 5 N·m (26 ± 4 lb ft)

(4) Install the gasket with the tab downward, as shown.

(5) Apply **4C-5599** Anti-Seize Compound to the threads of the locknuts and the bearing surface of the locknuts and the washers before assembly.



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i02837957

Exhaust Manifold

SMCS - 1059

S/N - 2KS1-UP

S/N - 8YF1-UP

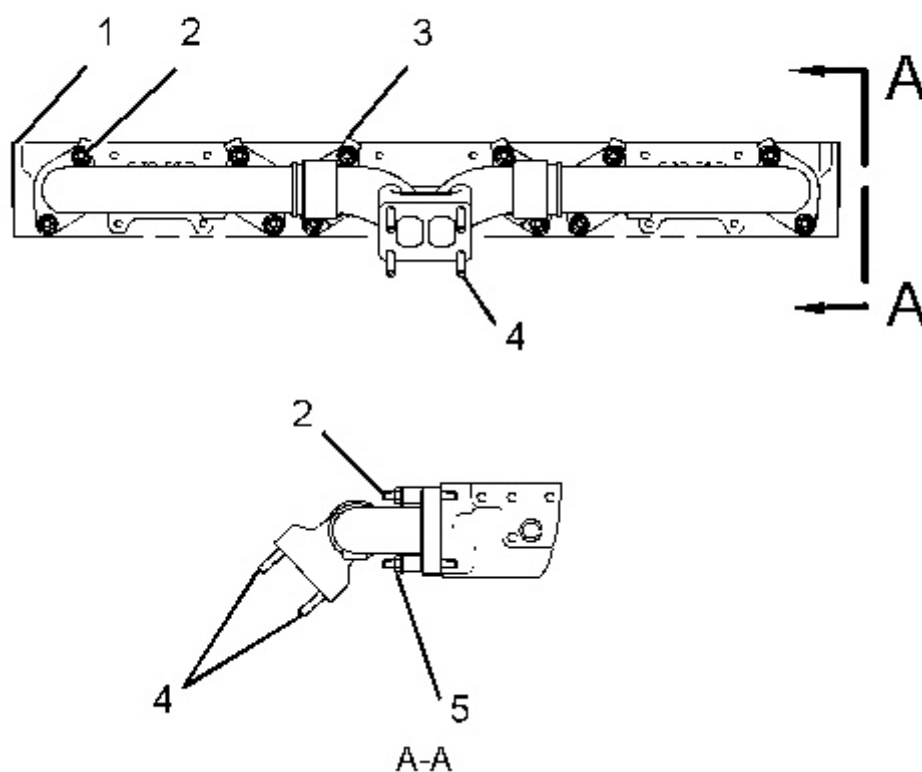


Illustration 1

g01414862

- (1) Cylinder head
 - (2) Torque for 12 taperlock studs ... $35 \pm 5 \text{ N}\cdot\text{m}$ ($26 \pm 4 \text{ lb ft}$)
 - (3) Install the gasket with the tab upward, as shown.
 - (4) Torque for four taperlock studs ... $35 \pm 5 \text{ N}\cdot\text{m}$ ($26 \pm 4 \text{ lb ft}$)
 - (5) Apply **4C-5599** Anti-Seize Compound to the threads of the locknuts and the bearing surface of the locknuts and the washers before assembly.
-



Specifications

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i02854067

Exhaust Manifold

SMCS - 1059

S/N - 2KS1-UP

S/N - 8YF1-UP

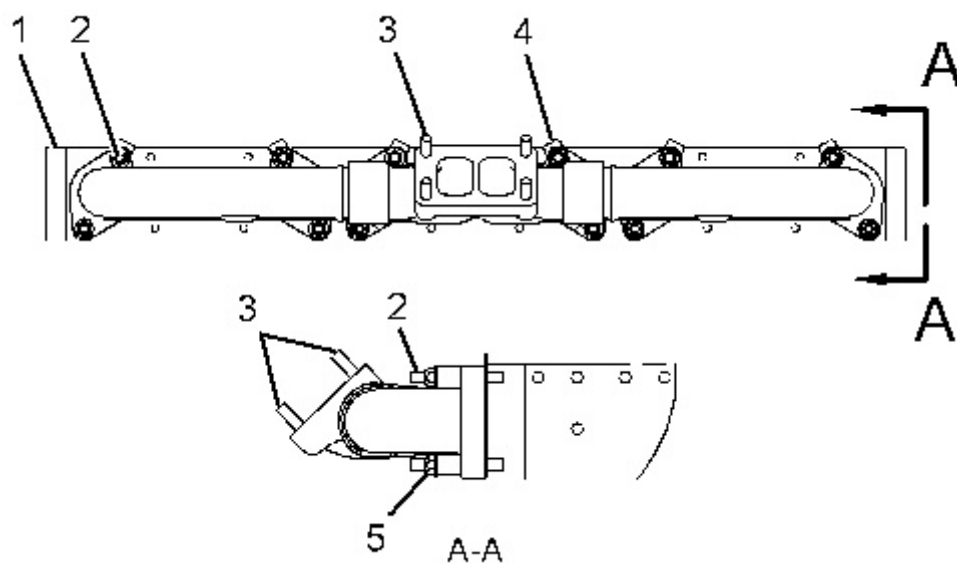


Illustration 1

g01421493

Typical example

(1) Cylinder head

(2) Torque for 12 taperlock studs ... $35 \pm 5 \text{ N}\cdot\text{m}$ ($26 \pm 4 \text{ lb ft}$)(3) Torque for four taperlock studs ... $35 \pm 5 \text{ N}\cdot\text{m}$ ($26 \pm 4 \text{ lb ft}$)

(4) Install the gasket with the tab upward, as shown.

(5) Apply **4C-5599** Anti-Seize Compound to the threads of the locknuts and the bearing surface of the locknuts and the washers before assembly.



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i02854118

Exhaust Manifold

SMCS - 1059

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

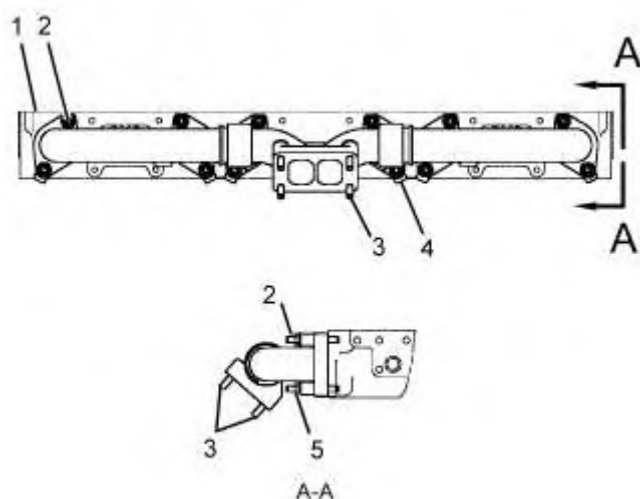


Illustration 1

g01421545

Typical example

(1) Cylinder head

(2) Torque for 12 taperlock studs ... $35 \pm 5 \text{ N}\cdot\text{m}$ ($26 \pm 4 \text{ lb ft}$)

- (3) Torque for four taperlock studs ... $35 \pm 5 \text{ N}\cdot\text{m}$ ($26 \pm 4 \text{ lb ft}$)
 - (4) Install the gasket with the tab downward, as shown.
 - (5) Apply **4C-5599** Anti-Seize Compound to the threads of the locknuts and the bearing surface of the locknuts and the washers before assembly.
-



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i02888041

Flywheel

SMCS - 1156

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

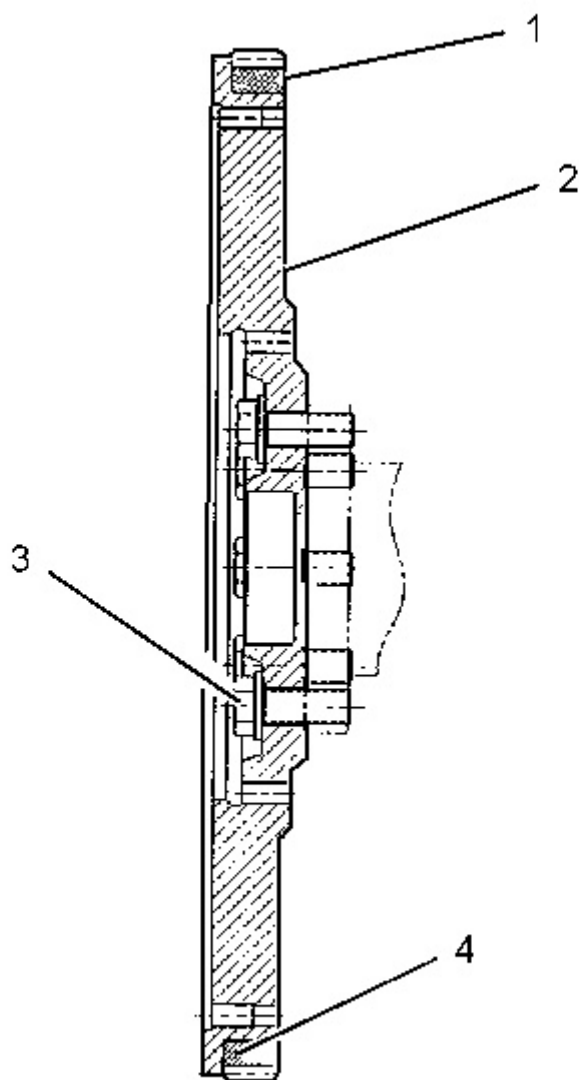


Illustration 1

g01416788

(2) Flywheel

Refer to the Testing And Adjusting, "Basic Block" topic for the proper procedure to check flywheel runout.

(1) Flywheel ring gear

The flywheel ring gear must be assembled against shoulder (4) of the flywheel. The maximum temperature of the ring gear for installation ... 315 °C (599 °F)

(3) Bolt

Apply **7M-7456** Compound to the bolt threads.

Tighten the bolts to the following torque. ... 300 ± 40 N·m (220 ± 30 lb ft)



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i02888139

Flywheel Housing - Rear Power Take-Off (RPTO)

SMCS - 1157

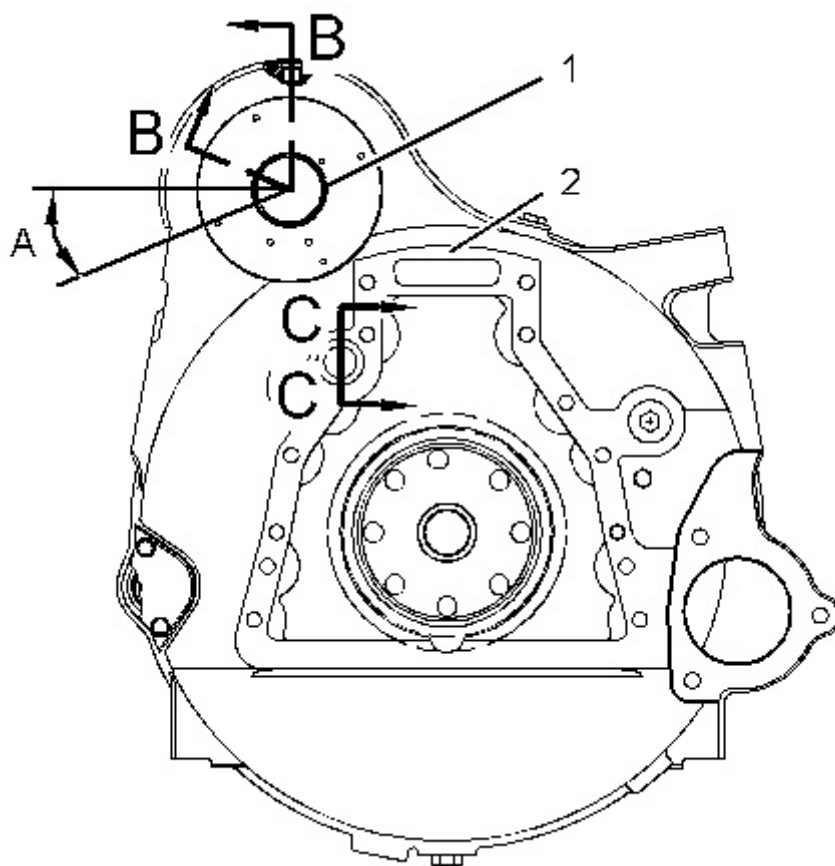


Illustration 1

g01437772

(2) Flywheel housing

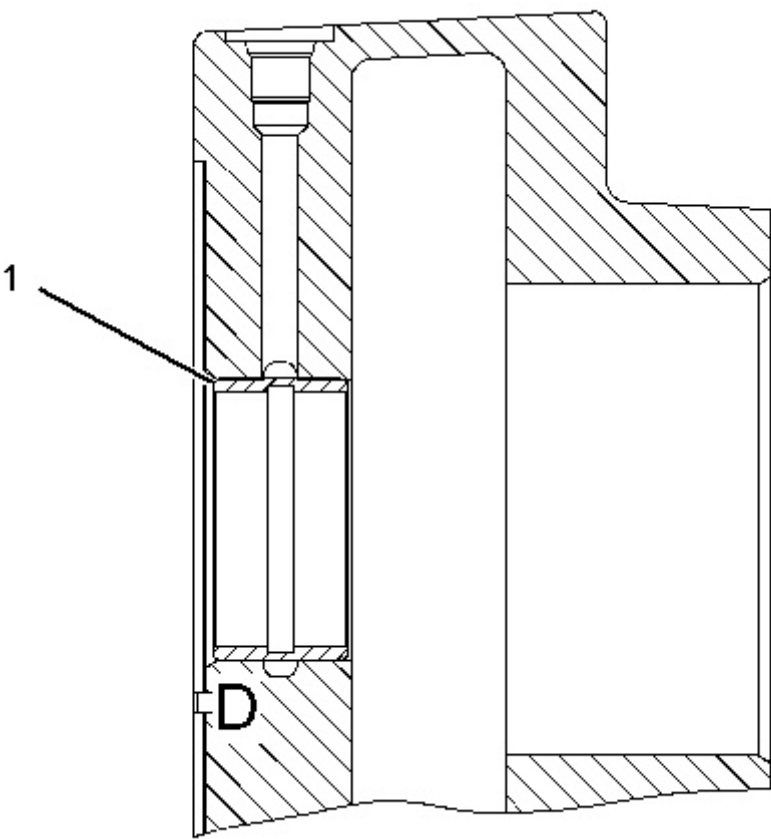


Illustration 2

g01437775

Section A-A

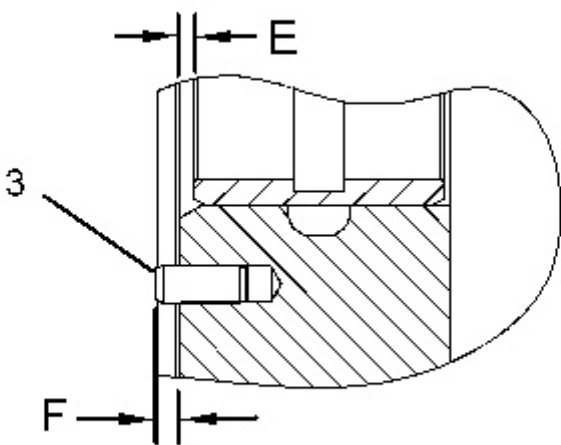


Illustration 3

g01437776

Detail D

(1) Bearing

The split in the bearing should be located at angle (A) 23 ± 10 degrees

The bearing should be installed to depth (E) 1.5 ± 0.5 mm (0.06 ± 0.02 inch)

(3) Dowel

The dowels should extend beyond the surface of the flywheel housing by distance (F) 3.0 ± 0.2 mm (0.12 ± 0.01 inch)

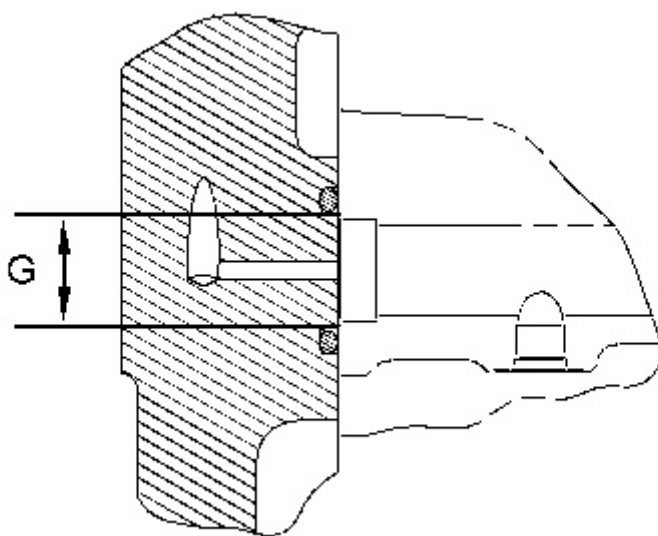


Illustration 4

g01437777

Section B-B

Apply **1U-8846** Sealant on the gasket surface of the flywheel housing except in area (G) .
Assemble the flywheel housing to the cylinder block and tighten within ten minutes of applying the sealant.



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i02888089

Flywheel Housing

SMCS - 1157

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

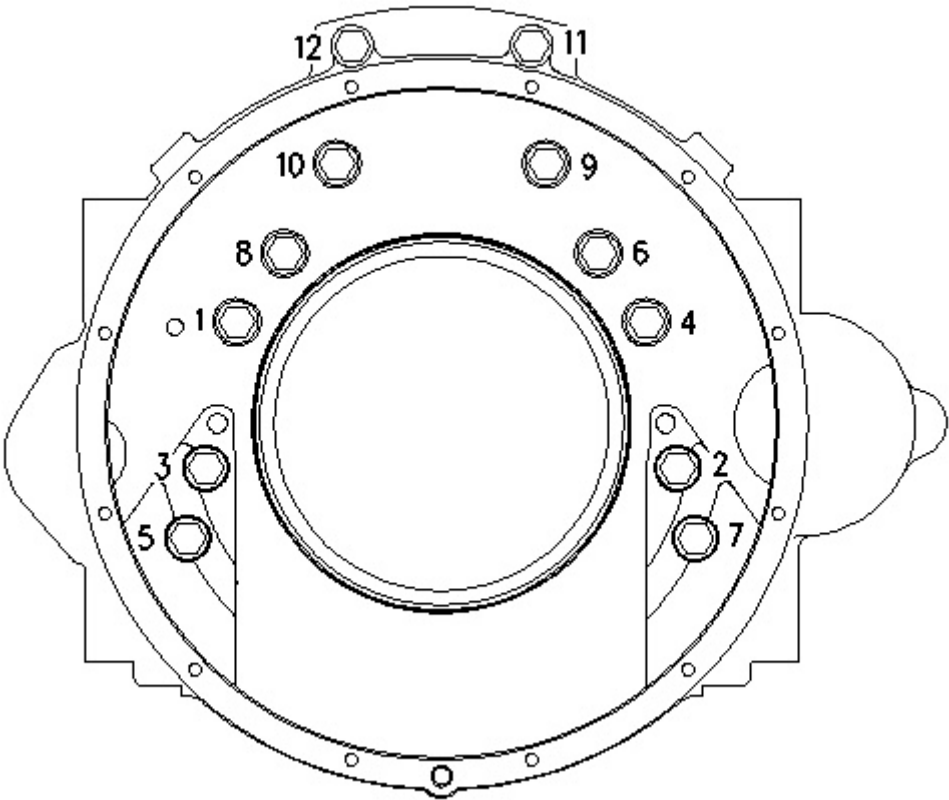


Illustration 1

g00871606

Torque sequence

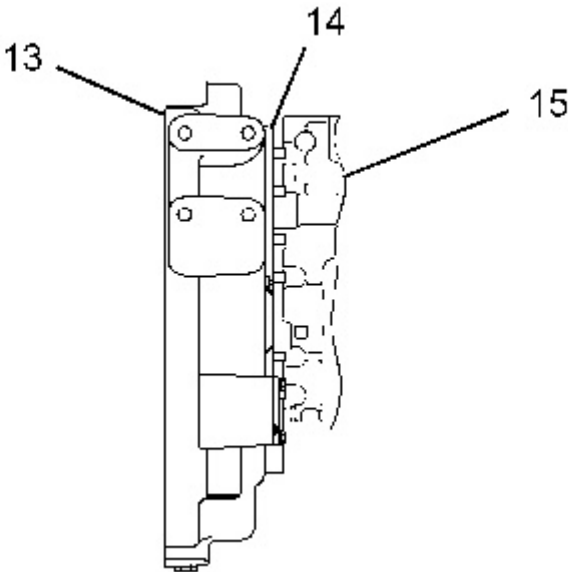


Illustration 2

g01437738

(13) Flywheel housing

(15) Cylinder block

(14) Apply **1U-8846** Sealant on the gasket surface of the flywheel housing. Assemble the flywheel housing to the cylinder block and tighten within ten minutes of applying the sealant. Use the torque sequence that is shown in Illustration 1 to tighten the bolts.



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i02888012

Flywheel

SMCS - 1156

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

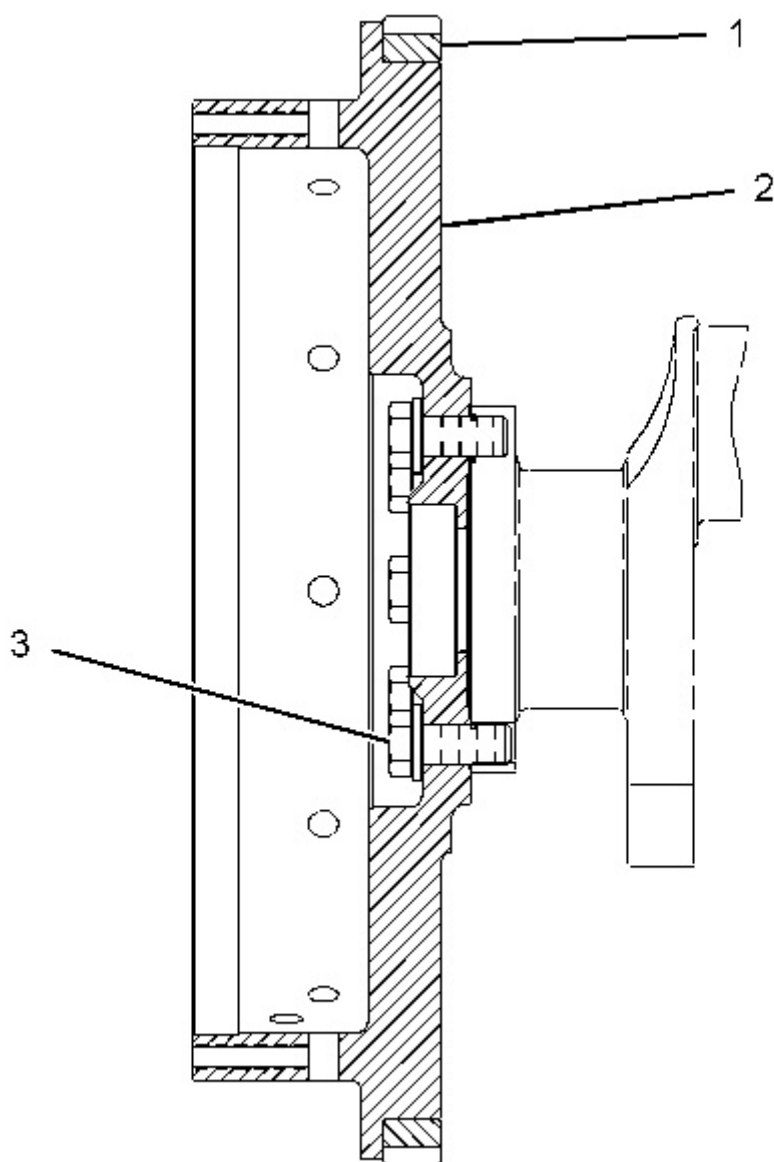


Illustration 1

g01437629

(1) Flywheel ring gear

(2) Flywheel

Refer to the Testing And Adjusting, "Basic Block" topic for the proper procedure to check flywheel runout.

(3) Bolt

Apply **7M-7456** Compound to the bolt threads.

Tighten the bolts to the following torque. ... 300 ± 40 N·m (220 ± 30 lb ft)



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i02877776

Fuel Lines

SMCS - 1274

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

Low Pressure Fuel System with Integral Pressure Regulator Valve

Type 1

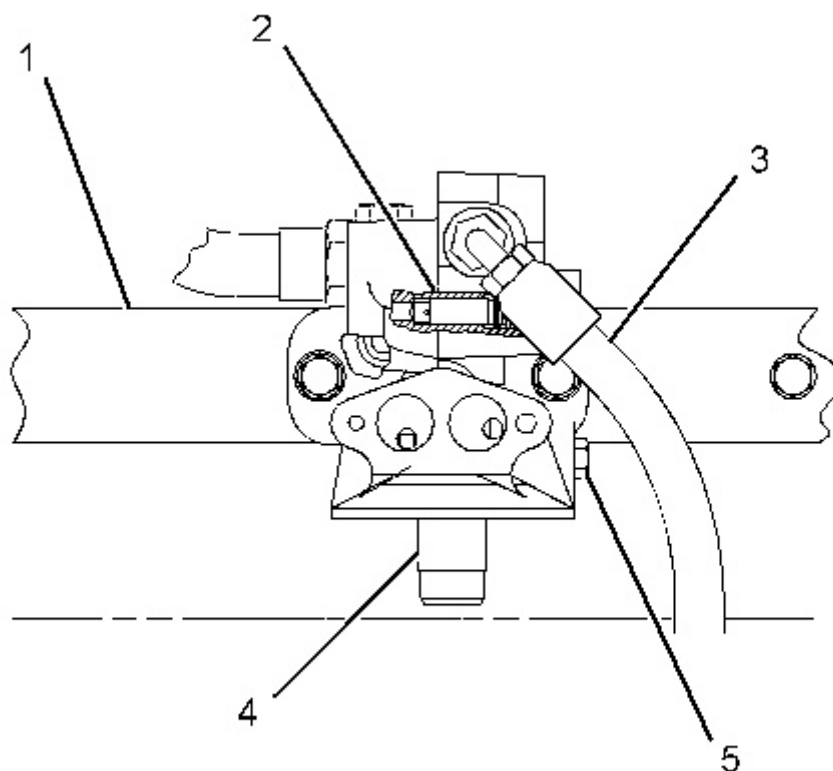


Illustration 1

g01432231

Fuel filter base with integral pressure regulator valve

Typical example

(1) Fuel manifold

(2) Integral pressure regulator valve

104-1729 Spring

Assembled length ... 18.37 mm (0.723 inch)

Operating length (minimum) ... 15.37 mm (0.605 inch)

Free length after test ... 30.2 ± 0.7 mm (1.19 ± 0.03 inch)

Outside diameter ... 11.29 mm (0.444 inch)

(3) Supply hose assembly

(4) Stud

Apply **155-0695** Thread Lock Compound on the taperlock end of the stud to the following length. ... 7.6 mm (0.30 inch)Tighten the stud to the following torque. ... 70 ± 15 N·m (50 ± 11 lb ft)

(5) Plug (filtered fuel pressure)

Type 2

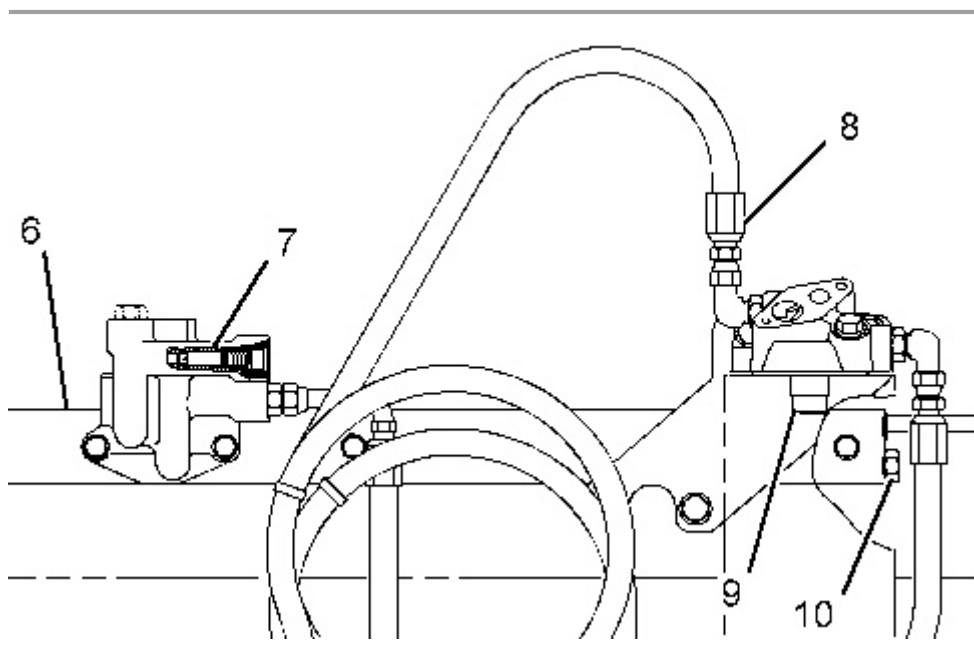


Illustration 2

g01432235

Fuel filter base and fuel adapter with integral pressure regulator valve

Typical example

(6) Fuel manifold

(7) Integral pressure regulator valve

104-1729 Spring

Assembled length ... 18.37 mm (0.723 inch)

Operating length (minimum) ... 15.37 mm (0.605 inch)

Free length after test ... 30.2 ± 0.7 mm (1.19 ± 0.03 inch)

Outside diameter ... 11.29 mm (0.444 inch)

(8) Supply hose assembly

(9) Stud

Apply **155-0695** Thread Lock Compound on the taperlock end of the stud to the following length. ... 7.6 mm (0.30 inch)

Tighten the stud to the following torque. ... 70 ± 15 N·m (50 ± 11 lb ft)

(10) Plug (filtered fuel pressure)

Low Pressure Fuel System with Pressure Regulator Assembly that is Self-Contained

Type 1

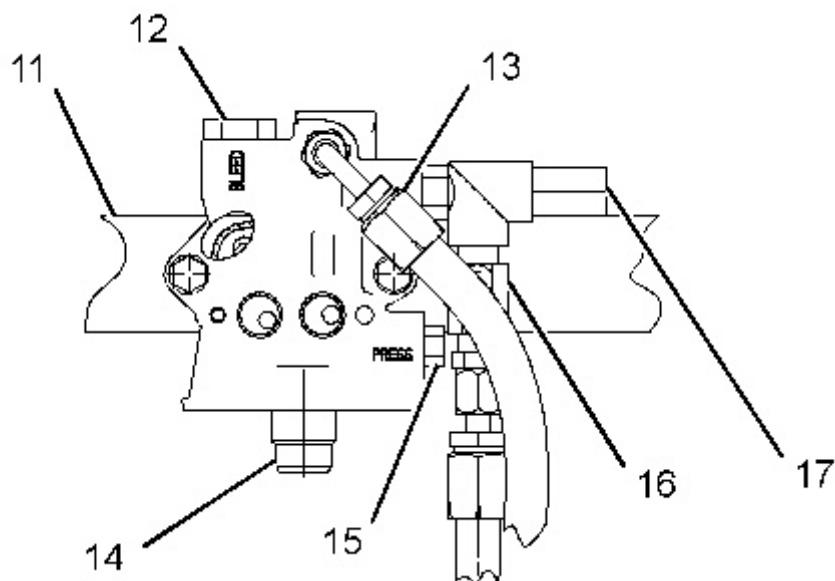


Illustration 3

g01432237

Fuel filter base with pressure regulator assembly that is self-contained

Typical example

(11) Fuel manifold

(12) Pressure regulator assembly

(13) Supply hose assembly

(14) Stud

Apply **155-0695** Thread Lock Compound on the taperlock end of the stud to the following length. ... 7.6 mm (0.30 inch)

Tighten the stud to the following torque. ... 70 ± 15 N·m (50 ± 11 lb ft)

(15) Plug (filtered fuel pressure)

(16) Ball valve assembly

(17) Backflow check valve

Type 2

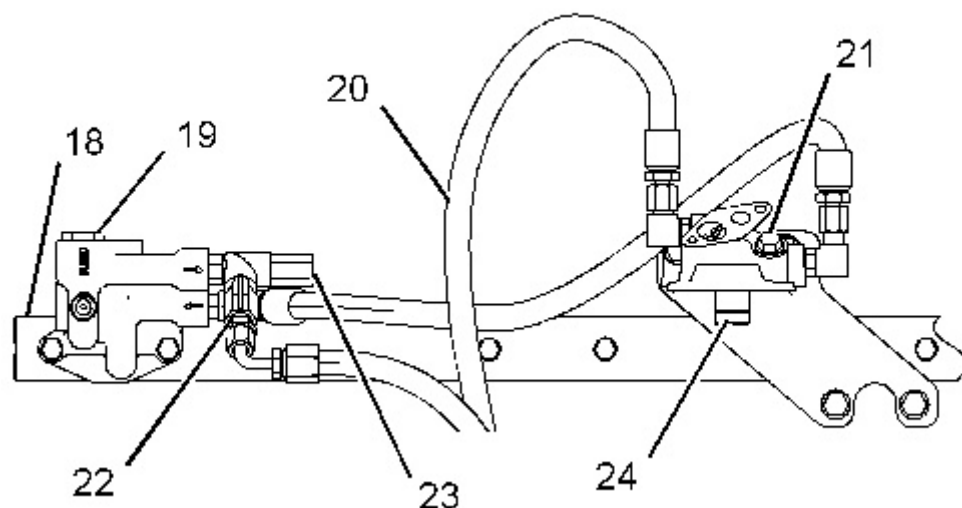


Illustration 4

g01432238

Fuel filter base and fuel adapter with pressure regulator assembly that is self-contained

Typical example

(18) Fuel manifold

(19) Pressure regulator assembly

(20) Supply hose assembly

(21) Plug (filtered fuel pressure)

(22) Ball valve assembly

(23) Backflow check valve

(24) Stud

Apply **155-0695** Thread Lock Compound on the taperlock end of the stud to the following length. ... 7.6 mm (0.30 inch)

Tighten the stud to the following torque. ... 70 ± 15 N·m (50 ± 11 lb ft)

Thermal Bypass Valve (if equipped)

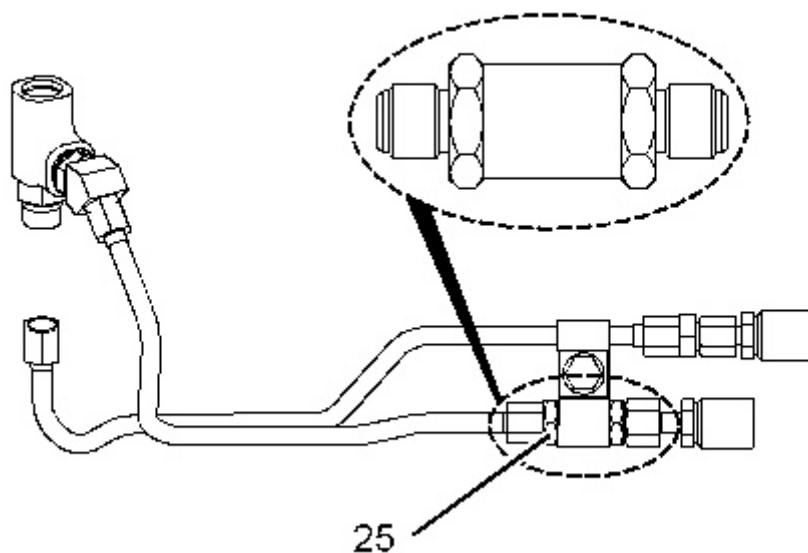


Illustration 5

g01432243

(25) The return fuel system may be equipped with a thermal bypass valve. The thermal bypass valve returns fuel that has been heated by the engine to the inlet of the fuel transfer pump. This valve is actuated by the temperature of the fuel.

The valve starts to close at the following temperature. ... 70 °C (158 °F)

The valve is fully closed at the following temperature. ... 80 °C (176 °F)

The maximum operating temperature of the valve is at the following temperature. ... 120 °C (248 °F)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

Date Updated -28/11/2007

i04385518

Fuel Pressure Sensor

SMCS - 1718

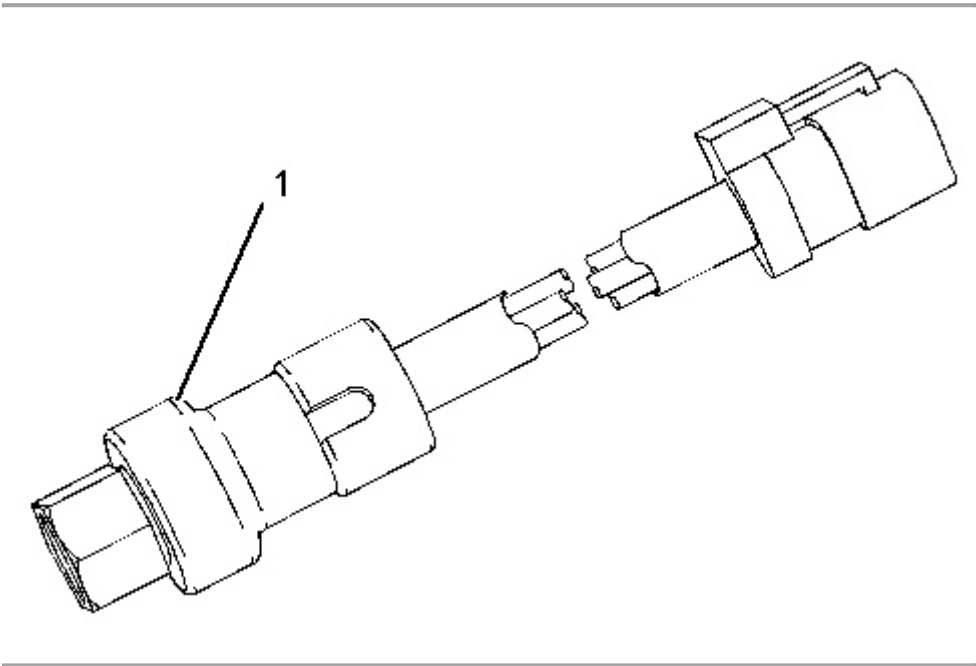


Illustration 1

g02554181

Typical example

Table 1

Specification for the 194-6725 Pressure Sensor Gp, and 320-3060 Pressure Sensor			
Item	Qty	Part	Specification Description
1	1	-	Torque to 10 ± 2 N·m (90 ± 18 lb in).
			Operating voltage is 5.00 ± 0.25 VDC.



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

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i02845094

Fuel Priming Pump

SMCS - 1258

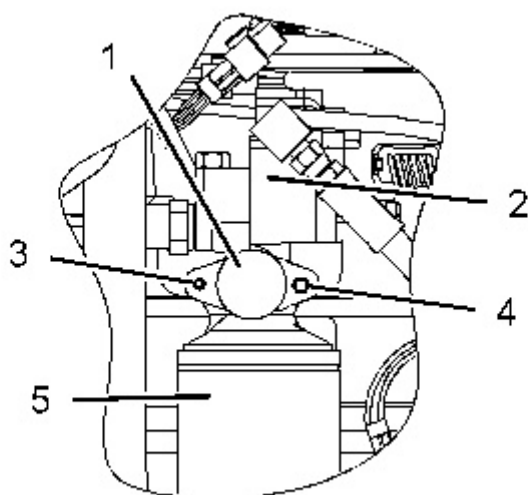


Illustration 1

g01296043

(1) Fuel priming pump assembly

(2) Fuel filter base

(5) Filter element

(3) Torque for bolt ... $13 \pm 3 \text{ N}\cdot\text{m}$ ($10 \pm 2 \text{ lb ft}$)(4) Torque for bolt ... $25 \pm 6 \text{ N}\cdot\text{m}$ ($18 \pm 4 \text{ lb ft}$)



Specifications
C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

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i04385495

Fuel Temperature Sensor

SMCS - 1922

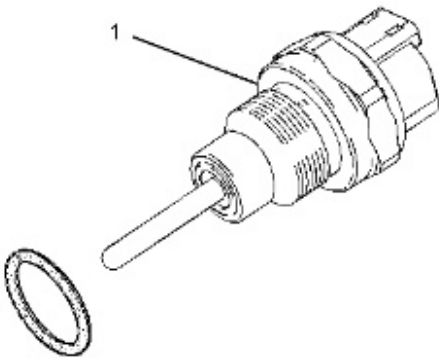


Illustration 1

g02574738

Table 1

Specification for the 130-9811 Temperature Sensor Gp, 197-8391 Temperature Sensor Gp, and 256-6453 Temperature Sensor			
Item	Qty	Part	Specification Description
1	1	-	Torque to 20 ± 3 N·m (175 ± 25 lb in).
			Output type is passive.



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

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i03596440

Fuel Transfer Pump

SMCS - 1256

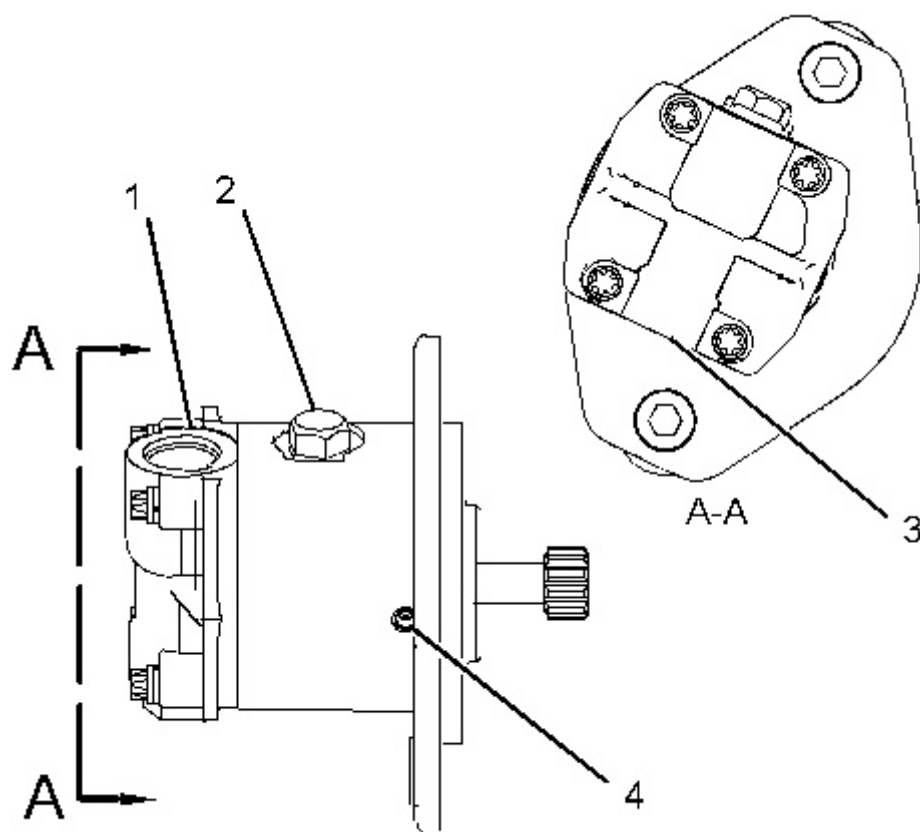


Illustration 1

g01430288

(1) Inlet port

(2) Pressure regulating valve

(3) Outlet port

(4) Drain hole

When the fuel transfer pump is using diesel fuel, the fuel transfer pump has the following specifications.

Run the pump at 2940 rpm. The full bypass pressure should be the following value. ... 716 to 786 kPa (104 to 114 psi)

Run the pump at 840 rpm. The flow at 550 kPa (80 psi) must be 3.0 L/min (0.8 US gpm).

Run the pump at 100 rpm with a blocked inlet for 30 seconds. The pump inlet must have a suction lift of 60.9 kPa (18 In Hg) minimum.

Run the pump at 2940 rpm. The flow at 650 kPa (94 psi) must be 4.5 L/min (1.19 US gpm).

When the pump is viewed from the drive end the rotation of the pump is counterclockwise.



Specifications

C-10 and C-12 Truck Engines

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i02847514

Gear Group (Front)

SMCS - 1206

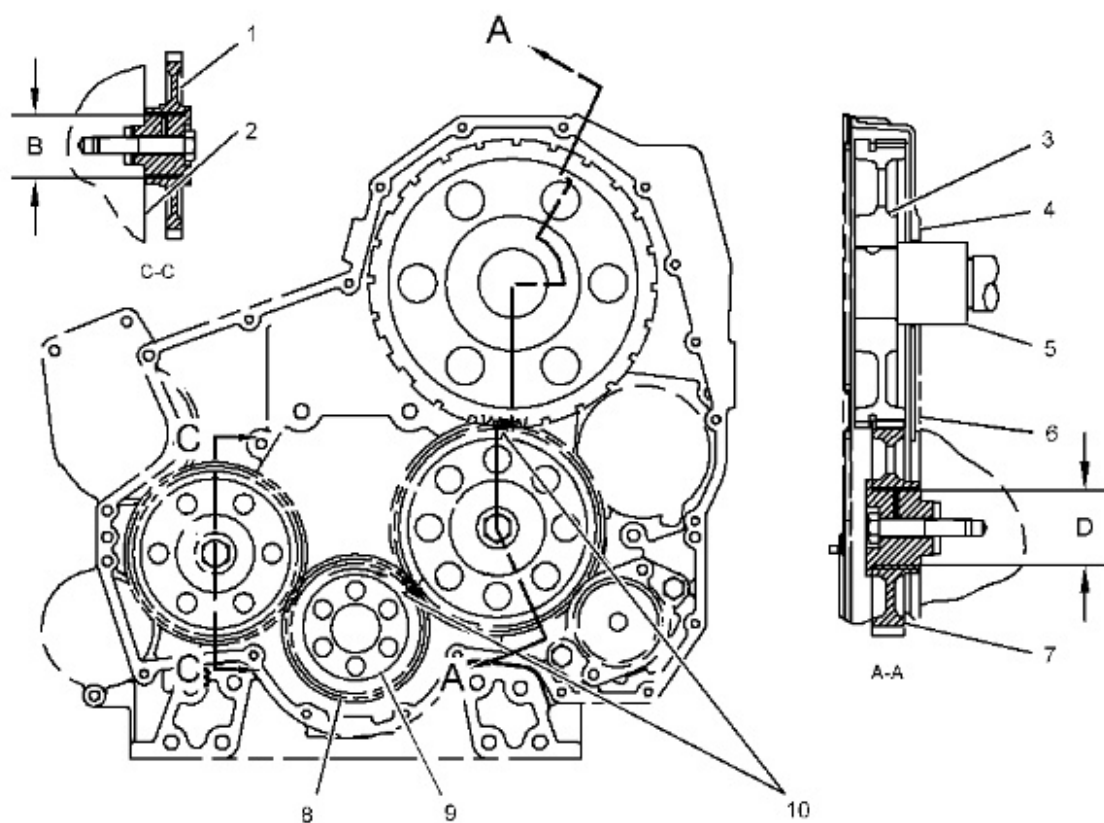


Illustration 1

g01419650

- (2) Front face of the block
- (3) Camshaft gear
- (4) Front timing gear housing

- (5) Camshaft
- (6) Front face of the block
- (8) Crankshaft gear
- (9) Crankshaft

(1) Idler gear

Bore in idler gear for the bearing ... 60.163 ± 0.015 mm (2.3686 ± 0.0006 inch)

Diameter of shaft for idler gear ... 55.047 ± 0.020 mm (2.1672 ± 0.0008 inch)

Installation depth of the bearing in the idler gear ... 1.00 ± 0.25 mm (0.039 ± 0.010 inch)

(B) Bore in bearing after assembly in the gear assembly ... 55.163 ± 0.041 mm (2.1718 ± 0.0016 inch)

(7) Idler gear assembly

Bore in cam idler gear for the sleeve bearing ... 74.452 ± 0.015 mm (2.9312 ± 0.0006 inch)

Diameter of shaft for cam idler gear ... 69.141 ± 0.008 mm (2.7221 ± 0.0003 inch)

Installation depth of the sleeve bearing in the cam idler gear ... 1.00 ± 0.25 mm (0.039 ± 0.010 inch)

(D) Bore in sleeve bearing after assembly in the idler gear assembly ... 69.205 ± 0.013 mm (2.7246 ± 0.0005 inch)

(10) Align the timing marks that are on the idler gear assembly with the timing marks that are on the crankshaft gear and the camshaft gear.



Specifications

C-10 and C-12 Truck Engines

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i02875871

Housing (Front)

SMCS - 1151

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

The front of the engine block has been modified. This modification requires the use of different bolts that attach the front housing to the engine block. The correct bolts need to be used to attach the front housing. Use Table 1 to identify your type of engine.

Table 1

	Engine Model	Serial Number (Type 1)	Serial Number (Type 2)
Industrial	3176C	2AW1-1328	2AW1329-UP
	3196	1DW1-627	1DW628-UP
	C-10	BCX1-268	BCX269-UP
	C-10	-	7131-UP
	C-10	-	CRJ1-UP
	C-12	CPN1-323	CPN324-UP
	C-12	BDL1-236	BDL237-UP
Machine	3176C	3PD1-450	3PD451-UP
	3176C	7ZR1-6108	7ZR6109-UP
	3196	5ED1-428	5ED429-UP

	3196	6AR276-2872	6AR2873-UP
	C-12	-	JAC1-UP
Marine	3176C	6BW333-500	6BW501-UP
	3196	2XR1-4150	2XR4151-42478
	C-12	-	9HP1-UP
	C-12	-	C1Z1-UP
	C-12	-	K1M1-UP
	C-12	-	NFL1-UP
Truck	C-10	2PN1-UP	-
	C-10	8YS1-UP	-
	C-10	3CS1-21950	3CS21951-UP
	C-12	1YN1-UP	-
	C-12	9NS1-UP	-
	C-12	2KS1-69602	2KS69603-UP
	C-12	SRK1-UP	
	C-12	CPD1-304	CPD305-UP
	C-12	9SM1-862	9SM863-UP

Table 1 is your guide for selecting your correct type of engine. If your engine is a Type 1 proceed to "Type 1". If your engine is a Type 2 refer to "Type 2". If you have a remanufactured engine then proceed to "Remanufactured Engines". The following illustrations will provide added information on the installation of the front housing.

Note: Check on the availability of a gasket kit for your application. The kit includes both the short bolts and the long bolts.

Type 1

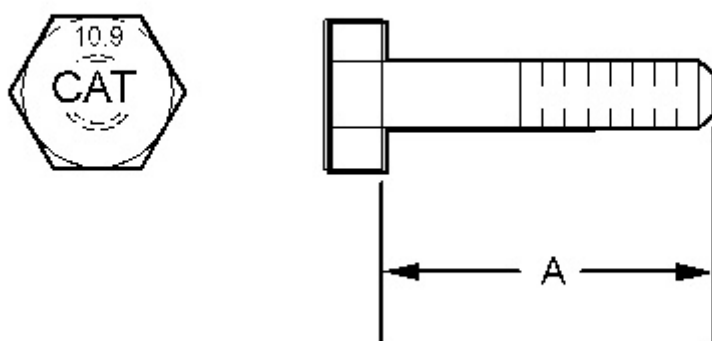


Illustration 1

g01408559

6V-2317 Bolt

(A) 30 mm (1.18 inch)

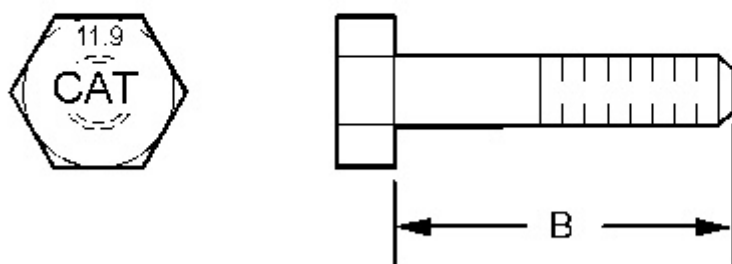


Illustration 2

g01408564

204-0712 Bolt

(B) 30 mm (1.18 inch)

The two bolts that are Type 1 are 30 mm (1.18 inch) long. Marks on the head of the bolt represent the grade of the bolt. These markings can be seen in Illustration 1 and Illustration 2. The **6V-2317**

Bolt is being replaced by **204-0712** Bolt . The **204-0712** Bolt has a torque value of $40 \pm 8 \text{ N}\cdot\text{m}$ ($30 \pm 6 \text{ lb ft}$). 13 bolts are required for the installation of the front housing.

Note: Check availability of the kit which includes the gasket and the bolts.

Type 2

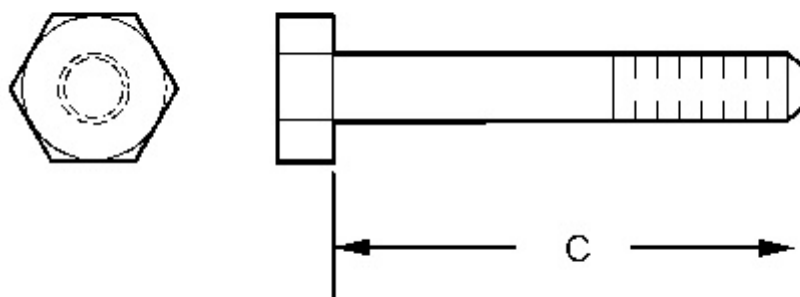


Illustration 3

g01408555

6V-5219 Bolt

(C) 40 mm (1.57 inch)

The **6V-5219** Bolt is 10 mm (0.39 inch) longer than the **204-0712** Bolt . The longer bolt is needed due to a deeper counterbore and the tapped hole. **Do Not use the 204-0712 Bolt or the 6V-2317 Bolt . This could result in damage to the threads. The torque on the 6V-5219 Bolt bolt is $35 \pm 8 \text{ N}\cdot\text{m}$ ($26 \pm 6 \text{ lb ft}$).** 13 bolts are required for the installation of the front housing.

Remanufactured Engines

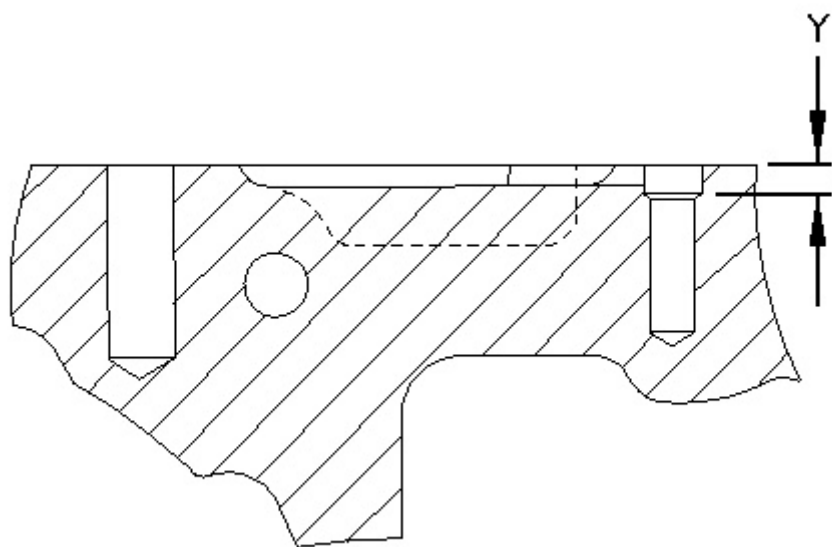


Illustration 4

g00838008

Depth of counterbore (Y) requires engines to use 204-0712 Bolt .

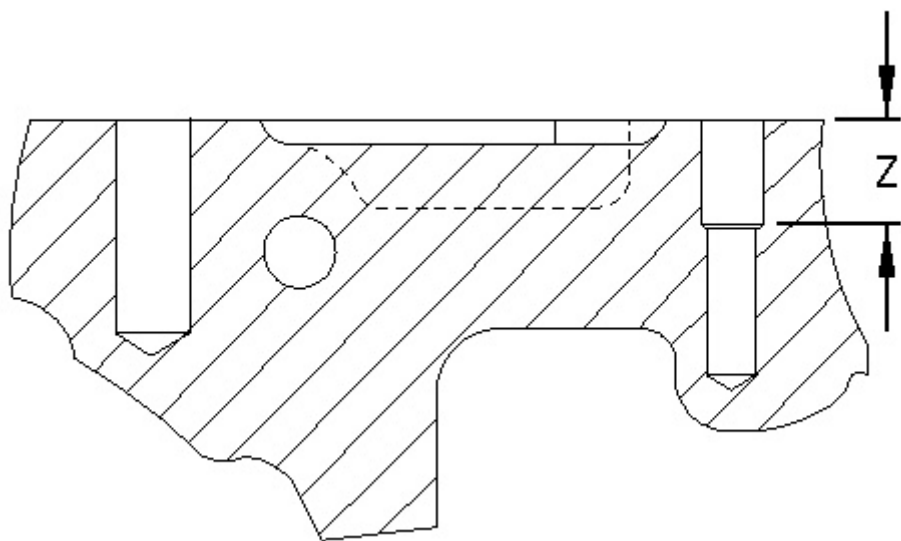


Illustration 5

g00838007

Depth of counterbore (Z) requires engines to use 6V-5219 Bolt .

Remanufactured engines require special attention in order to use the correct bolt. The counterbore in the engine block must be measured in order to determine the bolts that will be used. If your engine block has a counterbore (Y) of 5 mm (0.20 inch), refer to "Type 1". If your engine block has a counterbore (Z) of 15 mm (0.60 inch), refer to "Type 2".

Table 2		

Dimension	Counterbore Depth	Bolt
Y	5 mm (0.20 inch)	204-0712 (Type 1)
Z	15 mm (0.60 inch)	6V-5219 (Type 2)

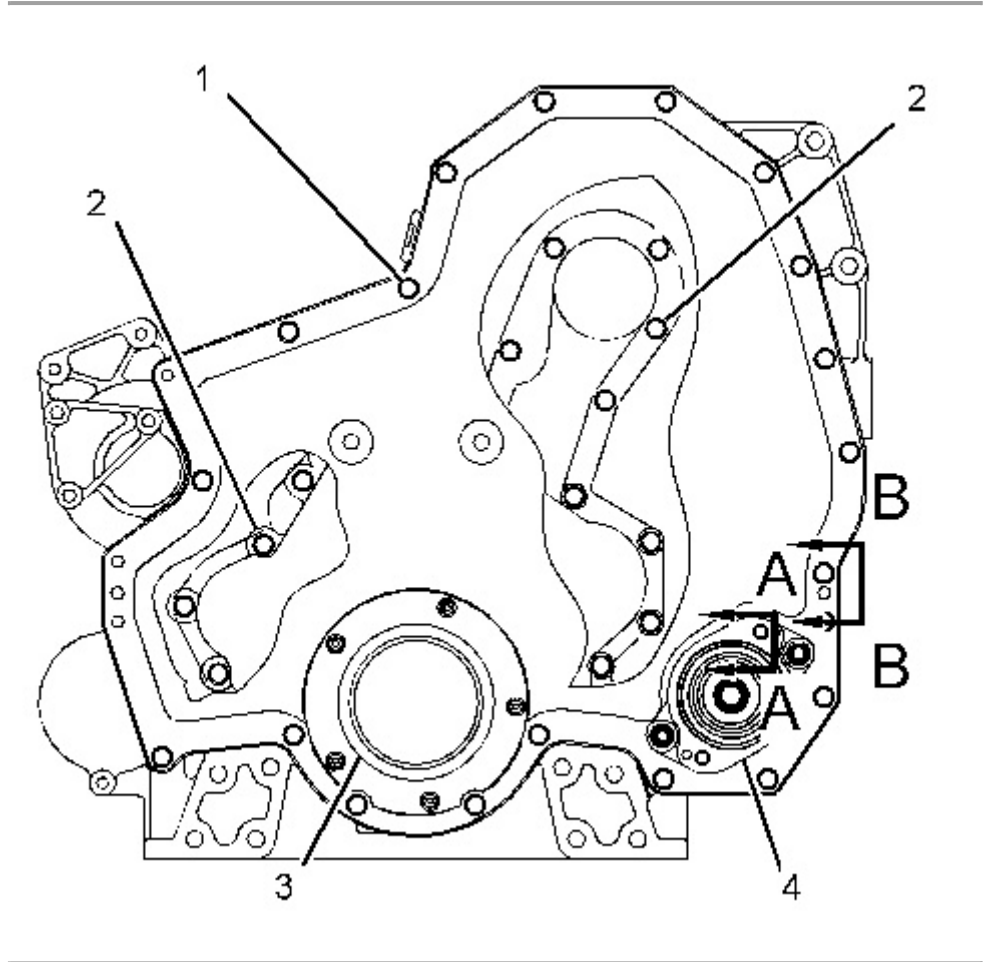


Illustration 6

g01301899

Typical example

(1) Bolt

Tighten the bolts to the following torque. ... 35 ± 8 N·m (26 ± 6 lb ft)

(2) Bolt

Tighten the **204-0712** Bolt 30 mm (1.18 inch) in length to the following torque. ... 40 ± 8 N·m (30 ± 6 lb ft)

Tighten the **6V-5219** Bolt to the following torque. ... 35 ± 8 N·m (26 ± 6 lb ft)

(3) O-ring seal and crankshaft front seal

The O-ring seal and the crankshaft front seal are part of the front housing. Lubricate the O-ring seal lightly. Use the lubricant that is being sealed.

(4) Accessory drive assembly

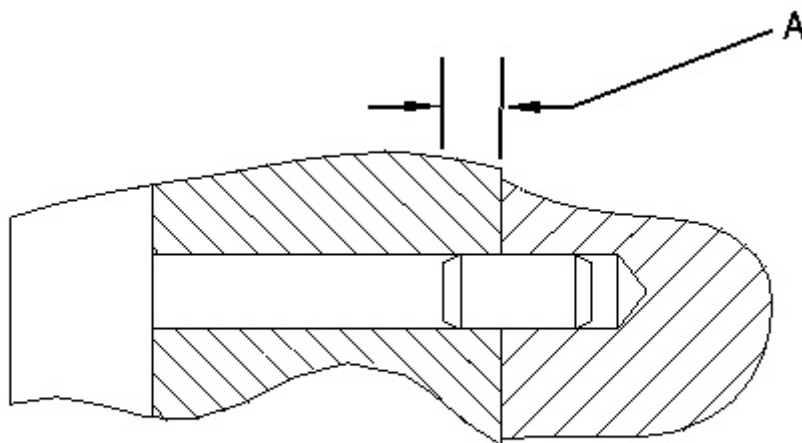


Illustration 7

g00837947

View A-A

(A) The dowel extends past the surface of the housing by the following distance. ... 5.0 ± 0.5 mm (0.20 ± 0.02 inch)

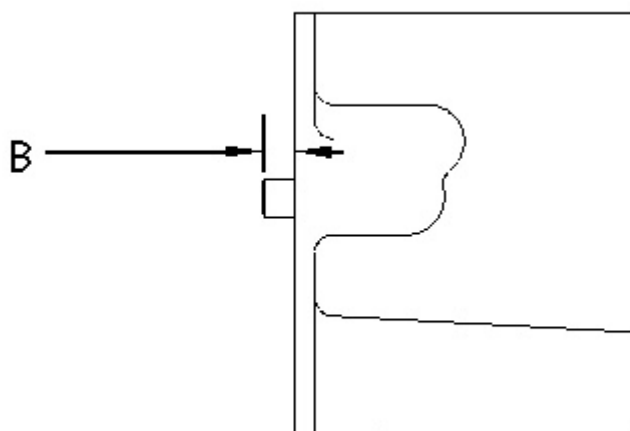


Illustration 8

g00837952

View B-B

(B) The dowel extends past the surface of the housing by the following distance. ... 10.0 ± 0.5 mm
(0.39 ± 0.02 inch)

Note: Apply **8T-9022** Silicone Gasket in order to fill the space at the connecting joint of the front housing, the engine oil pan, and the cylinder block.



Specifications

C-10 and C-12 Truck Engines

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i02886267

Inlet Manifold

SMCS - 1058

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

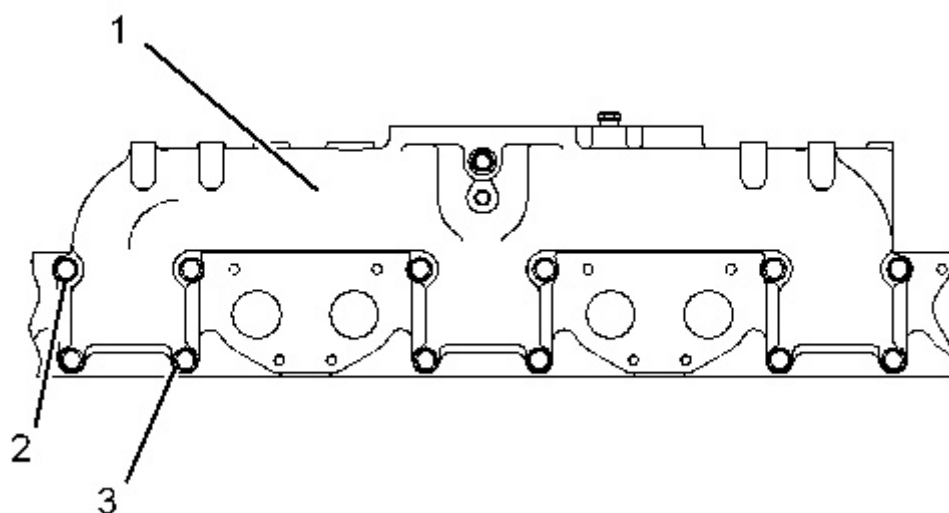


Illustration 1

g01436772

(1) Air inlet manifold

(2) Long bolts

(3) Short bolts

Two different lengths of mounting bolts are used. There are six long bolts and six shorter bolts that are used on this engine. Be sure that the correct length of bolt is installed in the correct hole.

Tighten the mounting bolts to the following torque: ... $55 \pm 10 \text{ N}\cdot\text{m}$ ($40 \pm 7 \text{ lb ft}$)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

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i02878585

Lifter Group

SMCS - 1209

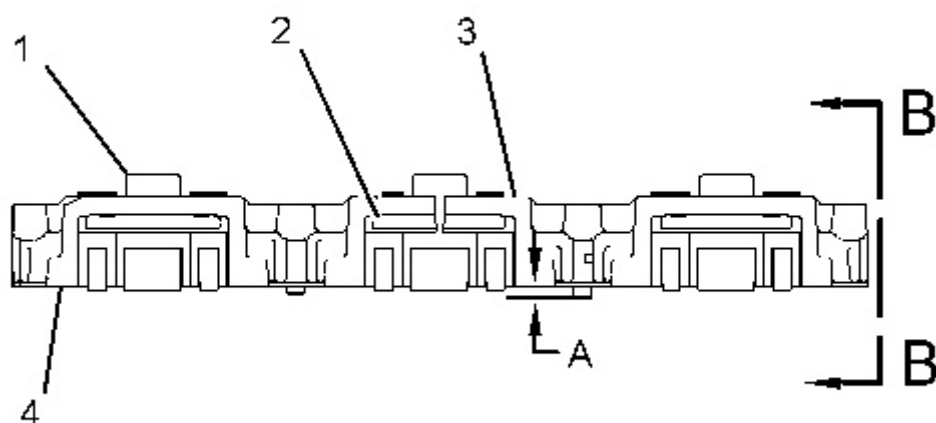


Illustration 1

g01432795

- (1) Electronic unit injector lifter
- (2) Valve lifter
- (3) Oil supply lifter bracket assembly
- (4) Lifter bracket assembly

(A) The dowels extend below the surface of the lifter bracket assembly. Distance ... 4.0 ± 1.0 mm (0.16 ± 0.04 inch)

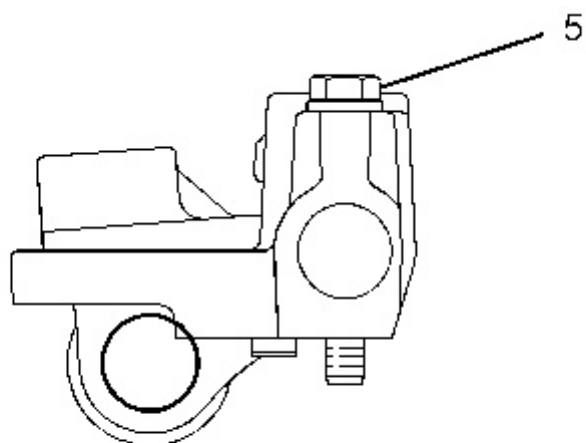


Illustration 2

g01433121

View B-B

(6) Tighten the bolts to the following torque. ... $30 \pm 3 \text{ N}\cdot\text{m}$ ($22 \pm 2 \text{ lb ft}$)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

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i02845841

Main Bearing Journal

SMCS - 1202; 1203

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

Table 1

Main Bearing Journal	
Original size journal	108.000 $\pm$ 0.020 mm (4.2520 $\pm$ 0.0008 inch)
Undersize journal 0.510 mm (0.0201 inch)	107.490 $\pm$ 0.020 mm (4.2319 $\pm$ 0.0008 inch)
Undersize journal 0.760 mm (0.0299 inch)	107.240 $\pm$ 0.020 mm (4.2220 $\pm$ 0.0008 inch)

Clearance between a new bearing and a new journal ... 0.081 to 0.181 mm (0.0032 to 0.0071 inch)

Table 2

Main Bearing Bore	
Main bearing bore (original size)	116.000 $\pm$ 0.013 mm (4.5669 $\pm$ 0.0005 inch)
Oversize bore in block 0.510 mm (0.0201 inch)	116.510 $\pm$ 0.013 mm (4.5870 $\pm$ 0.0005 inch)



Specifications

C-10 and C-12 Truck Engines

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i02838591

Piston and Rings

SMCS - 1214; 1215

S/N - 2KS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

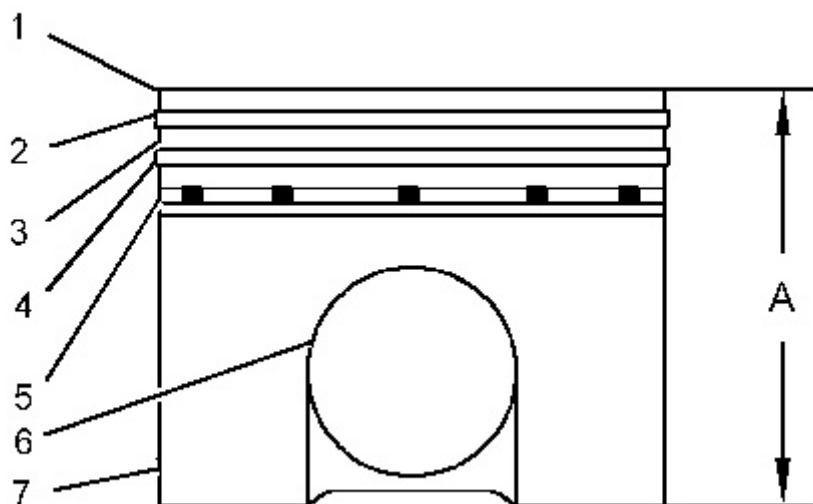


Illustration 1

g01416650

NOTICE

The piston pin plug prevents the piston pin from scuffing the cylinder wall. The piston pin plug does not retain the piston pin into the

assembly. Once the piston assembly is removed from the cylinder liner, the piston pin is free to fall out of the piston assembly at any time.

(1) The piston is symmetrical with a center crater. Lubricate the entire piston in zone (A) prior to assembly into the cylinder block. Use clean engine oil.

Top Piston Ring (2)

Install the piston ring with the side marked "UP-1" toward the top of the piston. The green stripe faces to the right of the piston ring end gap.

When a new piston ring is installed in a cylinder liner with a bore of 130 mm (5.1 inch), the clearance between the ends of the piston ring is the following value: ... 0.50 ± 0.10 mm (0.020 ± 0.004 inch)

Increase in clearance for the piston ring ... 0.03 mm (0.001 inch)

Increase in the cylinder liner bore ... 0.09 mm (0.004 inch)

Minimum depth of the groove in new piston for the top ring ... 5.43 mm (0.214 inch)

Thickness of new top piston ring ... 4.05 mm (0.159 inch)

Intermediate Piston Ring (4)

Install the piston ring with the side marked "UP-2" toward the top of the piston. The red stripe faces to the right of the piston ring end gap.

Width of groove in new piston for intermediate piston ring ... 3.052 ± 0.013 mm (0.1202 ± 0.0005 inch)

Minimum depth of groove in new piston for intermediate piston ring ... 5.43 mm (0.214 inch)

Thickness of new intermediate piston ring ... 2.950 ± 0.013 mm (0.1161 ± 0.0005 inch)

Clearance between groove and intermediate piston ring ... 0.076 to 0.128 mm (0.0030 to 0.0050 inch)

When a new piston ring is installed in a cylinder liner with a bore of 130 mm (5.1 inch), the clearance between the ends of the piston ring is the following value: ... 0.65 ± 0.10 mm (0.026 ± 0.004 inch)

Increase in the clearance for the intermediate piston ring ... 0.03 mm (0.001 inch)

Increase in the cylinder liner bore ... 0.09 mm (0.004 inch)

Oil Control Piston Ring (5)

The ends of the oil control piston ring should be a distance of 180 degrees from the ring end gap when the oil control piston ring is assembled. The white colored portion of the piston ring must be visible at the ring end gap.

Width of groove in new piston for oil control piston ring ... 4.042 ± 0.012 mm (0.1591 ± 0.0005 inch)

Minimum depth of groove in new piston for oil control piston ring ... 3.727 mm (0.1467 inch)

Thickness of new oil control piston ring ... 3.980 ± 0.010 mm (0.1567 ± 0.0004 inch)

Clearance between groove and oil control piston ring ... 0.030 to 0.080 mm (0.0012 to 0.0031 inch)

When a new piston ring is installed in a cylinder liner with a bore of 130 mm (5.1 inch), the clearance between the ends of the piston ring is the following value: ... 0.44 ± 0.15 mm (0.017 ± 0.006 inch)

Increase in clearance for the piston ring ... 0.03 mm (0.001 inch)

Increase in the cylinder liner bore ... 0.09 mm (0.004 inch)

After the piston rings have been installed, rotate the piston rings so that the end gaps are 120 degrees from each other.

Piston Pin Bore (6)

Note: Thoroughly lubricate the piston pin with clean engine oil prior to assembly.

Bore in piston skirt (7) for the piston pin ... 53.190 ± 0.005 mm (2.0941 ± 0.0002 inch)

Bore in the piston crown assembly (3) for the piston pin ... 53.205 ± 0.010 mm (2.0947 ± 0.0004 inch)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

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i02853435

Piston and Rings

SMCS - 1214; 1215

S/N - 2KS1-UP

S/N - 8YF1-UP

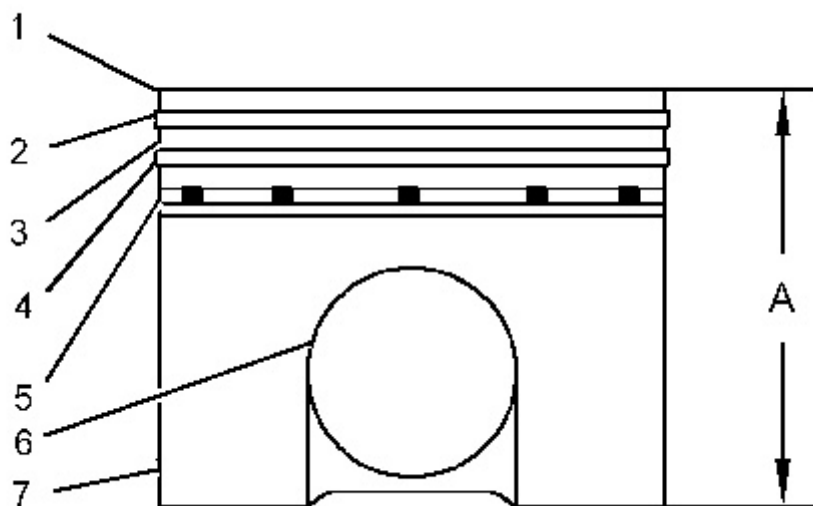


Illustration 1

g01416650

NOTICE

The piston pin plug prevents the piston pin from scuffing the cylinder wall. The piston pin plug does not retain the piston pin into the assembly. Once the piston assembly is removed from the cylinder liner, the piston pin is free to fall out of the piston assembly at any time.

(1) The piston is symmetrical with a center crater. Lubricate the entire piston in zone (A) prior to assembly into the cylinder block. Use clean engine oil.

Top Piston Ring (2)

Install the piston ring with the side marked "UP-1" toward the top of the piston. The green stripe faces to the right of the piston ring end gap.

When a new piston ring is installed in a cylinder liner with a bore of 130 mm (5.1 inch), the clearance between the ends of the piston ring is the following value: ... 0.515 ± 0.125 mm (0.0203 ± 0.0049 inch)

Increase in clearance for the piston ring ... 0.03 mm (0.001 inch)

Increase in the cylinder liner bore ... 0.09 mm (0.004 inch)

Minimum depth of the groove in new piston for the top ring ... 5.43 mm (0.214 inch)

Thickness of new top piston ring ... 3.93 mm (0.155 inch)

Intermediate Piston Ring (4)

Install the piston ring with the side marked "UP-2" toward the top of the piston. The red stripe faces to the right of the piston ring end gap.

Width of groove in new piston for intermediate piston ring ... 3.052 ± 0.013 mm (0.1202 ± 0.0005 inch)

Minimum depth of groove in new piston for intermediate piston ring ... 5.43 mm (0.214 inch)

Thickness of new intermediate piston ring ... 2.950 ± 0.013 mm (0.1161 ± 0.0005 inch)

Clearance between groove and intermediate piston ring ... 0.076 to 0.128 mm (0.0030 to 0.0050 inch)

When a new piston ring is installed in a cylinder liner with a bore of 130 mm (5.1 inch), the clearance between the ends of the piston ring is the following value: ... 0.65 ± 0.10 mm (0.026 ± 0.004 inch)

Increase in clearance for the intermediate piston ring ... 0.03 mm (0.001 inch)

Increase in the cylinder liner bore ... 0.09 mm (0.004 inch)

Oil Control Piston Ring (5)

The ends of the oil control piston ring should be a distance of 180 degrees from the ring end gap when the oil control piston ring is assembled. The white colored portion of the piston ring must be visible at the ring end gap.

Width of groove in new piston for oil control piston ring ... 4.042 ± 0.012 mm (0.1591 ± 0.0005 inch)

Minimum depth of groove in new piston for oil control piston ring ... 3.727 mm (0.1467 inch)

Thickness of new oil control piston ring ... 3.980 ± 0.010 mm (0.1567 ± 0.0004 inch)

Clearance between groove and oil control piston ring ... 0.030 to 0.080 mm (0.0012 to 0.0031 inch)

When a new piston ring is installed in a cylinder liner with a bore of 130 mm (5.1 inch), the clearance between the ends of the piston ring is the following value: ... 0.45 ± 0.15 mm (0.018 ± 0.006 inch)

Increase in clearance for the piston ring ... 0.03 mm (0.001 inch)

Increase in the cylinder liner bore ... 0.09 mm (0.004 inch)

After the piston rings have been installed, rotate the piston rings so that the end gaps are 120 degrees from each other.

Piston Pin Bore (6)

Note: Thoroughly lubricate the piston pin with clean engine oil prior to assembly.

Bore in piston skirt (7) for the piston pin ... 53.190 ± 0.005 mm (2.0941 ± 0.0002 inch)

Bore in the piston crown assembly (3) for the piston pin ... 53.205 ± 0.010 mm (2.0947 ± 0.0004 inch)



Specifications

C-10 and C-12 Truck Engines

Media Number -REN2234-08

Publication Date -01/11/2007

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i02851353

Piston Cooling Jet

SMCS - 1331

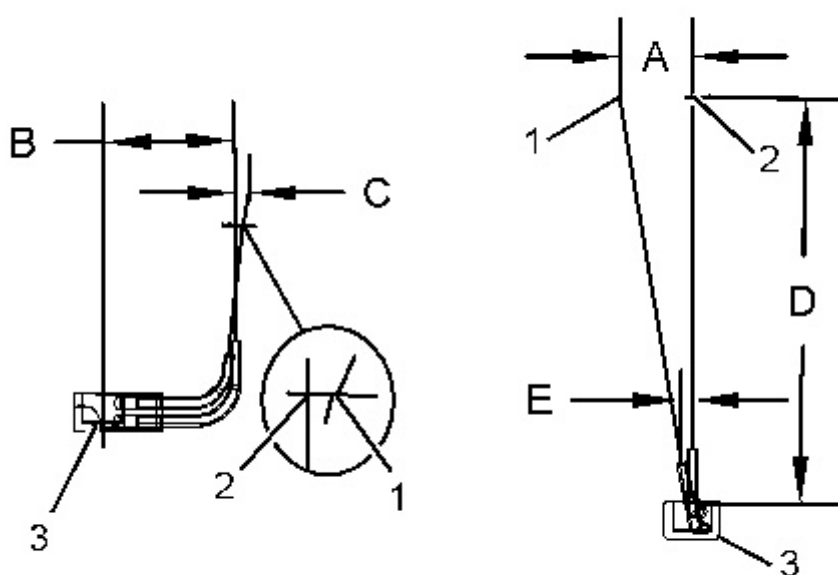


Illustration 1

g01416036

The piston cooling jets (3) must be checked for the location of the stream of oil. Insert a drill rod with a diameter of 2.0 mm (0.08 inch) into the orifice. This drill rod simulates the stream of oil under normal operating pressure. The drill rod must pass through a circle with a diameter of 10.0

mm (0.39 inch) at point (2) . Place the drill rod in the second orifice. The drill rod must pass through a circle with a diameter of 10.0 mm (0.39 inch) at point (1) . The circles are located at dimension (D) .

Use the following dimensions in order to locate point (2) and point (1) .

- Dimension (A) ... 38.44 ± 0.25 mm (1.513 ± 0.010 inch)
- Dimension (B) ... 52.30 ± 0.25 mm (2.059 ± 0.010 inch)
- Dimension (C) ... 15.46 ± 0.25 mm (0.609 ± 0.010 inch)
- Dimension (D) ... 250.00 ± 0.25 mm (9.842 ± 0.010 inch)
- Dimension (E) ... 4.4 mm (0.17 inch)

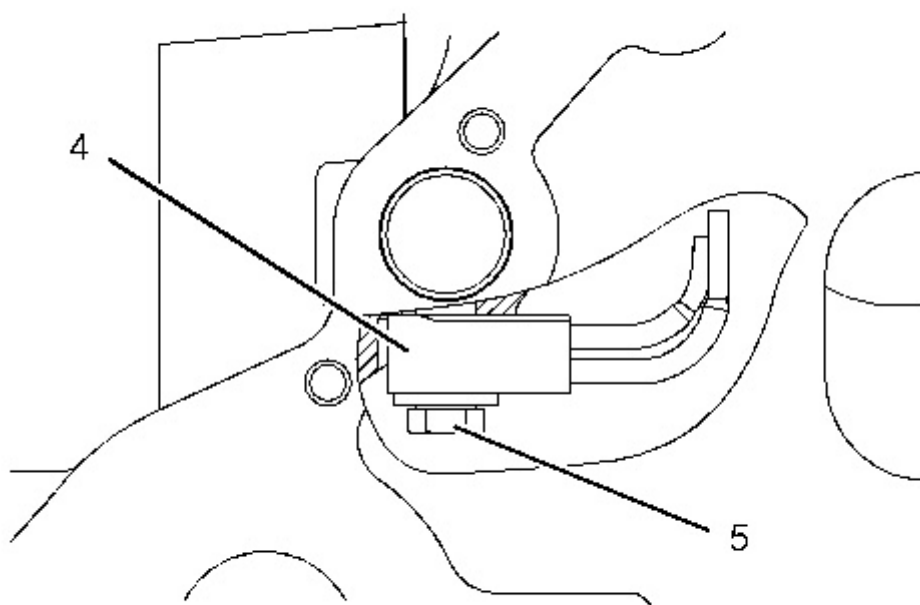


Illustration 2

g01325266

(4) Piston cooling jet

(5) Torque for the bolt ... 30 ± 7 N·m (22 ± 5 lb ft)



Specifications

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i02887867

Rear Power Take-Off (RPTO)

SMCS - 1165-RE

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

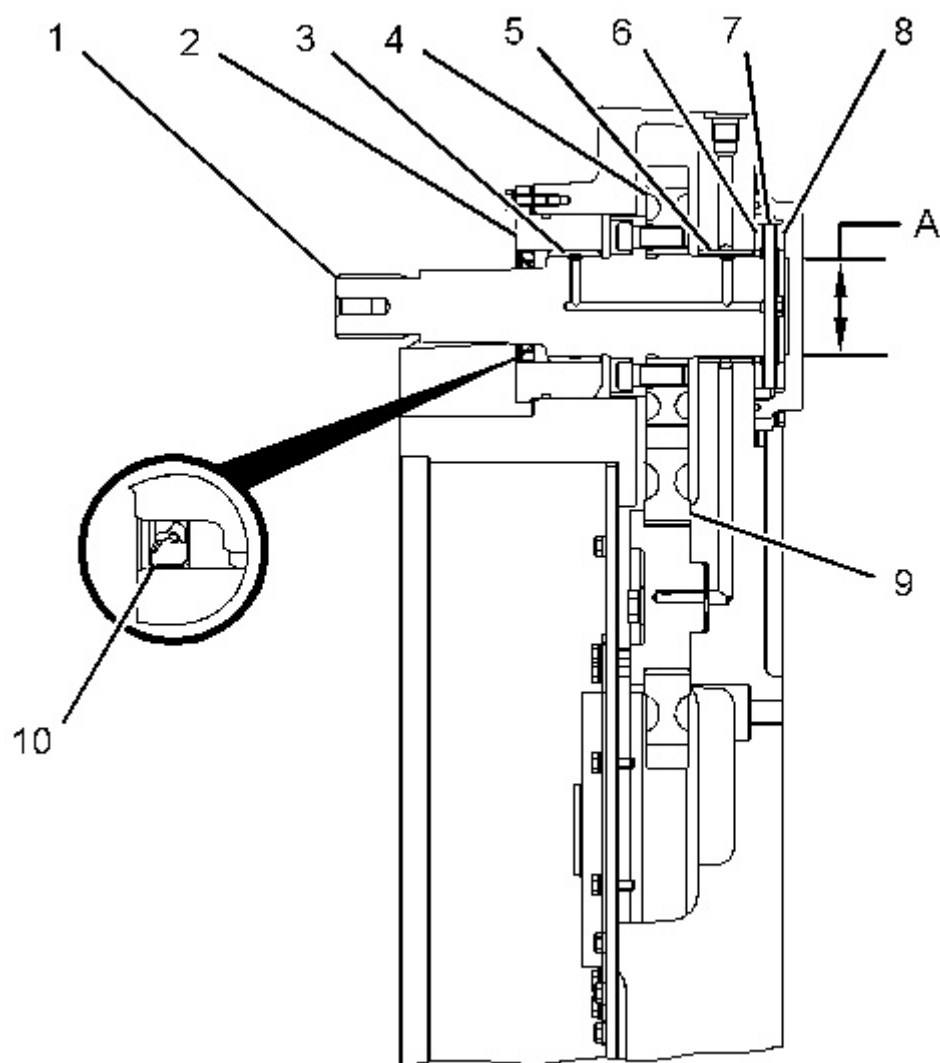


Illustration 1

g01437478

(2) Carrier bearing housing

(5) Bearing

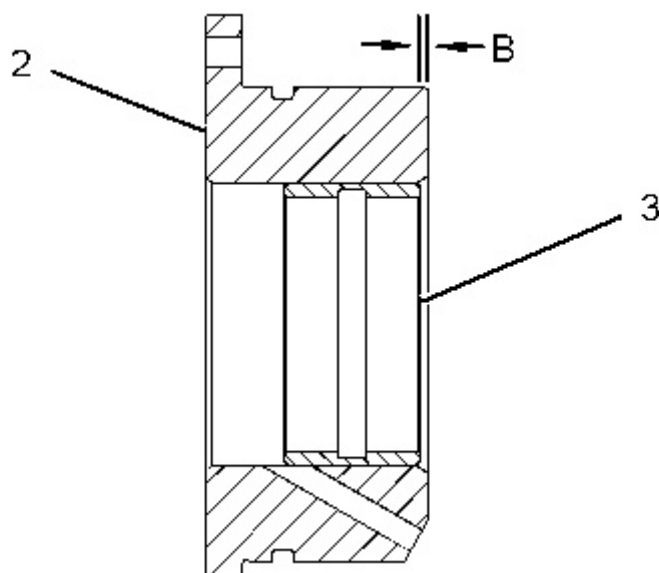


Illustration 2

g01437480

(1) Output shaft (PTO)

Diameter of the output shaft (A) ... 60.245 ± 0.025 mm (2.3718 ± 0.0010 inch)

End play of the output shaft ... 0.97 ± 0.43 mm (0.038 ± 0.017 inch)

(3) Carrier bearing

The carrier bearing should be installed to the following depth (B) in the carrier bearing housing. ... 1.6 ± 0.5 mm (0.06 ± 0.02 inch)

When the engine is viewed from the rear, the split in the bearing should be located counterclockwise from the center of the top bolt at the following angle. ... 112 ± 10 degrees

(4) Drive gear

Inside diameter of the drive gear ... 60.320 ± 0.025 mm (2.3750 ± 0.0010 inch)

The clearance between the output shaft (1) and the drive gear (4) ... 0.025 to 0.125 mm (0.0010 to 0.0050 inch)

(6) Thrust washer

Thickness of the thrust washer ... 4.690 ± 0.025 mm (0.1846 ± 0.0010 inch)

(7) Thrust plate

Thickness of the thrust plate ... 6.75 mm (0.266 inch)

(8) Thrust washer

Thickness of the thrust washer ... 4.690 ± 0.025 mm (0.1846 ± 0.0010 inch)

(9) Idler gear

Inside diameter of the idler gear ... 88.90 ± 0.013 mm (3.500 ± 0.0005 inch)

(10) Seal

Lubricate the sealing lip lightly with the lubricant that is being sealed.

Install the seal to the following depth. ... 2.5 ± 0.5 mm (0.10 ± 0.02 inch)

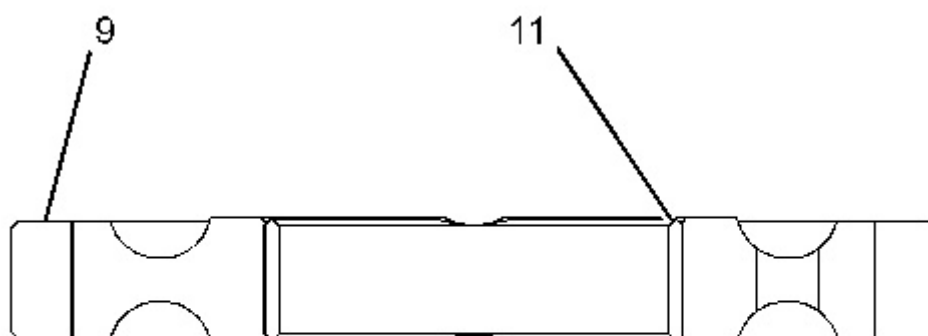


Illustration 3

g01437483

(11) Bearing

The bearing must not extend beyond either face of the idler gear.

The relief groove in the bearing must line up with the relief grooves in the idler gear within the following value. ... 2 degrees



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i02889818

Rocker Shaft

SMCS - 1102

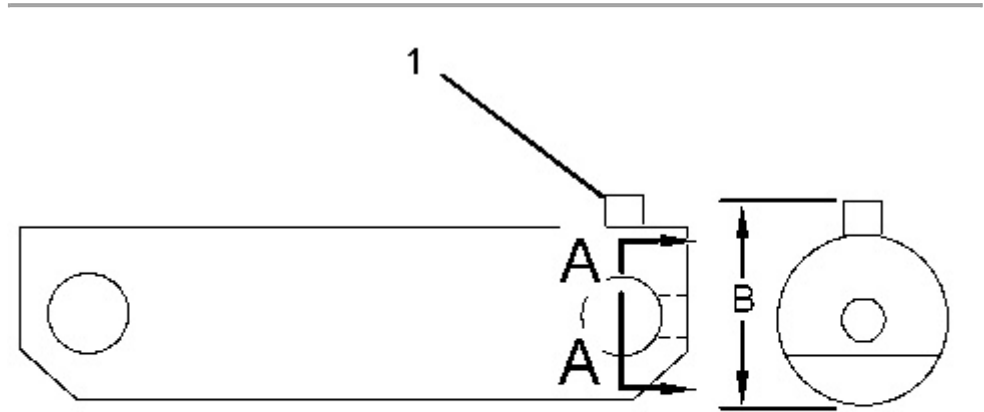


Illustration 1

g01439035

(1) Pin

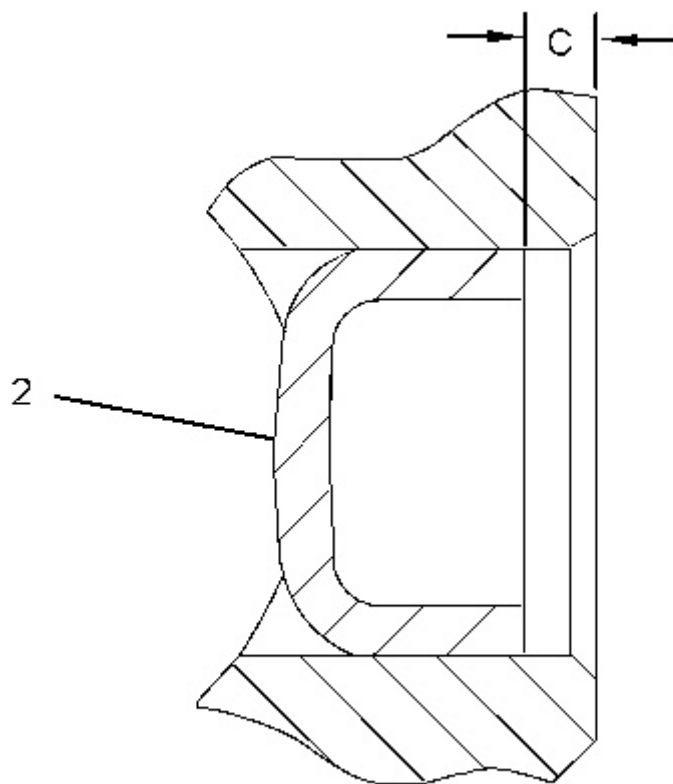


Illustration 2

g01439038

View A-A

(B) Distance from the top surface of the pin to the opposite side of the shaft ... 40.0 ± 1.0 mm
(1.57 ± 0.04 inch)

(2) The plug is installed into the end of the shaft.

(C) Install the plug to the following depth. ... 1.25 ± 0.25 mm (0.049 ± 0.010 inch)



Specifications

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i02896822

Speed/Timing Sensor

SMCS - 1907; 1912

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

Secondary Speed Sensor

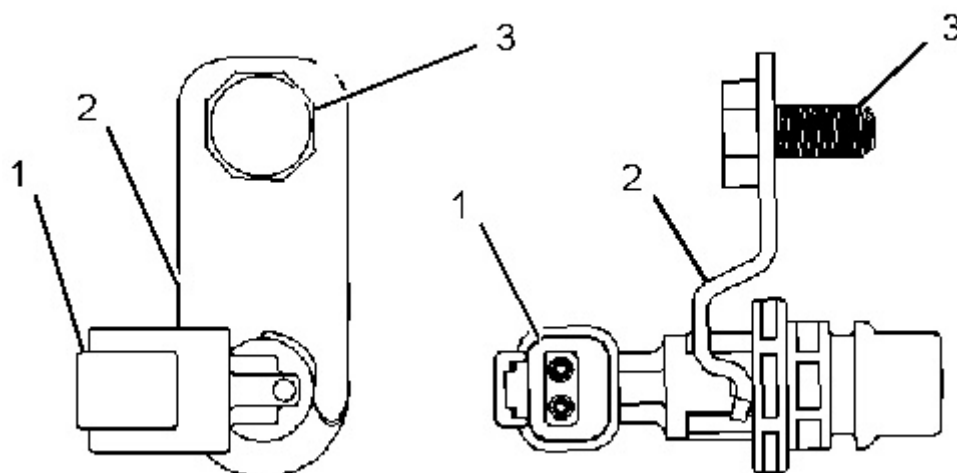


Illustration 1

g01432017

Camshaft position sensor

(1) Sensor

(2) Bracket

Ensure that the bracket is installed in the orientation that is shown.

(3) Tighten the bolt to the following torque: ... $28 \pm 7 \text{ N}\cdot\text{m}$ ($21 \pm 5 \text{ lb ft}$)

Ensure that the sensor is seated before the bolt is tightened.

Primary Speed Sensor

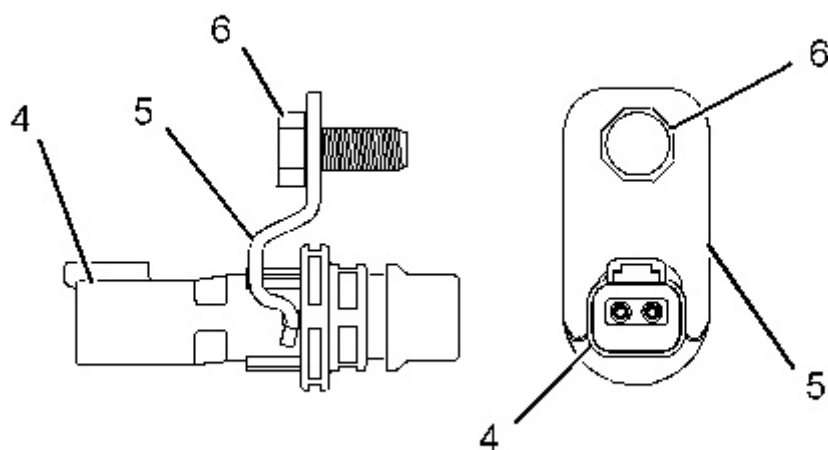


Illustration 2

g01432020

Crankshaft position sensor

(4) Sensor

(5) Bracket

Ensure that the bracket is installed in the orientation that is shown.

(6) Tighten the bolt to the following torque: ... $28 \pm 7 \text{ N}\cdot\text{m}$ ($21 \pm 5 \text{ lb ft}$)

Ensure that the sensor is seated before the bolt is tightened.



Specifications

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i02894588

Turbocharger

SMCS - 1052

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

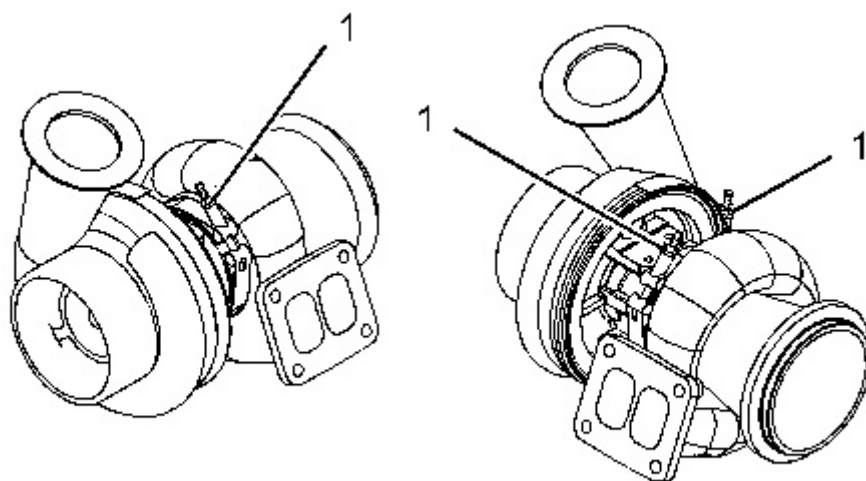


Illustration 1

g01440729

Typical example

(1) Clamp

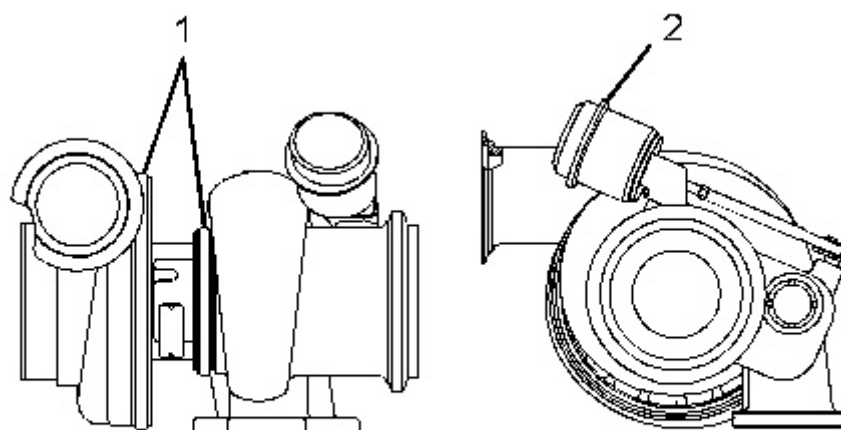


Illustration 2

g01440730

Typical example of a turbocharger with a wastegate

(1) Clamp

(2) Wastegate

(1) Clamps

To tighten clamps (1) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. ... $14 \pm 1 \text{ N}\cdot\text{m}$ ($10.3 \pm 0.7 \text{ lb ft}$)

2. Gently tap the clamp with a soft faced hammer.

3. Again tighten the clamp.

Tighten the clamp to the following torque. ... $14 \pm 1 \text{ N}\cdot\text{m}$ ($10.3 \pm 0.7 \text{ lb ft}$)

(2) Wastegate

Opening pressure of the wastegate ... 138 kPa (20 psi)

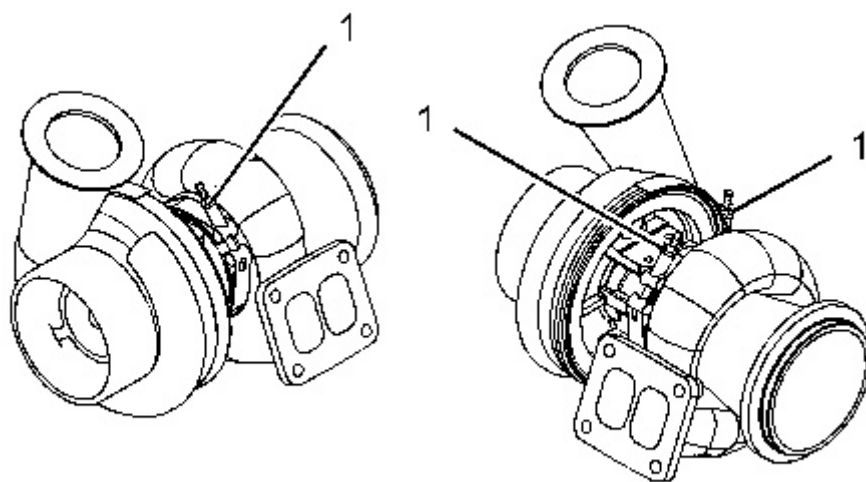


Illustration 3

g01440731

Typical example

(1) Clamp

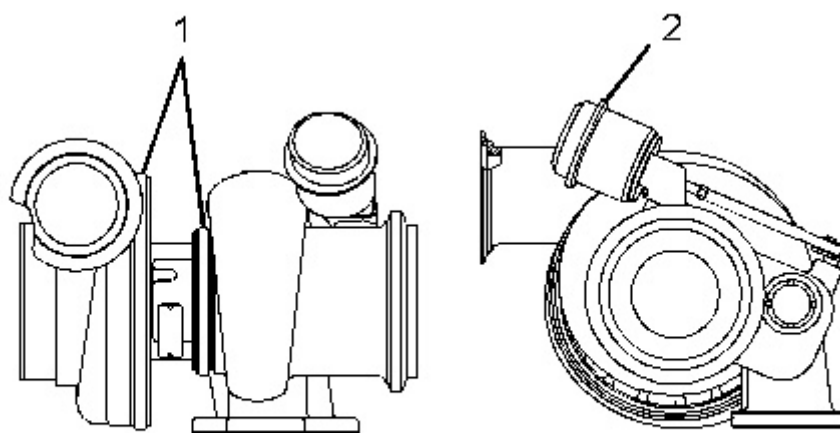


Illustration 4

g01440732

Typical example of a turbocharger with a wastegate

(1) Clamp

(2) Wastegate

(1) Clamps

To tighten clamps (1) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. ... $14 \pm 1 \text{ N}\cdot\text{m}$ ($10.3 \pm 0.7 \text{ lb ft}$)

2. Gently tap the clamp with a soft faced hammer.

3. Again tighten the clamp.

Tighten the clamp to the following torque. ... $14 \pm 1 \text{ N}\cdot\text{m}$ ($10.3 \pm 0.7 \text{ lb ft}$)

(2) Wastegate

Opening pressure of the wastegate ... 193 kPa (28 psi)

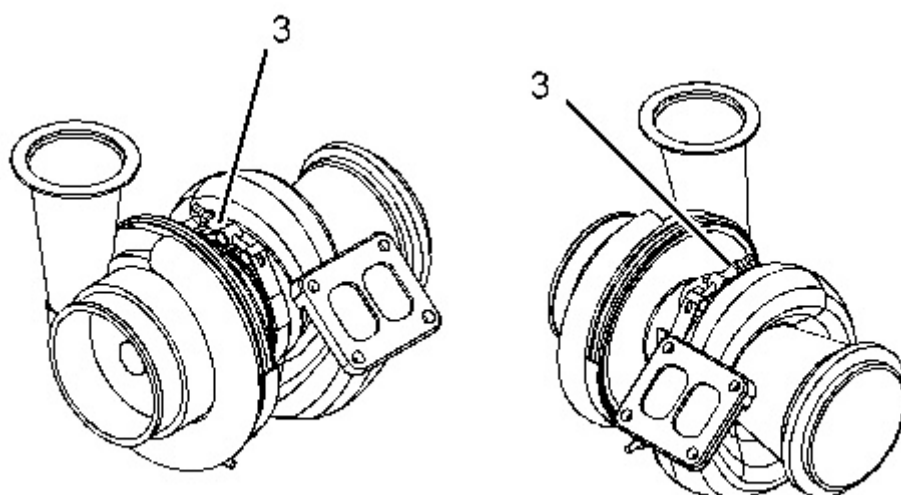


Illustration 5

g01440733

Typical example

(3) Clamp

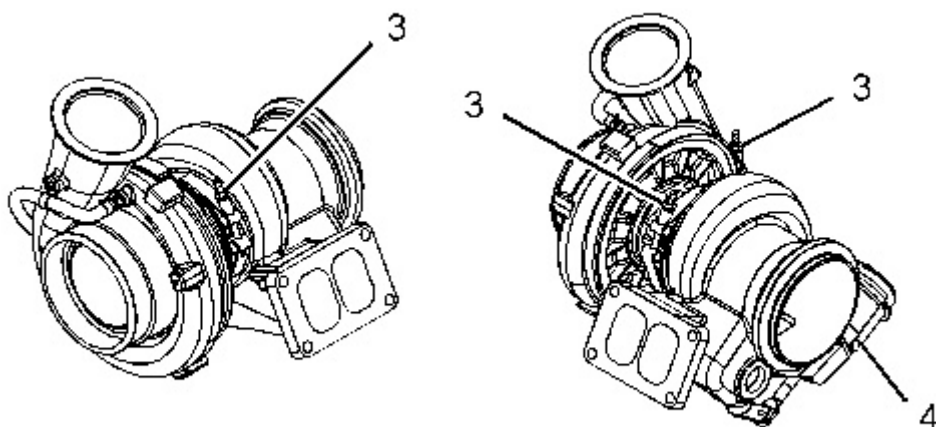


Illustration 6

g01440735

Typical example of a turbocharger with a wastegate

(3) Clamp

(4) Wastegate

(3) Clamp

To tighten the clamp (3) on the turbocharger, use the procedure that follows.

1. Tighten the clamp.

Tighten the clamp to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the clamp.

Tighten the clamp to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(4) Wastegate

Opening pressure of the wastegate ... 156 kPa (23 psi)

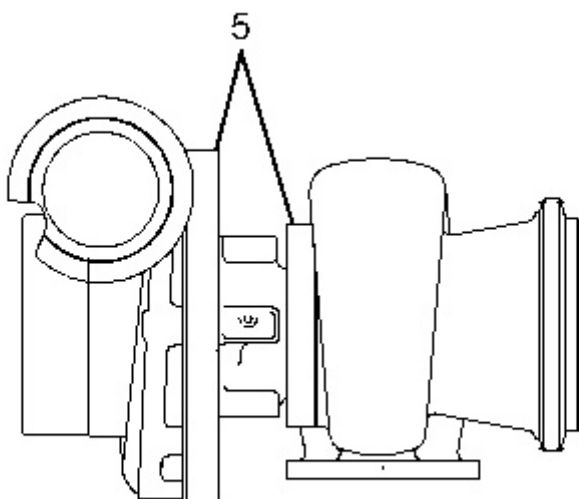


Illustration 7

g01440737

Typical example

(5) Clamp

(5) Clamps

To tighten clamps (5) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. ... $18 \pm 3 \text{ N}\cdot\text{m}$ ($13.3 \pm 2.2 \text{ lb ft}$)

2. Gently tap the clamp with a soft faced hammer.

3. Again tighten the clamp.

Tighten the clamp to the following torque. ... $18 \pm 3 \text{ N}\cdot\text{m}$ ($13.3 \pm 2.2 \text{ lb ft}$)

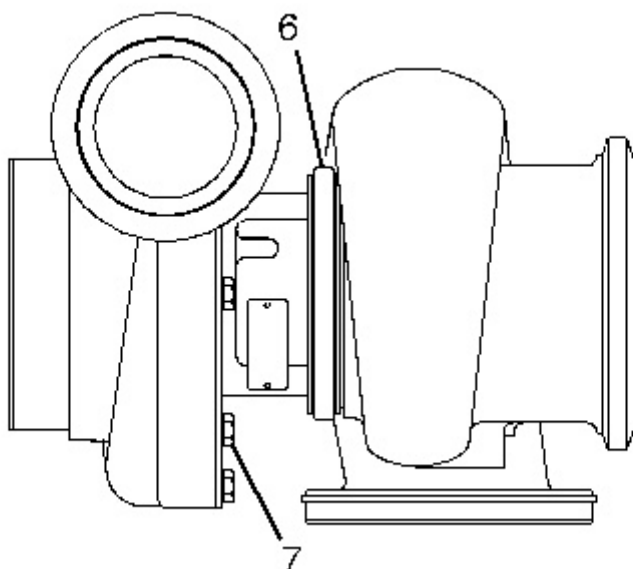


Illustration 8

g01440738

Typical example

(6) Clamp

(7) Bolt

(6) Clamp

To tighten the clamp (6) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp nut to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the clamp.

Loosen the clamp nut to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the clamp.

Tighten the clamp nut to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(7) Bolt

Tighten the bolts (7) for the compressor clamp assembly in opposite pairs. Tighten the bolts for the compressor clamp assembly to the following torque 17.5 ± 1 N·m (12.90 ± 0.70 lb ft).

Apply **5P-3931** Anti-Seize Compound to the nuts that are used to mount the turbocharger. Tighten the nuts to the following torque 70 ± 15 N·m (51.6 ± 11.0 lb ft).

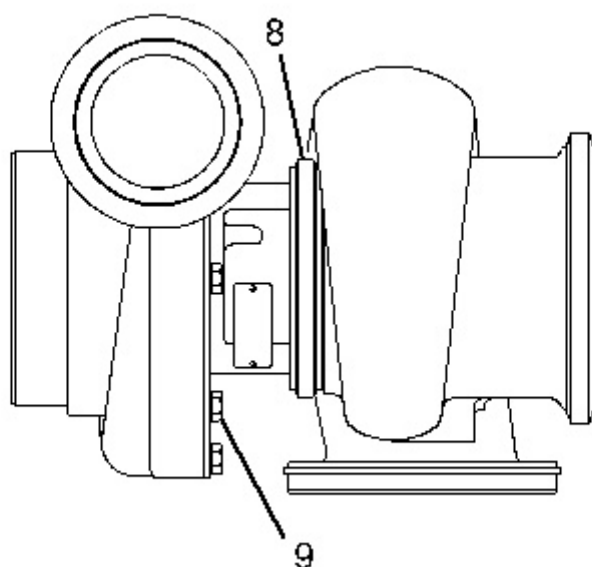


Illustration 9

g01440739

Typical example

(8) Clamp

(9) Bolt

(8) Clamp

To tighten the clamp (8) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the clamp.

Tighten the clamp to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(9) Bolt

Tighten the bolt to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

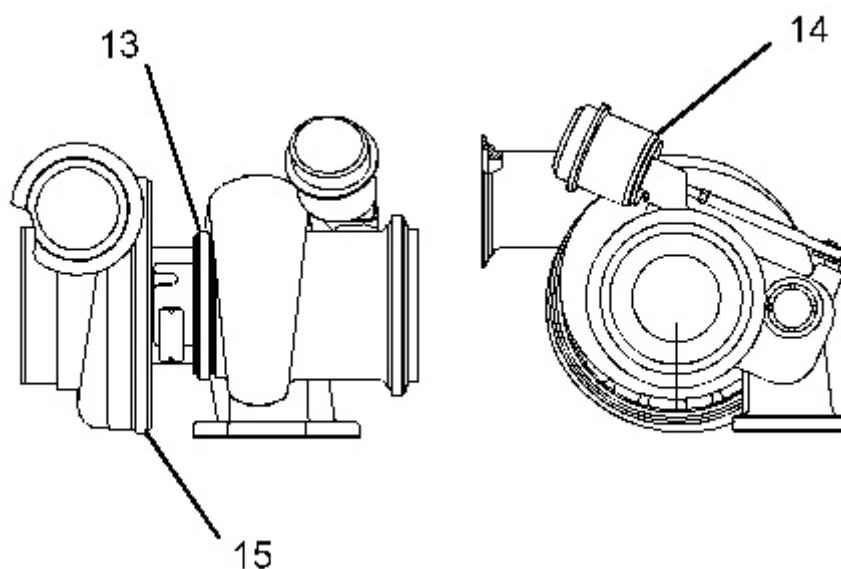


Illustration 10

g01440751

Typical example of a turbocharger with a wastegate

(10) Clamp

(11) Wastegate

(12) Clamp

(10) Clamp

To tighten the clamp (10) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the clamp.

Tighten the clamp to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(11) Wastegate

Opening pressure of the wastegate ... 156 kPa (23 psi)

(12) Clamp

Tighten the clamp (12) to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

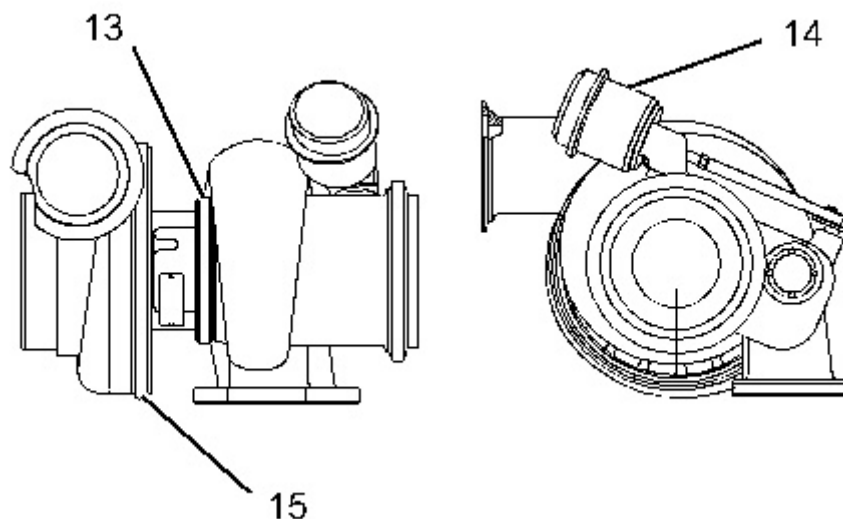


Illustration 11

g01440754

Typical example of a turbocharger with a wastegate

(13) Clamp

(14) Wastegate

(15) Clamp

(13) Clamp

To tighten the clamp (13) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the clamp.

Tighten the clamp to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(14) Wastegate

Opening pressure of the wastegate ... 165 kPa (24 psi)

(15) Clamp

Tighten the clamp to the following torque. ... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

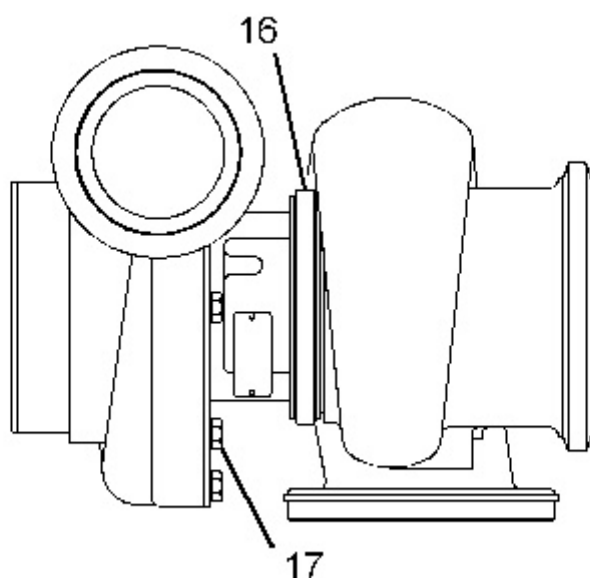


Illustration 12

g01440757

Typical example

(16) Clamp

(17) Bolt

(16) Clamp

To tighten the clamp (16) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the clamp.

Tighten the clamp to the following torque. ... 13 ± 1 N·m (9.6 ± 0.7 lb ft)

(17) Bolt

To tighten the Bolt (17) on the turbocharger, use the following procedure:

1. Tighten the bolt.

Tighten the bolt to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the bolt.

Loosen the bolt to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the bolt.

Tighten the bolt to the following torque. ... $13 \pm 1 \text{ N}\cdot\text{m}$ ($9.6 \pm 0.7 \text{ lb ft}$)

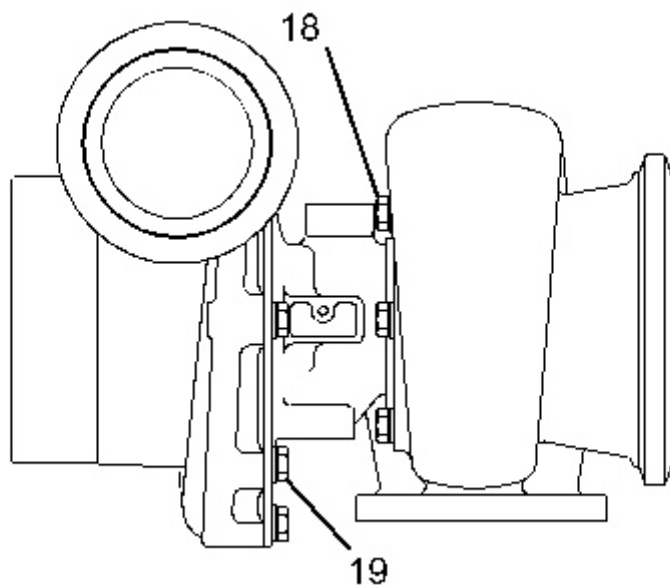


Illustration 13

g01440766

Typical example

(18) Bolt

(19) Bolt

(18) Bolt

1. Tighten the bolt.

Tighten the turbine clamp plate bolts (18) to the following torque. ... $15.8 \pm 0.6 \text{ N}\cdot\text{m}$ ($11.6 \pm 0.4 \text{ lb ft}$)

(19) Bolt

1. Tighten the bolt.

Tighten the compressor clamp plate bolts (19) in opposite pairs. Tighten the compressor clamp plate bolts to the following torque. ... $17.5 \pm 1 \text{ N}\cdot\text{m}$ ($12.9 \pm 0.7 \text{ lb ft}$)

Apply **5P-3931** Anti-Seize Compound to the nuts that are used to mount the turbocharger. Tighten the nuts to the following torque $70 \pm 15 \text{ N}\cdot\text{m}$ ($51.6 \pm 11.0 \text{ lb ft}$).



Specifications

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i02093112

Valve Mechanism Cover

SMCS - 1107

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

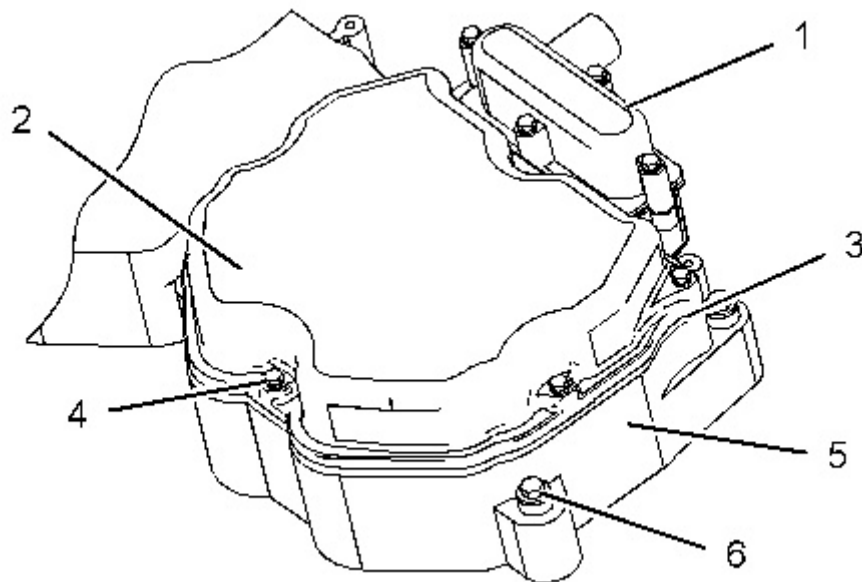


Illustration 1

g01067262

(1) Breather assembly

The conditions that follow will allow the crankcase pressure to be less than 0.4 kPa (0.06 psi).

Inside diameter of fumes disposal tube ... 31.8 mm (1.25 inch)

Blowby per hour ... 19.82 cubic meters

(2) Valve mechanism covers

Cut the seal (3) to length for each of the three valve mechanism covers. Coat the joint surface with **3S-6252** Sealant.

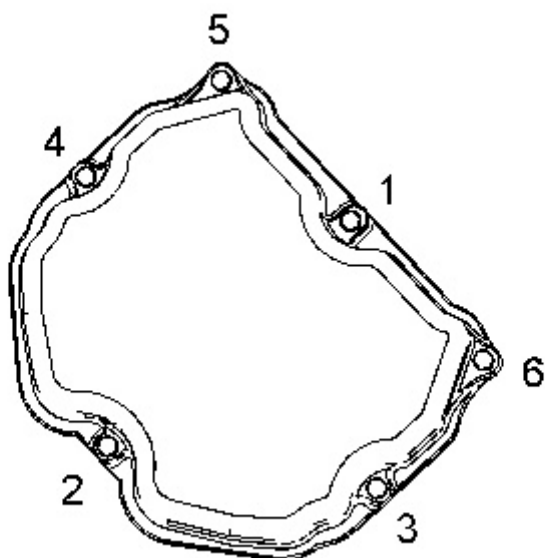


Illustration 2

g01067307

(4) Tighten bolts 1 through 6 in a numerical sequence to the following torque. ... 12 ± 3 N·m (105 $\pm$ 27 lb in)

Again tighten bolts 1 through 6 in a numerical sequence to the following torque. ... 12 ± 3 N·m (105 $\pm$ 27 lb in)

(5) Valve mechanism cover base

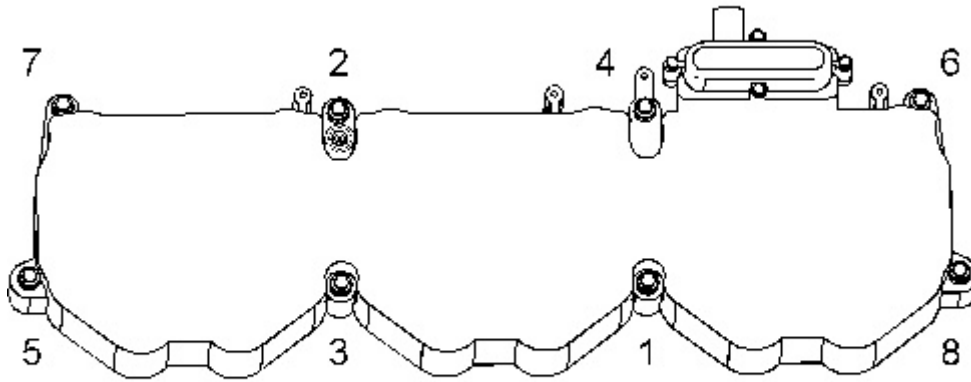


Illustration 3

g01067433

(6) Tighten bolts 1 through 8 in a numerical sequence to the following torque. ... $30 \pm 3 \text{ N}\cdot\text{m}$ ($22 \pm 2 \text{ lb ft}$)



Specifications
C-10 and C-12 Truck Engines
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Publication Date -01/11/2007

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i02879480

Valve Mechanism

SMCS - 1102

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

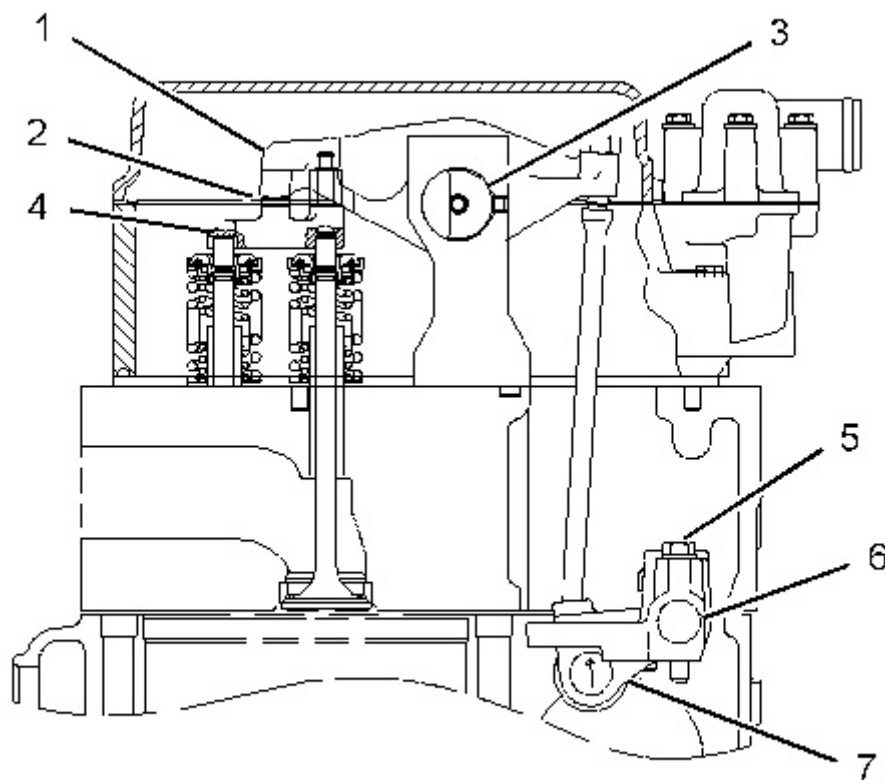


Illustration 1

g01416165

(1) Valve rocker arm

(4) Valve bridge

(7) Valve lifter

(2) Valve lash (engine stopped)

Inlet valves ... 0.38 ± 0.08 mm (0.015 ± 0.003 inch)Exhaust valves ... 0.64 ± 0.08 mm (0.025 ± 0.003 inch)**Note:** After setting the valve lash, tighten the adjustment screw locknut. Recheck the valve lash.Torque for the adjustment screw locknut ... 30 ± 7 N·m (22 ± 5 lb ft)

(3) Rocker arm shaft

Diameter of new rocker arm shaft ... 34.000 ± 0.010 mm (1.3386 ± 0.0004 inch)(5) Tighten the bolts to the following torque. ... 30 ± 3 N·m (22 ± 2 lb ft)

(6) Lifter group shaft

Diameter of new lifter group shaft ... 20.000 ± 0.010 mm (0.7874 ± 0.0004 inch)

Bore diameter in the body of the valve lifter ... 20.050 ± 0.015 mm (0.7894 ± 0.0006 inch)



Specifications

C-10 and C-12 Truck Engines

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i02887021

Vibration Damper and Pulley

SMCS - 1205

S/N - 2KS1-UP

S/N - 3CS1-UP

S/N - 8YF1-UP

S/N - CPD1-UP

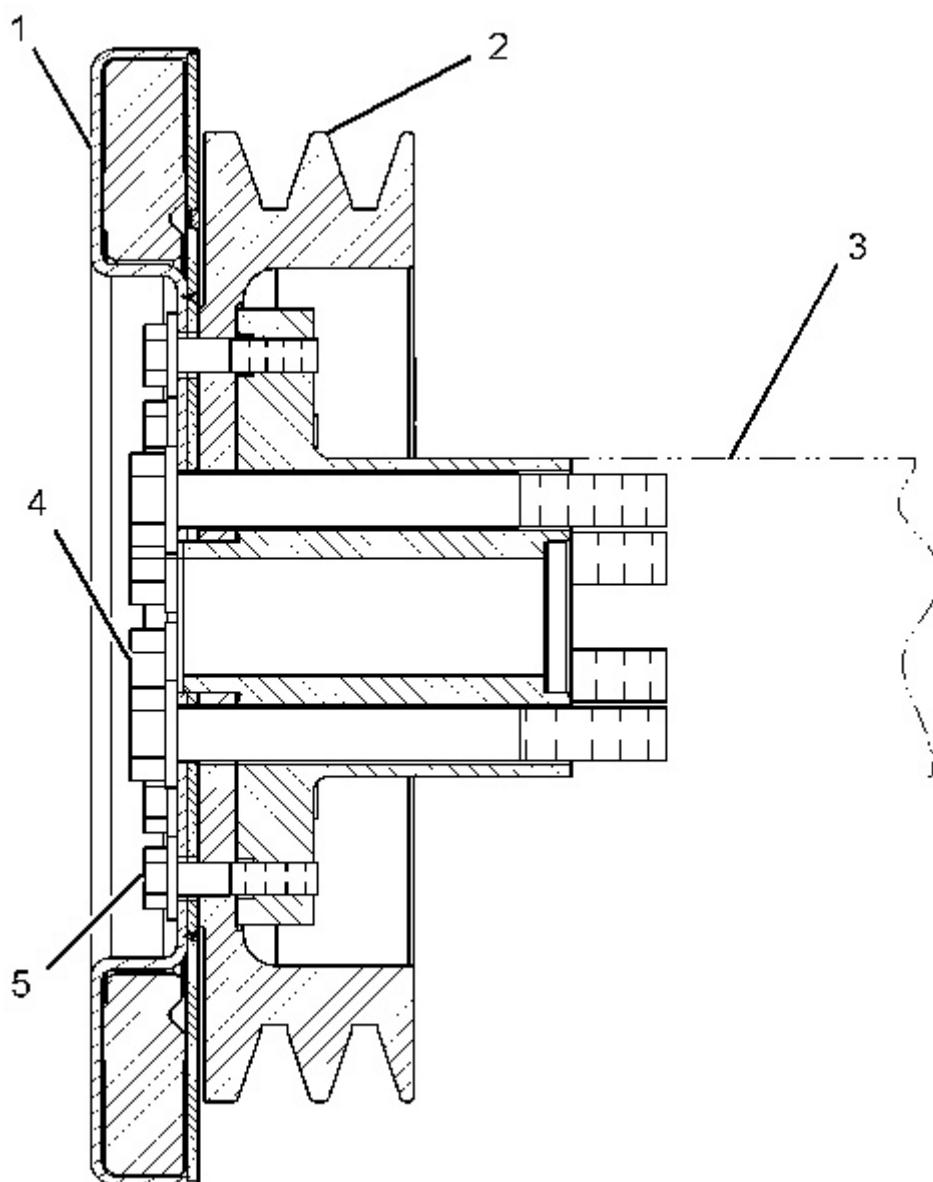


Illustration 1

g01437116

(1) Damper assembly

(2) Pulley

(3) Crankshaft

(4) Tighten the bolts to the following torque. ... $300 \pm 40 \text{ N}\cdot\text{m}$ ($220 \pm 30 \text{ lb ft}$)(5) Tighten the bolts to the following torque. ... $70 \pm 15 \text{ N}\cdot\text{m}$ ($50 \pm 11 \text{ lb ft}$)



Specifications

C-10 and C-12 Truck Engines

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i02886855

Water Pump

SMCS - 1361

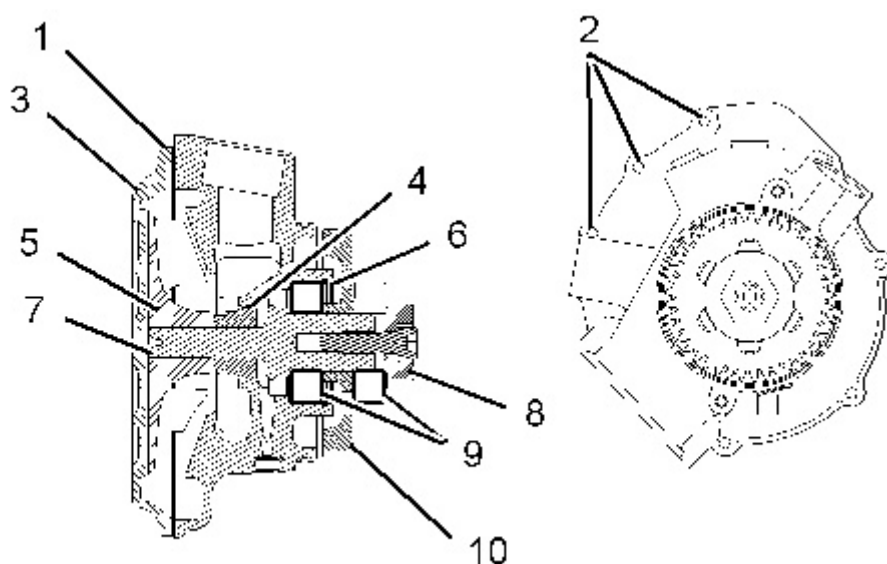


Illustration 1

g01431206

Water pump

- (1) Gasket
- (2) Mounting holes for the impeller cover
- (4) Seal assembly
- (5) Impeller
- (6) Retaining ring
- (7) Shaft

- (8) Bearing retaining ring
- (9) Bearing
- (10) Water pump drive gear

(3) Impeller cover

Tighten the bolts (3) on the impeller cover to the following torque: ... $18 \pm 3 \text{ N}\cdot\text{m}$ ($13 \pm 2 \text{ lb ft}$)

Note: The impeller is pressed onto shaft (7) . The face of the impeller is flush to the end of the shaft.

The clearance between the impeller and the water pump housing is $0.94 \pm 0.47 \text{ mm}$ ($0.037 \pm 0.019 \text{ inch}$).

Note: Clearance between the impeller and the water pump housing is preset once the impeller is pressed onto the shaft.

The clearance between the water pump housing and the water pump drive gear is 2.8 mm (0.11 inch).

Note: Clearance between the water pump housing and the water pump drive gear is preset once the bearings are tightened against the water pump drive gear.



Specifications

C-10 and C-12 Truck Engines

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Water Temperature Regulator

SMCS - 1355

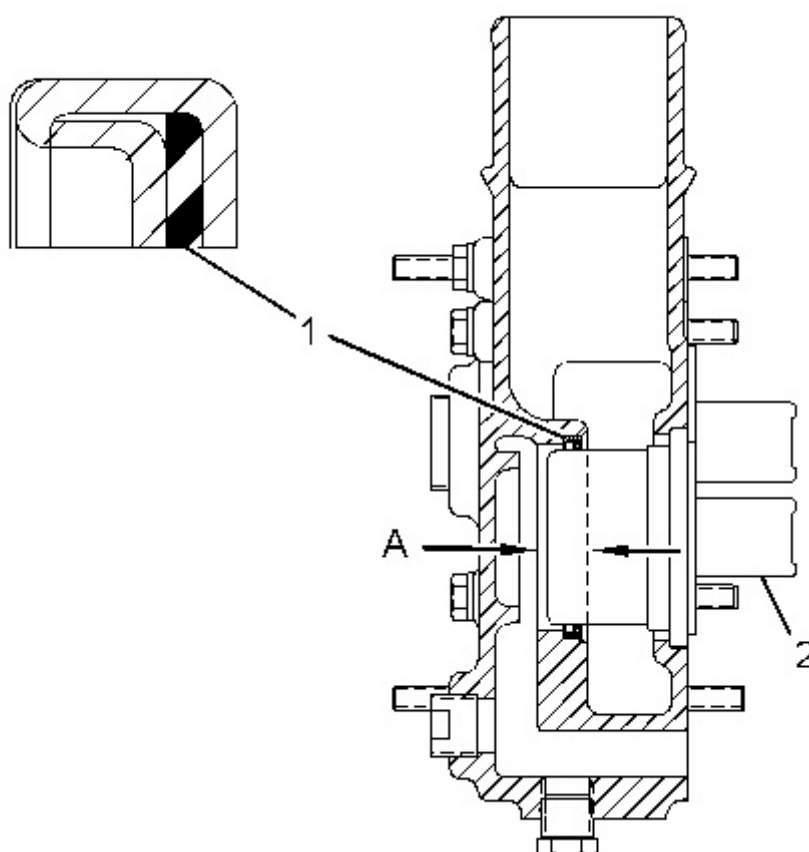


Illustration 1

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(1) Install the lip type seal into the water temperature regulator housing. Lubricate the sealing lip on the lip type seal with a small amount of antifreeze.

(A) Minimum stroke of the water temperature regulator at fully open temperature ... 10.4 mm
(0.41 inch)

(2) Fully open temperature of the water temperature regulator ... 98 °C (208 °F)



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